Bleeding after birth

Course on prevention, diagnosis and treatment of postpartum haemorrhage



jhpiego
Saving lives. Improving health. Transforming futures.

FIGO
International Federation of Gynecology and Obstetrics

International Confederation of Midwives

Laerdal
helping save lives

UNFPA

human reproduction programme hrp
research for impact
UNDP · UNFPA · UNICEF · WHO · WORLD BANK

World Health Organization

Bleeding after birth: course on prevention, diagnosis and treatment of postpartum haemorrhage.

ISBN 978-92-4-011583-5 (print version)

ISBN 978-92-4-011584-2 (electronic version)

© World Health Organization 2025

Some rights reserved. This work is available under the Creative Commons Attribution-NonCommercial-ShareAlike 3.0 IGO licence (CC BY-NC-SA 3.0 IGO; https://creativecommons.org/licenses/by-nc-sa/3.0/igo).

Under the terms of this licence, you may copy, redistribute and adapt the work for non-commercial purposes, provided the work is appropriately cited, as indicated below. In any use of this work, there should be no suggestion that WHO endorses any specific organization, products or services. The use of the WHO logo is not permitted. If you adapt the work, then you must license your work under the same or equivalent Creative Commons licence. If you create a translation of this work, you should add the following disclaimer along with the suggested citation: "This translation was not created by the World Health Organization (WHO). WHO is not responsible for the content or accuracy of this translation. The original English edition shall be the binding and authentic edition".

Any mediation relating to disputes arising under the licence shall be conducted in accordance with the mediation rules of the World Intellectual Property Organization.

Suggested citation. Bleeding after birth: course on prevention, diagnosis and treatment of postpartum haemorrhage. Geneva: World Health Organization; 2025. Licence: CC BY-NC-SA 3.0 IGO.

Cataloguing-in-Publication (CIP) data. CIP data are available at http://apps.who.int/iris.

Sales, rights and licensing. To purchase WHO publications, see http://apps.who.int/bookorders. To submit requests for commercial use and queries on rights and licensing, see http://www.who.int/about/licensing.

Third-party materials. If you wish to reuse material from this work that is attributed to a third party, such as tables, figures or images, it is your responsibility to determine whether permission is needed for that reuse and to obtain permission from the copyright holder. The risk of claims resulting from infringement of any third-party-owned component in the work rests solely with the user.

General disclaimers. The designations employed and the presentation of the material in this publication do not imply the expression of any opinion whatsoever on the part of WHO concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. Dotted and dashed lines on maps represent approximate border lines for which there may not yet be full agreement.

The mention of specific companies or of certain manufacturers' products does not imply that they are endorsed or recommended by WHO in preference to others of a similar nature that are not mentioned. Errors and omissions excepted, the names of proprietary products are distinguished by initial capital letters. All reasonable precautions have been taken by WHO to verify the information contained in this publication. However, the published material is being distributed without warranty of any kind, either expressed or implied. The responsibility for the interpretation and use of the material lies with the reader. In no event shall WHO be liable for damages arising from its use.

Contents

Acknowledgements iv
Acronyms and abbreviations v
Plan and prepare to facilitate a course 1
Using this Flip Chart to facilitate learning 1b

1. Saving lives at birth 2
Main causes of bleeding after birth 3
Provide respectful care and prevent infection 4
Prepare for birth, check equipment and medications 5
Assess and record 6

2. Prevention and early diagnosis of PPH 7
Give a uterotonic within 1 min 8
Start measuring blood loss 9
How accurate is visual estimation of blood loss? 10
Be alert. If PPH, start bundle 11
Cut the cord and perform controlled cord traction 12
Check tone, massage if soft, check bleeding 13
Check placenta 14
Check for tears and repair 15
Continue care for woman and baby 16
Scenario practice 1 17
Empty bladder and repeat controlled cord traction 18
Remove manually 19

3. Treatment with the PPH bundle 20
Call for help 21
Start PPH bundle 22
Examine and address causes 23
Scenario practice 2 24
Scenario practice 3 25
Scenario practice 4 26

4. Continued treatment if bleeding persists 27
Arrange transfer or surgery 28
Rapidly apply NASG 29
Skills practice – NASG 30
Continue massage, catheterize bladder, remove clots/debris 31
Start second IV, give additional uterotonics and TXA second dose, assess and record 32
Scenario practice 5 33

5. Advanced interventions and post-PPH care 34
Bimanual compression 35
Insert uterine balloon tamponade 36
Repair deep tears 37
Provide post-PPH care 38

6. Ongoing practice and quality improvement 39
References 39b

Acknowledgements

The World Health Organization (WHO) acknowledges the contributions of individuals and partner organizations to the development of Bleeding after birth: course on prevention, diagnosis and treatment of postpartum haemorrhage. WHO thanks Jhpiego, an international non-profit health organization affiliated with Johns Hopkins University, for its leadership in updating the Bleeding after Birth Complete 3.0 module from the Helping Mothers Survive programme. This updated module, led by Cherrie Lynn Evans, was developed in response to updated WHO recommendations and emerging evidence on the new diagnostic criteria for earlier diagnosis of postpartum haemorrhage. The original module was developed by Jhpiego's Technical Leadership Office in collaboration with Laerdal Global Health, and built on the educational design of Helping Babies Breathe, created by the American Academy of Pediatrics and Laerdal Global Health.

The development of the technical content in this update was initiated by Susheela Engelbrecht, Cherrie Lynn Evans and Jennifer Stevens from Jhpiego, in collaboration with Teesta Dey and Ioannis Gallos from WHO Department of Sexual, Reproductive, Maternal, Child, Adolescent Health and Ageing, who ensured alignment with WHO guidelines and current evidence. WHO acknowledges the substantial technical contributions provided by Jolly Beyeza-Kashesya, Anne Beatrice Kihara and Alison Wright from the International Federation of Gynecology and Obstetrics (FIGO); Jacqueline Dunkley-Bent, Zab Frankling, Liselotte Kweekel, Stephanie Marriott and Kate Stringer from the International Confederation of Midwives (ICM); and Muna Abdullah, Manuel Couffignal, Catherine Howard Taylor, Chisato Masuda, Isabelle Moreira, Catrin Schulte-Hillen and Duncan Shikuku from the United Nations Population Fund (UNFPA). WHO also thanks Camila Barrera Daza, Bjørn Mike Boge and the Laerdal Global Health design team for their contributions to the educational design of the module.

Special thanks are extended to reviewers from each of the six WHO regions: Hadil Ali-Masri (Ministry of Health, Occupied Palestinian Territory, including east Jerusalem); Ferdousi Begum (Ibrahim Medical College and Birdem Hospital, Bangladesh); Brendan Carvalho (Stanford University School of Medicine, United States of America); Catherine Deneux-Tharaux (INSERM, Université Paris Cité, France); Caroline Homer (Burnet Institute, Australia); and Angela Nguku (White Ribbon Alliance, Kenya). WHO further acknowledges the contributions of its Regional Advisors for Maternal and Newborn Health, whose insights helped ensure the relevance and applicability of the course across diverse health system contexts: Adeniyi Aderoba (WHO Regional Office for Africa); Suzanne Serruya (Pan American Health Organization/WHO Regional Office for the Americas); Karima Gholbzouri (WHO Regional Office for the Eastern Mediterranean); Oleg Kuzmenko (WHO Regional Office for Europe); Chandani Anoma Jayathilaka (WHO Regional Office for South-East Asia); and Delgermaa Vanya (WHO Regional Office for the Western Pacific). WHO expresses its sincere gratitude to the health workers in Afghanistan, Democratic Republic of the Congo, Guinea, India, Kenya, Malawi, Nigeria, United Republic of Tanzania, and Zambia, whose participation in testing and refining the course materials was instrumental to its development.

This work was funded by the University of Birmingham, Jhpiego, Laerdal Foundation and UNDP-UNFPA-UNICEF-WHO-World Bank Special Programme of Research, Development and Research Training in Human Reproduction (HRP), a co-sponsored programme executed by WHO. The views of the funding bodies have not influenced the content of this course.

Acronyms and abbreviations

ANC	antenatal care
BP	blood pressure
CCT	controlled cord traction
CS	caesarean section
FBC	full blood count
FIGO	International Federation of Gynecology and Obstetrics
Hb	Haemoglobin
HSC	heat-stable carbetocin
ICM	International Confederation of Midwives
IM	intramuscular
IV	intravenous
IU	international units
LDHF	low-dose, high-frequency
NASG	non-pneumatic anti-shock garment
NS	normal saline
OSCE	objective structured clinical examination
PO	oral administration
PPH	postpartum haemorrhage
PR	per rectum
RL	Ringer's Lactate
SBAR	situation, background, assessment, recommendation
SI	shock index
SL	sublingual
TXA	tranexamic acid
UNFPA	United Nations Population Fund
UBT	uterine balloon tamponade
WHO	World Health Organization

For the facilitator

Plan and prepare to facilitate a course

This course is for all health workers who care for women at birth. All women should give birth in facilities with the capacity to treat postpartum haemorhage (PPH), and administer intravenous (IV) medicines.

Plan and coordinate

- Coordinate with leadership and local organizers well in advance.
- Use the supplementary sample documents – like agendas, room layouts, checklists and evaluations. Download and modify these documents to fit your needs.
- Review clinical data with facility management to identify strengths and gaps.

Adapt the course to the needs

This course is for all health workers who care for women at birth. All women should give birth in facilities with the capacity to treat PPH, including administration of IV medicines.

- In facilities with no operating room and which cannot provide advanced care, women not responding to the PPH bundle should be transferred quickly. Adapt the course based on these conditions.
- Consider excluding from the course, if not available or relevant:
  - non-pneumatic anti-shock garment (NASG) – pages 29 and 30
  - uterine balloon tamponade (UBT) – page 36
  - repairing deep vaginal and cervical tears – page 37.
- Define the agenda based on the content you will be covering.

Prepare yourself to facilitate

- Review the agenda.
- Familiarize yourself with all pages in this flipchart.
- Rehearse in advance how you will deliver the course and how you will best engage the participants.
- Practise each of the demonstrations and simulations.

Set up the course for active participation

The number of participants in a course may vary, but ensure you can divide into small groups of 6 participants to 1 facilitator to ensure the quality of practice activities.

Prepare the learning materials

In addition to this Flip Chart, use these resources to support learning and ensure an impactful course delivery. Make sure you know them well, and have them ready for the course.

Action plans

Two large posters showing the steps to take during care, to be used during the course and as a job aid for clinical practice.

Provider guide

A booklet for every health worker to use during the course and keep afterwards, for continued practice.

Assessments

As part of the learning process, the course includes a pre-course and post-course knowledge check, and six objective structured clinical examinations (OSCEs).

Videos

Demonstration videos are available to view online and download. Use them during the course and share with participants for continued practice.

Prepare to assess

- Assessments support learning and help you identify gaps and needs.
- Ensure participants complete knowledge assessments before and after the course.
- Check skills after the course using OSCEs.

Enable continued practice

Support the facilities to embrace continued practice after the course.

- Make the Action Plans available and visible in the facility.
- Appoint two participants from each facility to become **practice coordinators** and support their peers.
- Ensure participants have the equipment, space and time to carry out low-dose, high-frequency (LDHF) practice at the facilities –using the activities in the Provider Guide.

The Bleeding after birth training course aims to translate standards into care.

It incorporates the latest WHO recommendations related to PPH[1,2] and includes links to key references and resources for training.

Find all learning materials and supplementary sample documents at

www.who.int/tools/bleeding-after-birth

For the facilitator

Using this Flip Chart to facilitate learning

The Flip Chart is a guide for the facilitator of a course. It includes all you need to facilitate learning in an engaging an participatory way. Place the Flip Chart with the illustration pages facing the participants, and the text facing you. The content is divided into sections with different learning methodologies, explained below.

Explain

These sections cover key knowledge the participants will need.

- Use your own words, be concise and avoid just reading.
- To engage participants, prepare how you will deliver explanations in advance.

Demonstrate

Show participants the actions they need to take, and prepare them to practice.

- For some steps, videos are available.
- Always emphasize and model respectful care and good communication.

Practise

Give participants time to repeat newly learned or refreshed skills for each step.

- Spend more time practicing than explaining.
- Encourage self-reflection and debriefing with the team.

Discuss

Encourage participants to share and explore what is done in their facility. Find ways to overcome barriers and put new skills into practice.

Ask

Ask questions to review and reinforce learning. Create a safe space and ensure all participants participate.

Scenario practice

Reinforce skills learned in a simulation practice.

- Ensure every group has all the equipment on the checklist.
- Encourage self-reflection and feedback.
- One or two participants will take the role of health workers, using only the Action Plan and Postpartum Monitoring Form for guidance.
- Ensure the rest follow the actions on the Action Plan. Rotate participants on the next scenario.
- You will act as the woman, operate the simulator, give prompts as needed, and provide vitals and other information when participants asses for them.
- Alternatively, a participant could be the woman with your guidance.

Assess knowledge

Assess the participants' knowledge before and after the course.

- Use the knowledge checks available as available as supplementary material. Find them at www.who.int/tools/bleeding-after-birth.
- Use the results from the pre-course test to tailor facilitation to the needs.

Assess skills

Assess the participants' skills after the course using OSCEs.

- Make sure to plan enough time to assess all participants.
- There are six OSCE cases available as supplementary material. Find them at www.who.int/tools/bleeding-after-birth.

Equipment checklist

For each group of 6 participants and 1 facilitator

Learning materials

- ☐ 1 Flip Chart
- ☐ 1 set of Action Plans
- ☐ 6 Provider Guides (1 per participant)

Simulation equipment

- ☐ PPH simulator with placenta and baby
- ☐ Simulated blood

Postpartum equipment

- ☐ Blood loss measurement tool
- ☐ Postpartum monitoring form
- ☐ Curved forceps
- ☐ Ring forceps
- ☐ Scissors
- ☐ Ties/haemostats or clamps
- ☐ Cloths and hat for newborn
- ☐ Suction bulb
- ☐ Bag and mask
- ☐ NASG – if using
- ☐ UBT – if using

General equipment

- ☐ Personal Protective Equipment
- ☐ Blood pressure cuff and stethoscope
- ☐ Syringes and needles
- ☐ Blood collection tubes
- ☐ Speculum
- ☐ Urinary catheter and bag

Mock medications and IV

- ☐ Oxytocin or other uterotonics
- ☐ Tranexamic acid (TXA)
- ☐ Antibiotic
- ☐ Pain medication
- ☐ IV catheter
- ☐ Crystalloid IV fluids
- ☐ Tape

Additional materials

- ☐ Knowledge assessments and OSCEs sheets
- ☐ Postpartum monitoring forms
- ☐ Demonstration videos on a device with good sound

MODULE 1

Saving lives at birth



MODULE 1

Saving lives at birth



Facilitation note

Before you start

- Have all equipment. Refer to page 1b in this flipchart.
- Prepare a co-facilitator or volunteer participant to wear the simulator and act as the woman during demonstrations, throughout the course.

Start the course

- Begin with the pre-course knowledge check.
- At the end of this page you may include a surprise role-play of a normal birth that progresses to PPH and leads to death.

Explain

- There are two people who need our care: the woman and her newborn.
- Survival of the baby can depend on survival of the woman.
- This course helps you master competencies to prevent, diagnose and treat PPH.
- All women should give birth in facilities with the capacity to treat PPH and administer IV medicines.

Continued learning for health workers

- This course is for all health workers who care for women at birth.
- We will use different learning methods:
  - discussions
  - demonstrations
  - hands-on practice
  - feedback
  - case scenarios for simulation.
- After the course, short, frequent drills and other exercises at the facility will strengthen and maintain skills.

Learning materials

- **Action Plans** are for clinical training, drills and as a job aid:
  - Action Plan 1 – Prevention and early diagnosis of PPH
  - Action Plan 2 – Treatment of PPH.
- **Provider Guide,** a personal resource for all participants for continued practice after the course. It includes:
  - LDFH activities for quality inprovement.
  - tools like medication information and Postpartum Monitoring Form
  - key knowledge and references.

Role-play (optional)

Suddenly start a role-play interrupting the course.

- A woman walks into the room about to give birth, asking for help.
- She gives birth, but bleeds excessively and you cannot stop it. She dies.
- Consider redoing this role-play at the end of the course and have her survive.

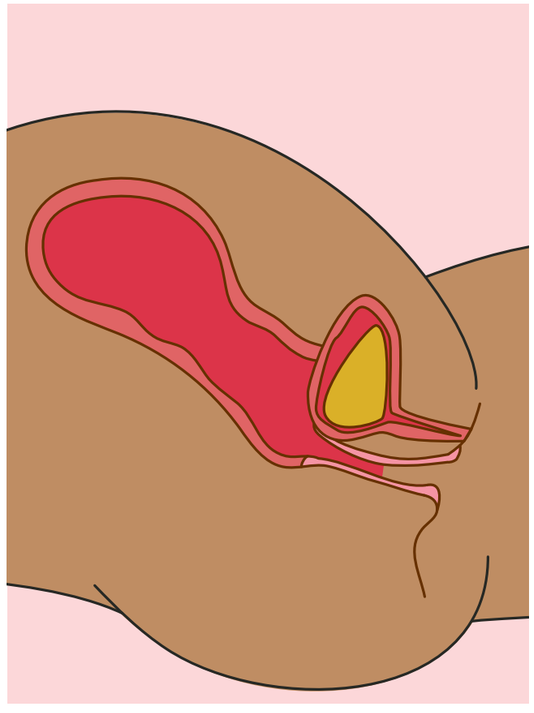
Discuss

1. Have you ever seen someone die from bleeding after birth? What happened?
2. Was there anything that could have been done differently or if the woman had been somewhere else?

Main causes of bleeding after birth

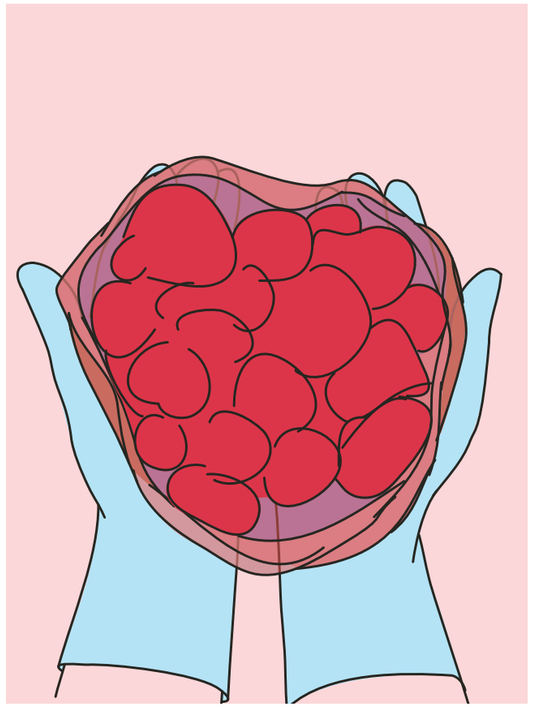
Tone

Soft uterus



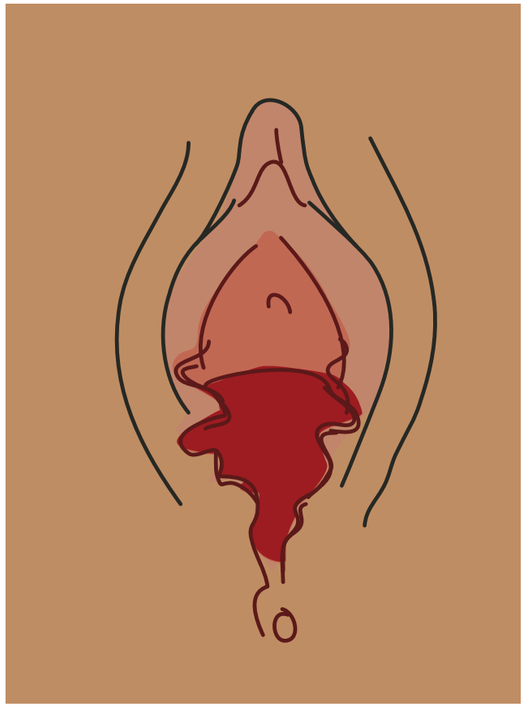
Tissue

Retained placenta or fragments



Trauma

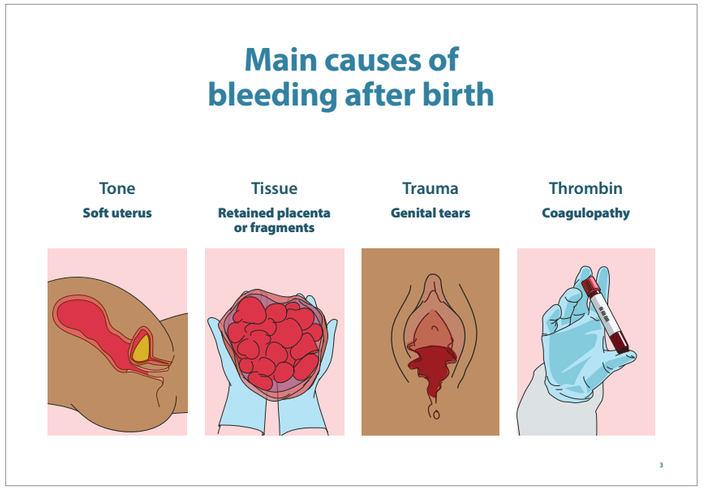
Genital tears



Thrombin

Coagulopathy





Explain

- All women are at risk for life-threatening bleeding.
- Be vigilant for early warning signs.
- Dangerous bleeding can be a slow, steady stream or a large gush.
- Respond quickly if a woman has signs that PPH may be likely.

The main causes of bleeding are:

1. Tone – accounts for about 70% of PPH cases

- If the uterus does not contract, blood continues to pump into the uterus.
- Monitor women closely. A uterus that is contracted can later become atonic.
- A full bladder can prevent the uterus from contracting.
- Risks:
  - large babies >4500 g
  - previous PPH
  - long labours
  - shoulder dystocia
  - anaemia
  - multiple pregnancy.

2. Tissue – accounts for about 20% of PPH cases

- Tissue left behind can cause atony and infection.
- Tissue includes:
  - retained placental tissue
  - amnion
  - blood clots
  - placenta praevia
  - invasive placenta.

3. Trauma – accounts for about 10% of PPH cases

- Includes laceration, hematoma, uterine inversion and rupture.
- Risks:
  - large babies
  - female genital mutilation
  - episiotomies – should not be done routinely.
  - augmented or obstructed labour
  - caesarean birth
  - vacuum or forceps birth.

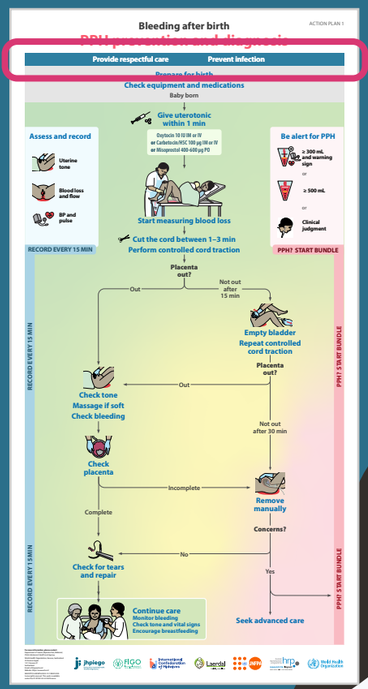
4. Thrombin (coagulopathy) – accounts for about 1% of PPH cases

- Risks:
  - severe pre-eclampsia or eclampsia
  - disseminated intravascular coagulopathy
  - placental abruption
  - invasive placenta
  - history of coagulopathy
  - sepsis.

Ask

1. Who is at risk for postpartum bleeding?
   - All women.
2. What is the most common cause of bleeding?
   - Atony or a uterus that will not contract.

Provide respectful care and prevent infection





Facilitation note

- Begin this page introducing Action Plan 1.
- As you explain respectful care and infection prevention, point out were they are on the Action Plan.

Explain

Provide respectful care

- All good relationships begin with **respect and trust**.
- This is important in healthcare – people trust us to help them.
- During complications, respectful care also involves being a calming presence.

Key points for respectful care

- Women make their own informed choices, even when unexpected things happen.
- Women feel safe and cared for.
- Women have privacy, even during an emergency.
- Women are enabled to have a companion of their choice.

Respectful communication

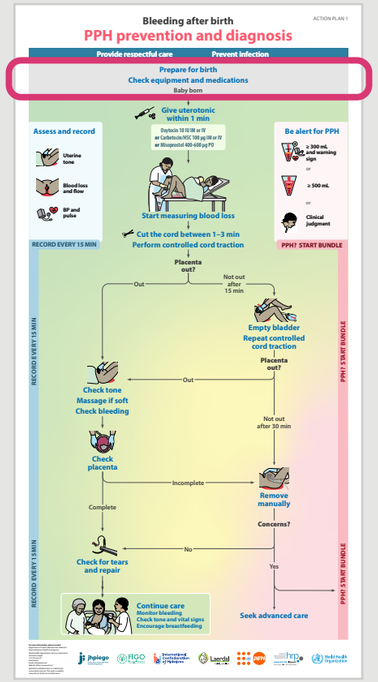
- **Introduce yourself** by name, and use the woman's name.
- **Inform the woman** and her family of progress, options, and concerns throughout care.
- **Listen** without interrupting, and confirm you understand her.
- **Explain early** if transfer for advanced care is needed.
- **Always be honest**:
  - admit if you don't know something
  - maintaining trust matters more than appearing knowledgeable.

Prevent Infection

- Wash hands and wear sterile gloves.
- Wear apron, mask, and eye shield to protect yourself.
- To protect the baby:
  - wear double gloves before birth
  - remove the soiled pair before clamping and cutting the cord.
- Properly dispose of or process contaminated supplies and equipment. Follow protocols.

Discuss

1. How does respectful care save lives?
2. How do you support a woman to make her own choices, especially when something unexpected happens?
3. What changes, big or small, could improve respectful care in your daily work?
4. Are there any challenges with infection prevention at your facility?



Prepare for birth, check equipment and medications



Explain

- Preparing the woman and the bedside helps prevent and treat PPH.

During antenatal care (ANC) – address anaemia to prevent PPH

Prevent anaemia

- Give all pregnant women:
  - 30–60 mg of elemental iron and
  - 400 µg of folic acid
  - see formulations on page 37 of the Provider Guide.

Assess for anaemia

- Full blood count (FBC) is recommended.
- If unavailable, use a haemoglobino-meter (preferred over colour scale).

Address anaemia

- Increase elemental iron if haemoglobin (Hb) is:
  - <11 g/dl in the 1st or 3rd trimester, or
  - <10.5 g/dl in the 2nd trimester
- Give 120 mg elemental iron a day until Hb reaches ≥11 g/dl.
- Consider IV iron before labour if:
  - iron deficiency anaemia is confirmed, and
  - oral iron is not tolerated, and
  - rapid response is needed.

Prepare your PPH emergency kit

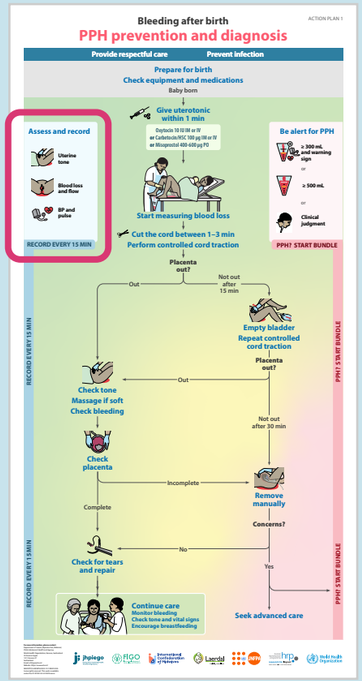
- Have an emergency trolley or box to manage complications quickly.
- Check daily if it is stocked and available. See page 13 of the Provider Guide for a list.

During labour

- Encourage frequent urination.
- Based on the woman's preference – reduce perineal trauma using massage, warm compresses, and a hands-on technique to deliver the head or breech.
- Avoid episiotomies. If medically indicated:
  - use medio-lateral incision for lower risk of injury to the anal sphincter.
  - continuous suturing repair is preferred to interrupted sutures.

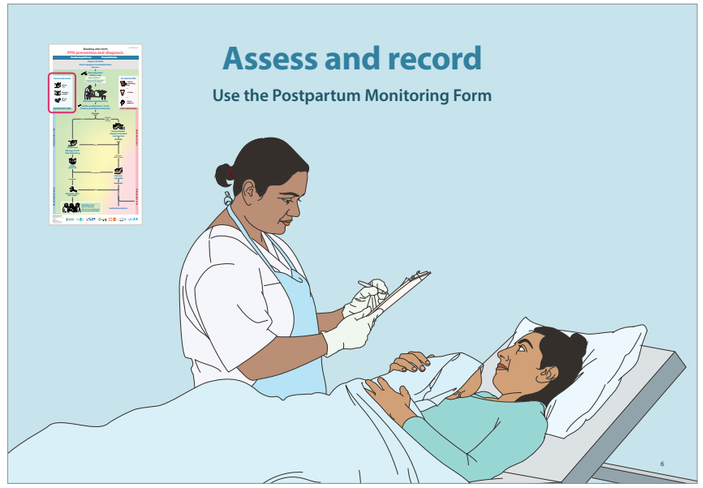
Discuss

1. During ANC, how do you test for anaemia?
   - Allow participants to describe their experience and reflect on recommendation.
2. At your facility, do you have all the PPH emergency supplies listed on page 13 of the Provider Guide?
   - If no or if incomplete, recommend the activity of page 12 of the Provider Guide later at the facility.



Assess and record

Use the Postpartum Monitoring Form



Explain

As you used the Labour Care Guide[3] during labour, **use the Postpartum Monitoring Form** to:

- **Assess** and record often, as you provide care:
  - consider all information, history, prior complications and other risk factors
  - review trends in current care.
- **Identify** possible diagnosis, based on your assessment and records.
- **Communicate and act**, based on your diagnosis:
  - What care does the woman need?
  - Call a more senior health worker?
  - Advanced care or transfer?

Assess and record repeatedly

- Before birth, fill in any risk factors for PPH.
- From the birth onwards, check regularly for signs of PPH.
- Record **four times in the first hour** after birth to help diagnose PPH quickly.
- First assessment:
  - when placenta is delivered, or
  - 15 min after birth if placenta is not yet out
  - this is the most critical, most PPH happens in the first 20 min after birth.
- Repeat every 15 min for the first hour, or until stable.

What should you check?

- Uterine tone.
- Volume of blood loss and flow using measurement tool (reviewed on next page).
- Blood pressure (BP) and pulse.

Warning vital signs include:

- Pulse >100.
- Systolic BP <100, diastolic BP <60.
- Shock index (SI) >1.

What is SI?

- Quick bedside tool to identify early signs of shock.
- SI = Pulse ÷ Systolic BP
- Warning. SI >1 may indicate shock.
- Example: Pulse 110 / systolic BP 100 = SI 1.1 = warning sign

> **Quick rule of thumb:**
> if pulse > systolic BP, then SI >1 and cause for warning.

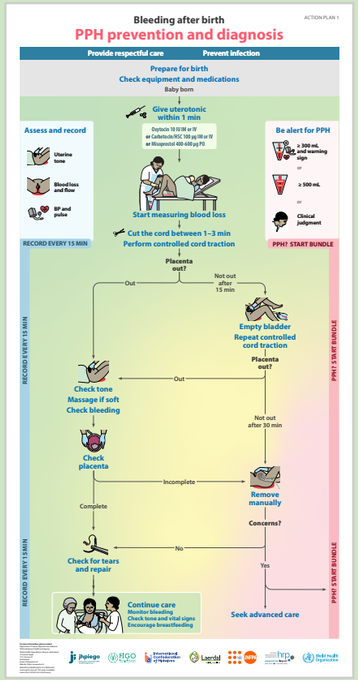
Demonstrate

Refer to the Postpartum Monitoring Form in Provider Guide page 8.

- Ask participants to review the form quietly for 1 min
- Quickly demonstrate how to use each section of the form.

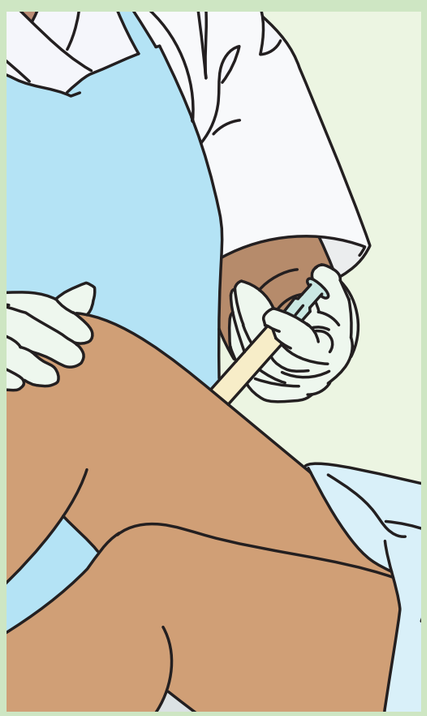
Discuss

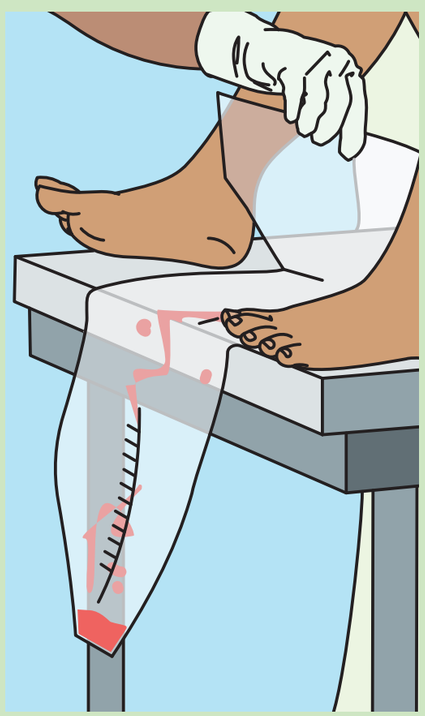
1. Form groups of 4 to discuss (3 min):
   - what is familiar?
   - what might be harder to use?
2. A person from each group shares group's observations in 1 min.

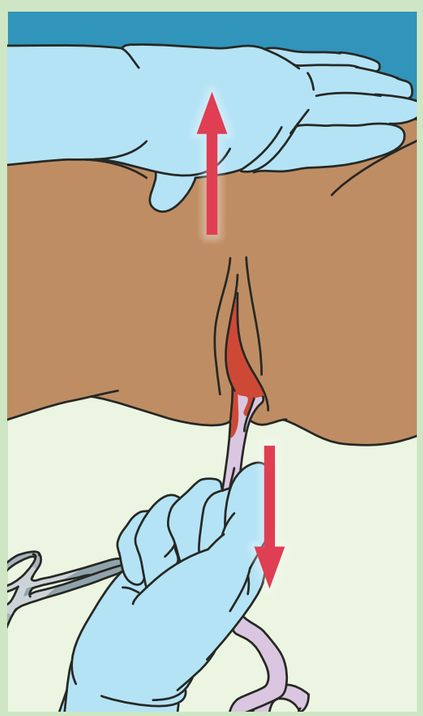


MODULE 2

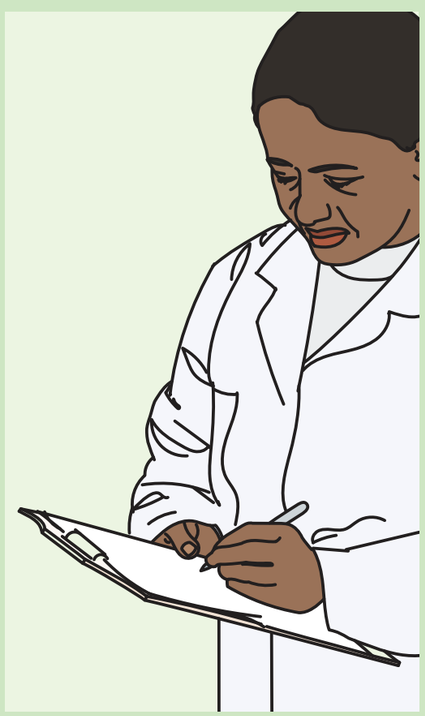
Prevention and early diagnosis of PPH

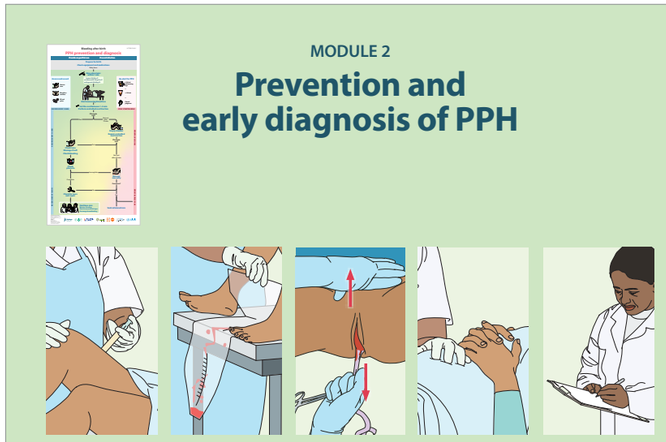












Facilitation note

- Point to the steps for PPH prevention on Action Plan 1 as you explain them.

Explain

Prevent PPH

Actively preventing PPH ensures:

— a shorter third stage of labour
— less blood loss
— fewer cases of PPH.

Steps to prevent and identify PPH

1. **Give a uterotonic** to prevent PPH in both vaginal and caesarean births after the last baby is born.
2. **Place the blood measurement tool** and monitor blood loss closely.
3. **Deliver placenta** with controlled cord traction (CCT) after clamping and cutting the cord.
4. **Check uterine tone** and massage if soft.

Once the woman is stable

- Check the placenta for completeness.
- Check for and repair tears.
- Assess and record tone, bleeding and vital signs.
- Keep the blood loss monitoring tool in place for the first hour for all women. If a woman continues to bleed, leave the tool in place for another hour.

Continue care for woman and baby

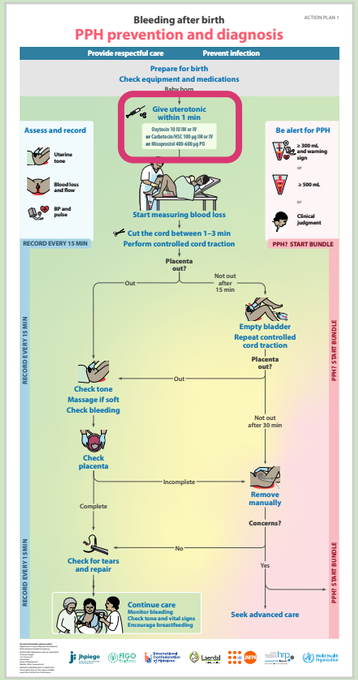
The time after birth is critical for both the woman and her baby.

- Ensure the baby is breathing well and kept warm.
- Keep them together and skin-to-skin for the first hour.
- Encourage breastfeeding.
- If the baby needs resuscitation and you are alone:
  — care for the baby
  — give the uterotonic and check tone as soon as possible.

If the woman arrives in your care after giving birth elsewhere: provide care based on her signs, symptoms, and any information about previous treatment.

Ask

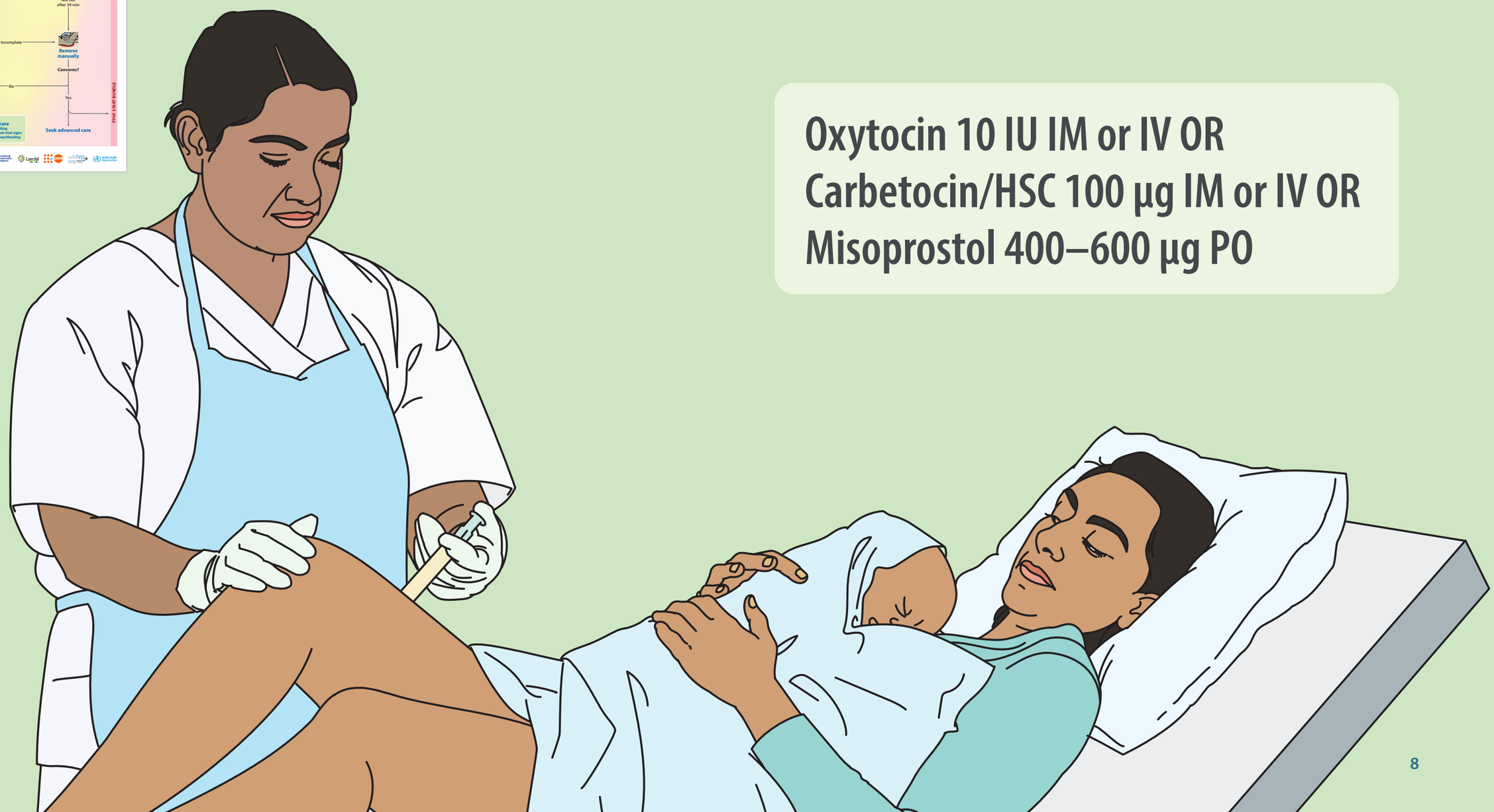
1. What is the most important step to prevent PPH?
   —Give a uterotonic immediately after the last baby is born.
2. Do you check the placenta for completeness immediately after birth?
   —No. Always check uterine tone and bleeding first. If the uterus is firm and there is no bleeding, then closely examine the placenta.

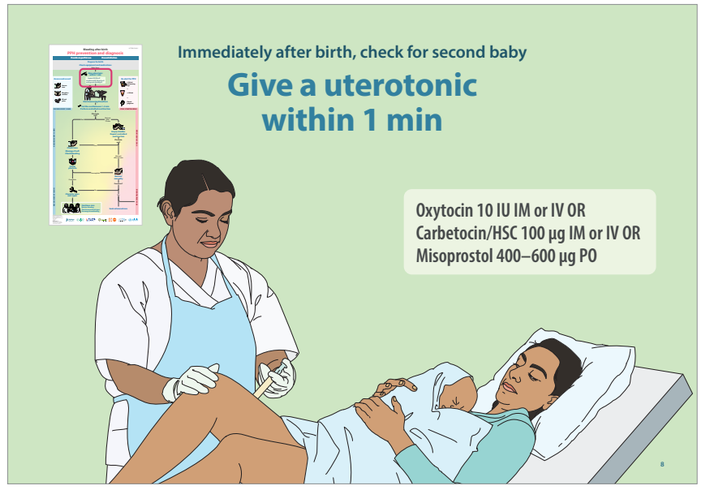


Immediately after birth, check for second baby

Give a uterotonic within 1 min

Oxytocin 10 IU IM or IV OR
Carbetocin/HSC 100 µg IM or IV OR
Misoprostol 400–600 µg PO





Facilitation note

- Focus on the medications available in the participant's facility.
- Refer participants to the medication chart on page 6, in the Provider Guide, for a detailed chart on medications.

Discuss

- What uterotonics are available in your facility and what is used for prevention?

Explain

- **Always prepare a uterotonic before birth.**
- Give uterotonic within 1 min after the last baby is born.

Never give a uterotonic before last baby is born. Always check for another baby.

Uterotonics for prevention

Oxytocin

- 10 international units (IU) intramuscular (IM) or intravenous (IV).
- IV preferred if IV already in place: 10 IU diluted and given slowly over 1–2 min.
- **Use and storage:**
  - can also be used for treatment
  - **WHO's preferred choice** for all women
  - refrigerate (2–8°C)
  - if cold storage can't be guaranteed, use carbetocin/HSC or misoprostol.

Carbetocin and heat-stable carbetocin (HSC)

- 100 µg (1 mL) IM or IV.
- If IV, administer slowly over 1 min.
- **Use and storage:**
  - only for prevention of PPH
  - **never** for induction or augmentation, can cause uterine rupture
  - carbetocin – refrigerate (2–8°C)
  - HSC – no refrigeration needed, protect from light.

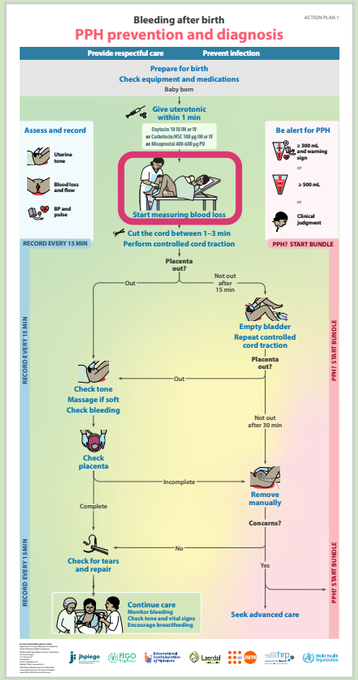
Misoprostol

- 400–600 µg orally (PO) sublingually (SL) or per rectum (PR) based on woman's preference.
- **Use and storage:**
  - no refrigeration needed
  - protect from humidity
  - for prevention and treatment.

Do not use for prevention of PPH: TXA, injectable prostaglandin, or ergometrine alone or in combination with oxytocin.

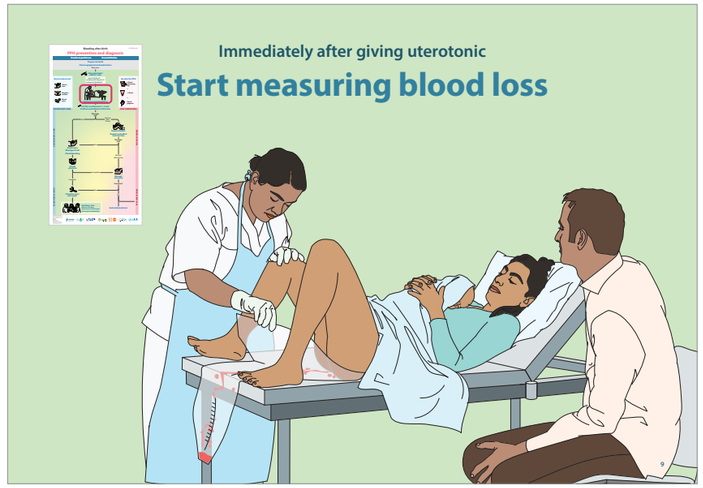
Special considerations for women at high risk of PPH

- Some women are at high risk of PPH – if they have anaemia, prior PPH, caesarean birth, female genital mutilation, sepsis, multiple pregnancy, placenta praevia, macrosomia with a birthweight of more than 4500 g, or shoulder dystocia.
- For women at high risk, you may use:
  - a combination of oxytocin 10 IU IM or IV and misoprostol 400–600 µg PO
  - Find details in Provider Guide, page 6.



Immediately after giving uterotonic

Start measuring blood loss



Facilitation note

- During the course, use the blood loss measurement tool available at the facility.
- If there is no blood loss measurement tool, clinical judgment is still used to trigger the PPH bundle.

Explain

- Measure blood loss objectively to quickly diagnose and treat PPH.
- A blood loss measurement tool is recommended **for all women,** both for vaginal and caesarean births.

Blood loss measurement tools

- For vaginal birth, use a tool that:
  - easily shows volume of blood loss
  - collects all blood and avoids spillage
  - ensures the woman's comfort and choice of birth position.
- Examples:
  - drapes with a calibrated funnel
  - trays with calibrated lines
  - other innovations.

Place the tool

- Immediately after the baby is born and you have given the uterotonic.
- For non-supine position birth:
  - assist her to a supine position
  - place baby on her chest
  - give uterotonic, then place tool.

Measure blood loss accurately

- Sweep blood and clots into the tool.
- If using a drape:
  - hang the funnel over the edge of the bed, or
  - lift the funnel so blood collects at the bottom.
- Check continuously to diagnose PPH quickly:
  - amount of blood in the tool and
  - blood flow.
- First recording of blood loss volume on the Postpartum Monitoring Form:
  - after the placenta delivers, or
  - 15 min after birth if placenta is not out.

> **Prevent infection:** follow protocols for safe disinfection, sterilisation or disposal of tools.

Demonstrate and practise

1. Place the tool

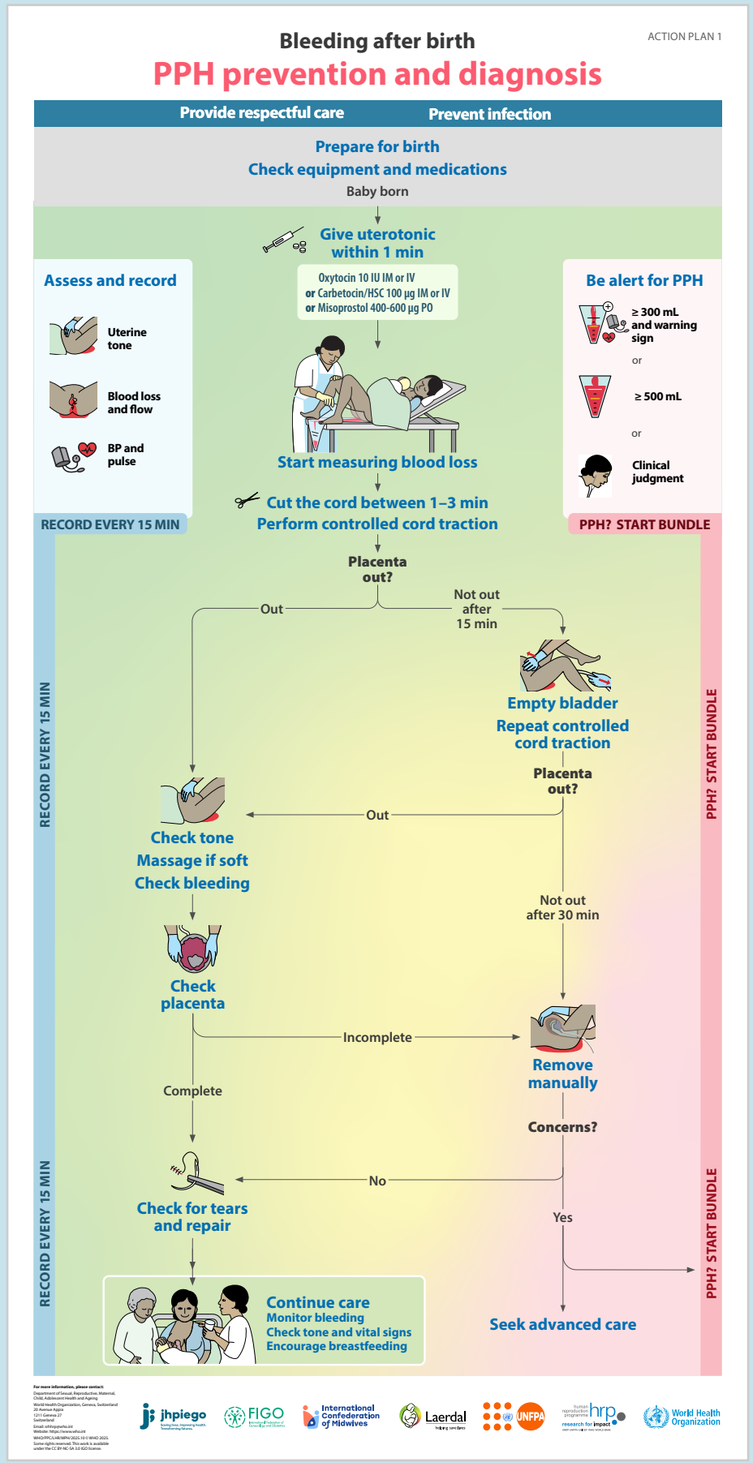
- Ask two participants to be the woman and the health worker, and demonstrate placing the tool.
- Make sure they place it correctly.
- Open blood tank to a heavy flow.
- Demonstrate how to sweep blood and clots into the tool.

2. Monitor blood loss

- When 300 mL blood in tool, ask a participant to read out the volume of blood lost.
- Show where to record it on the Postpartum Monitoring Form.

Discuss

1. Have you used a blood loss measuring tool?
2. What was challenging when placing the tool and checking the blood volume?
3. What can you do to make it easier?



Blood loss assessment activity

How accurate is visual estimation of blood loss?

Blood loss assessment activity

How accurate is visual estimation of blood loss?

1. Prepare

Set up four stations with containers holding different amounts of blood, labelled A, B, C, D:

A. White towel or cloth, with 600 mL.

B. Gauze bandage, with 100 mL.

C. Liquid in a basin, with 300 mL.

D. Blood clot (red fruit jam), with 500 mL.

2. Read these instructions

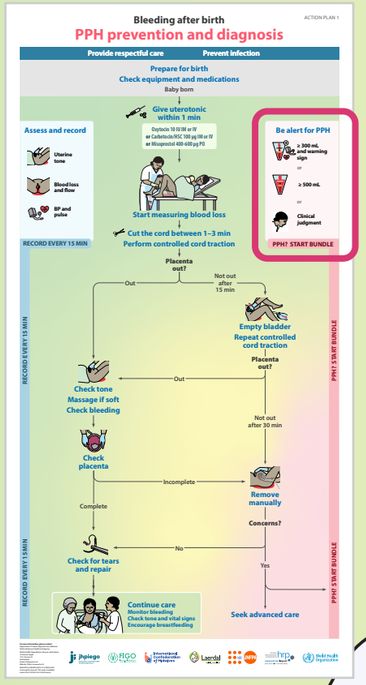
- Go for a short break to walk by stations.
- Estimate the blood loss at each station and write it down.

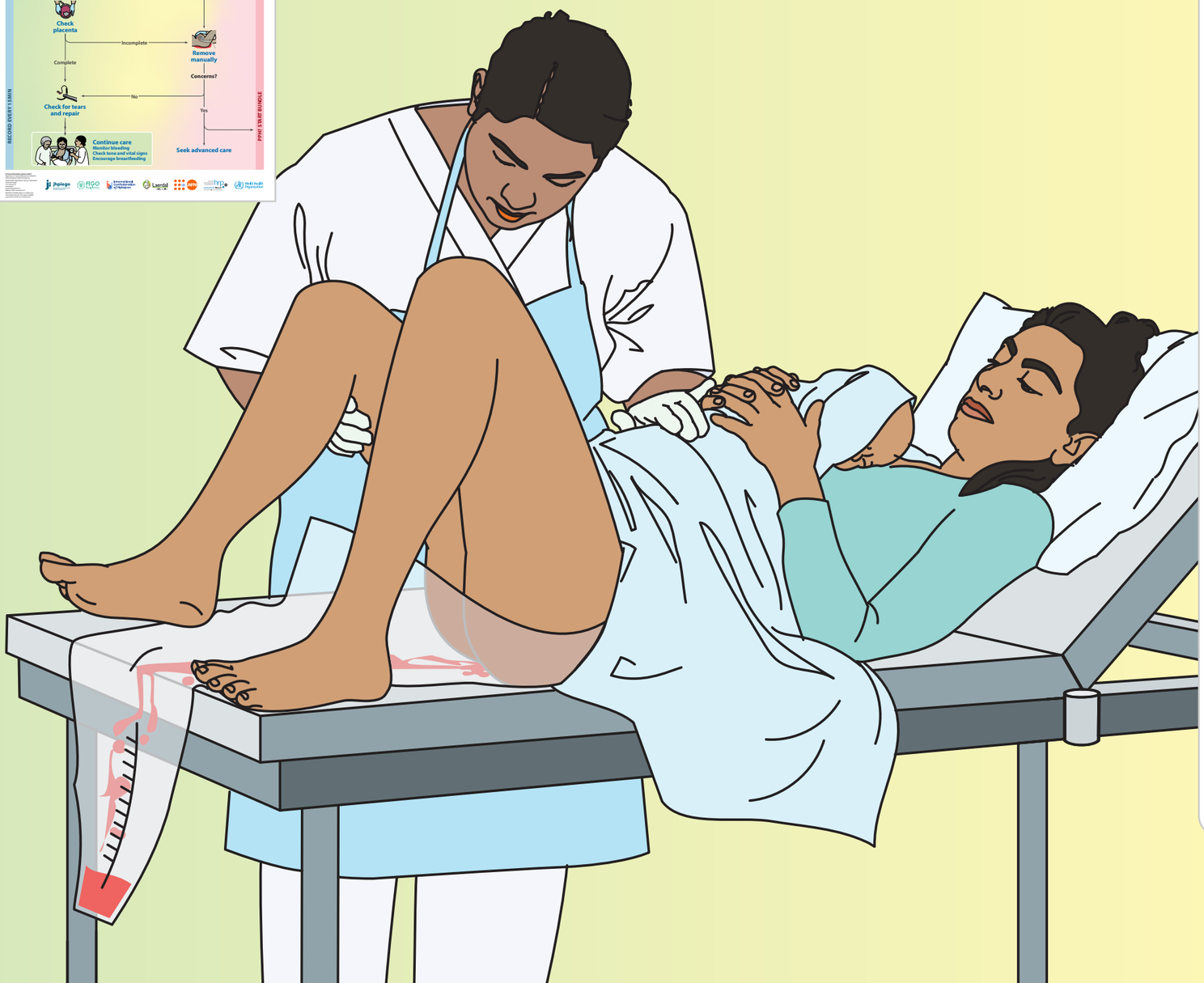
3. Discuss

- What were your estimates?
- Are they different among the group?
- What are the consequences of underestimating?
- Why is objective measurement of blood loss recommended?

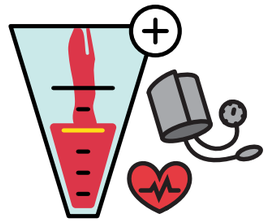
4. Explain

- The purpose of this activity is to create awareness of the importance of measuring blood loss objectively.
- Visual estimation of blood loss is unreliable and may delay diagnosis and treatment of PPH.
- This is why using a tool to measure blood loss is key to early diagnosis and treatment of PPH.
- Decision-making is guided by:
  - blood loss and rate of flow
  - uterine tone, and
  - the woman's vital signs.
- All women are at risk for life-threatening bleeding.
- Dangerous bleeding can be:
  - a slow, constant trickle
  - a heavy flow
  - a large gush
  - large blood clots expelled.



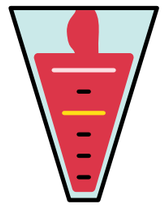


Be alert.
If PPH, start bundle



Blood loss ≥ 300 mL and warning signs

or

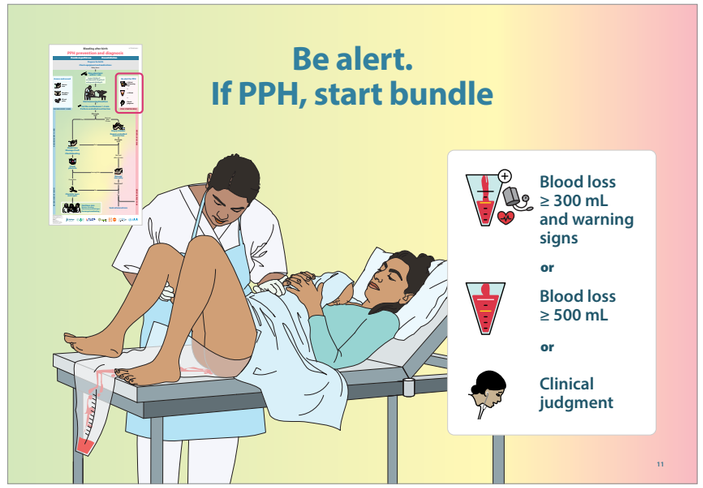


Blood loss ≥ 500 mL

or



Clinical judgment



Facilitation note

As you describe the definition of PPH, point to the Action Plan.

Explain

From the moment the baby is born, continuously assess for any signs for PPH.

Assess and record to diagnose PPH

- Start recording when placenta delivers, or 15 min after birth if placenta not delivered:
  - uterine tone
  - flow and amount of blood in the measurement tool
  - BP, pulse and SI.
- Repeat at least **every 15 min** for the first hour or until stable.

Diagnose PPH and start treatment bundle if

1) ≥300 mL of blood loss and a warning sign such as:

- Atonic uterus, heavy blood flow, large clots, or constant trickle of blood.
- Change in vital signs:
  - pulse >100 bpm
  - systolic BP < 100 mmHg
  - diastolic BP < 60 mmHg
  - SI >1.

2) ≥500 mL or more of blood loss

Start the PPH bundle immediately anytime blood loss reaches 500 mL, even if the woman has no other concerning signs.

3) Your clinical judgment

Any amount of blood loss but you are concerned because:

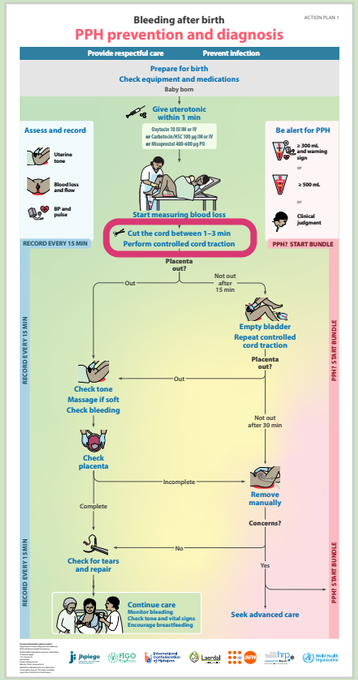
- Rate of blood loss and high risk for PPH:
  - previous PPH
  - twins
  - prolonged labour
  - pre-eclampsia
  - infection
  - retained placenta
  - augmented labour
  - history – e.g. severe anaemia.
- The woman was transferred to you after PPH.

Treatment with the PPH bundle

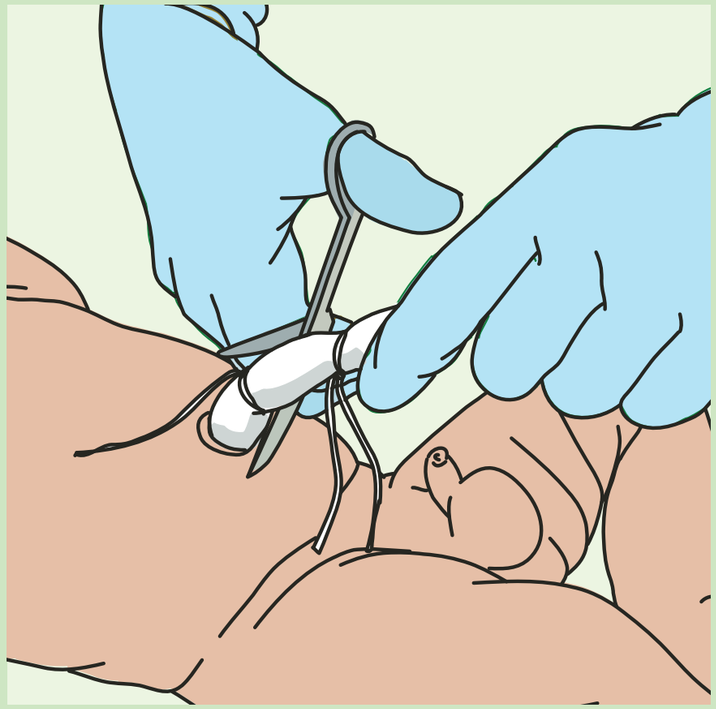
- The PPH bundle is the first line of treatment for PPH.
- It is a set of interventions that, together, improve outcomes from PPH.
- We will cover the bundle in Action Plan 2.

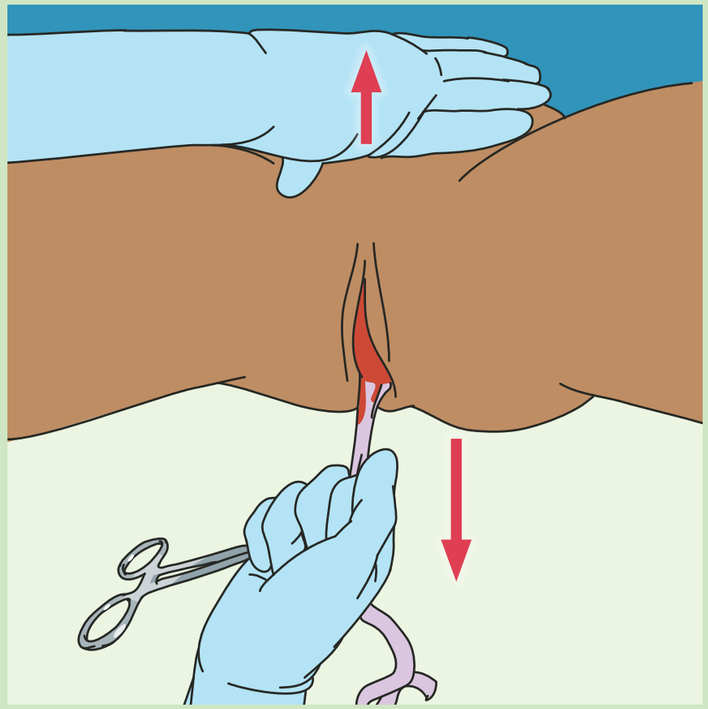
Discuss

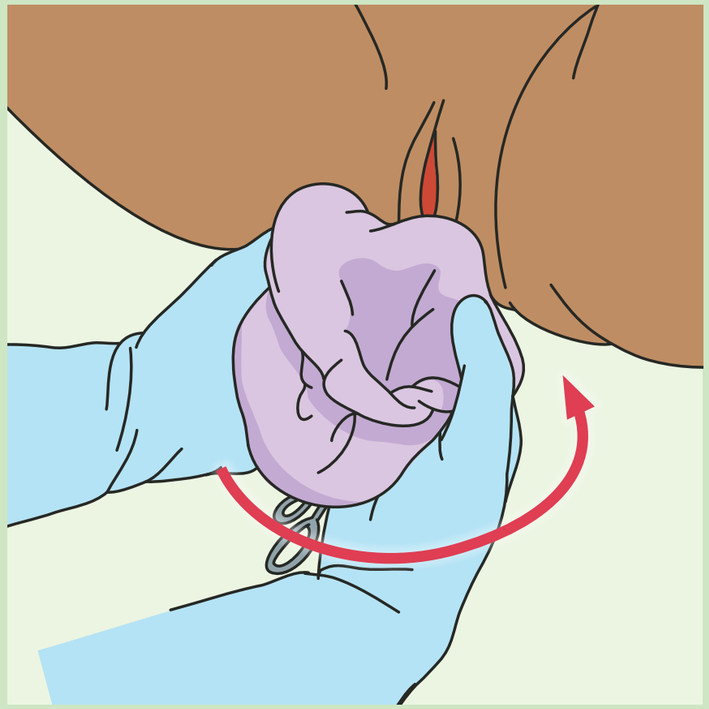
1. How will you diagnose PPH if a blood loss measurement tool is not available?

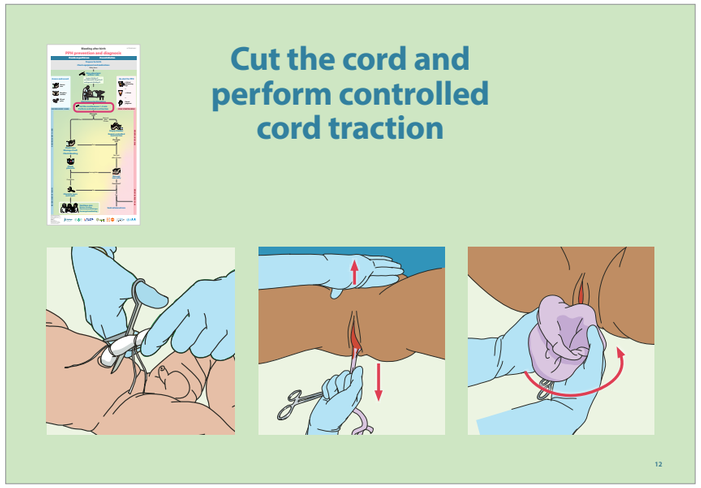


Cut the cord and perform controlled cord traction









Explain

Cutting the cord

- Waiting 1–3 min to cut the cord helps the baby with:
  - increased blood volume to support lung expansion
  - more red blood cells to avoid anaemia.
- Cut immediately and get help if:
  - the woman is bleeding heavily
  - the baby is not breathing well.

Prevent infection

- Use sterile or disinfected instruments.
- Change gloves or remove outer pair if double-gloved, before cutting the cord.

CCT

- Should only be done by skilled health workers.
- Begin CCT after signs of placental separation – gush of blood or lengthening of the cord.
- Apply CCT only during contractions.
- Never pull hard on the cord.
- Stop if resistance is felt to avoid tearing off the cord and uterine inversion.

Demonstrate

☐ Explain your actions respectfully.

☐ Prevent infection.

☐ Clamp and cut cord.

☐ Wait 1–3 min after birth if both woman and baby are stable.

☐ Place clamps. Cut cord between clamps.

Perform CCT

☐ Clamp the cord close to perineum.

☐ Look for signs of separation.

☐ During a contraction:

- gently pull down on the cord
- provide counter traction just above the pubic bone to stabilize the uterus.

☐ Do not pull suddenly or in other directions.

☐ Release traction between contractions.

☐ Stop if you notice resistance.

Deliver the placenta

☐ When visible at the opening of the vagina:

- hold placenta with both hands and turn in one direction.
- continue until placenta and all membranes are delivered.

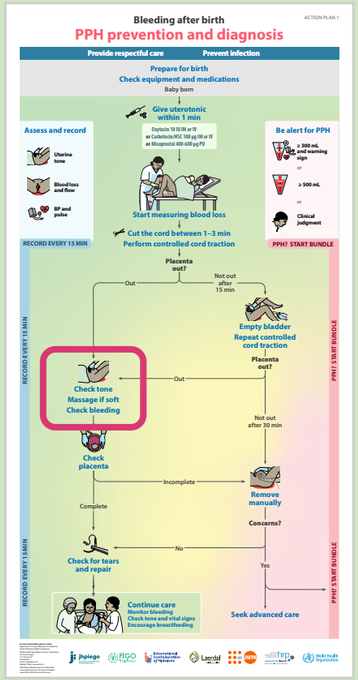
☐ Place placenta in bowl/basin.

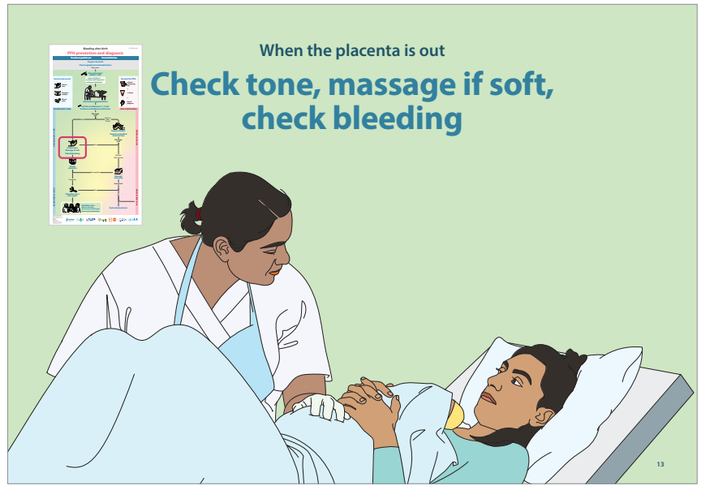
☐ Check uterine tone immediately.

If heavy bleeding, but placenta is not out. Start PPH treatment bundle immediately, then move to manual removal. We will discuss this more later.

When the placenta is out

Check tone, massage if soft, check bleeding





Explain

A soft uterus is the #1 cause of PPH

- Always check tone immediately after the placenta is delivered.
- Monitor and massage the uterus when soft to help prevent PPH.
- Uterine tone can change quickly.
- **Check the uterus at least every 15** min for the first hour after birth.

Assessing tone

- A hard uterus feels like your forehead and does not need massage.
- A soft uterus feels like the tip of your nose and needs massage.
- The uterus should be firm at midline, and approximately level or below the umbilicus.

Soft uterus

- If the fundus rises, is off to the side or becomes soft again:
  - it may be full of clots
  - the bladder may be full.
- Massaging helps contract and expels blood clots.
- Encourage frequent urination. A full bladder may slow uterine contraction.

If PPH, start treatment bundle immediately.

Demonstrate

Check tone

☐ Place hand on abdomen around the umbilicus to find the uterus.

☐ Check – it should feel like your forehead.

If soft

☐ Firmly rub the top of the uterus, wrapping towards the back.

☐ Guard the uterus with your other hand while you massage.

☐ Massage to cause a contraction and express clots.

☐ Sweep blood and clots into the measurement tool.

☐ Encourage urination.

☐ Catheterize if bladder is full and she is unable to urinate.

☐ Recheck uterine tone, massage if soft.

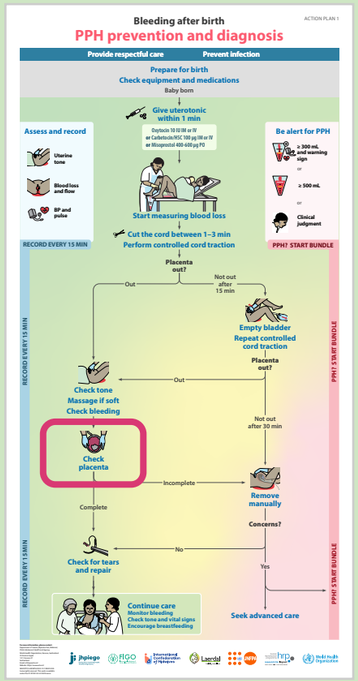
Assess and record

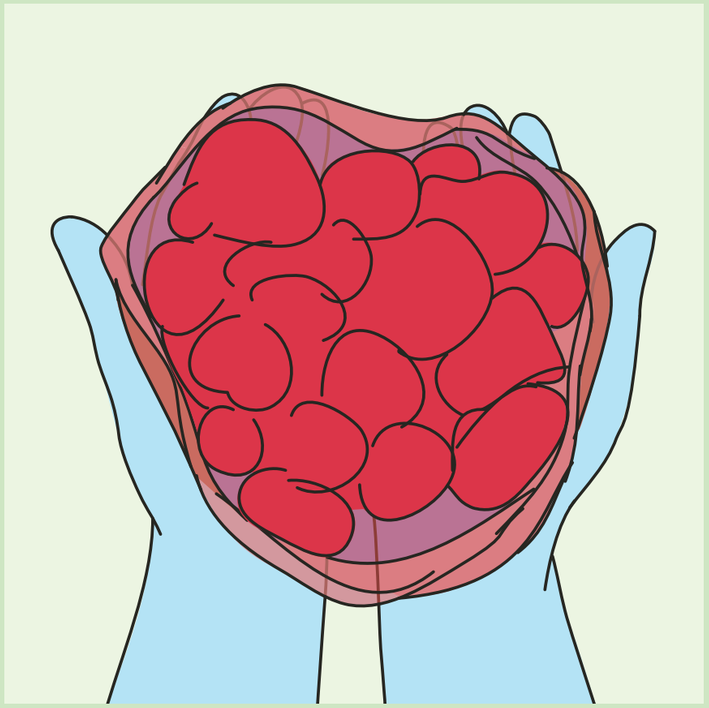
☐ Assess and record on the Postpartum Monitoring Form.

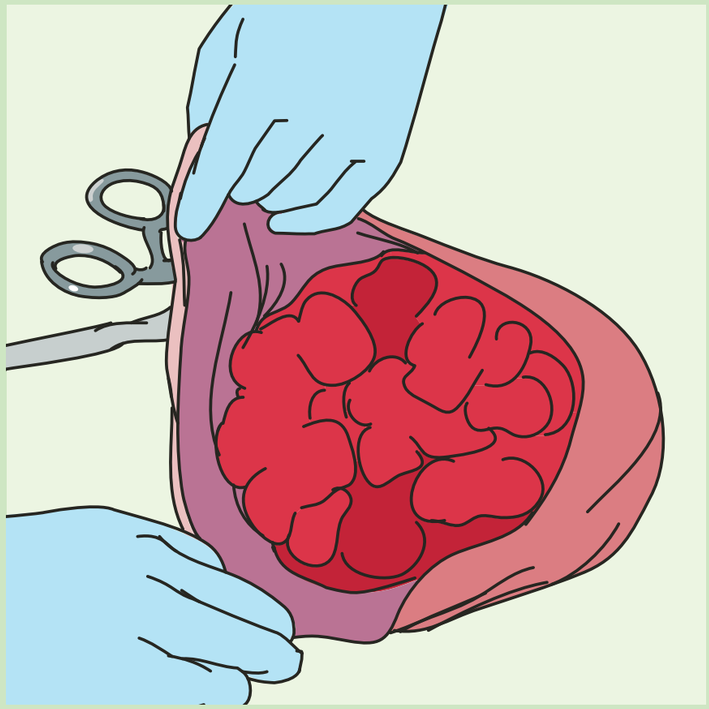
Ask

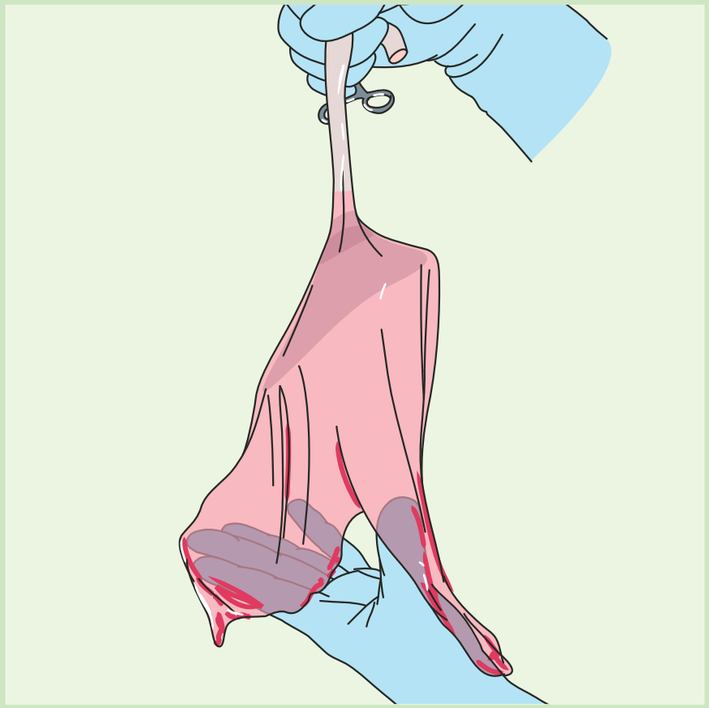
1. True or false: once the uterus contracts or gets hard, it will always stay hard.
   - False – A uterus can lose tone and begin to bleed.
2. Why might a uterus not contract?
   - There may be retained tissue, a ruptured uterus, or the woman's bladder may be full.

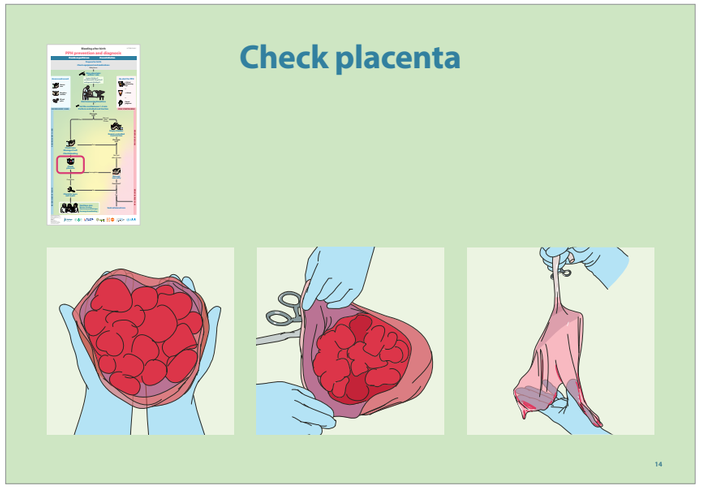
Check placenta











Explain

After delivering the placenta

- **First, ensure good uterine tone and no PPH.**
- Then check the placenta and membranes for completeness – identify missing lobes or fragments.
- Any retained tissue can cause haemorrhage and infection – even a piece of membrane.
- If the placenta is not complete, missing lobes must be removed manually. We will review this later.

If PPH, start treatment bundle immediately.

Demonstrate

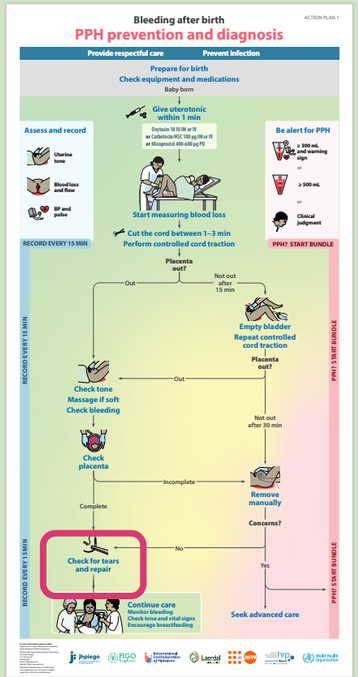
Show video **Examining the placenta** (4 min) or demonstrate using the simulator.

- ☐ Cup the placenta in your hands, maternal side facing up.
- ☐ Check if the lobes fit together and if any are missing.
- ☐ Hold the placenta upside down by the cord.
- ☐ With opposite hand, spread out the membranes to recreate the sac.
- ☐ Check the membranes for missing pieces.

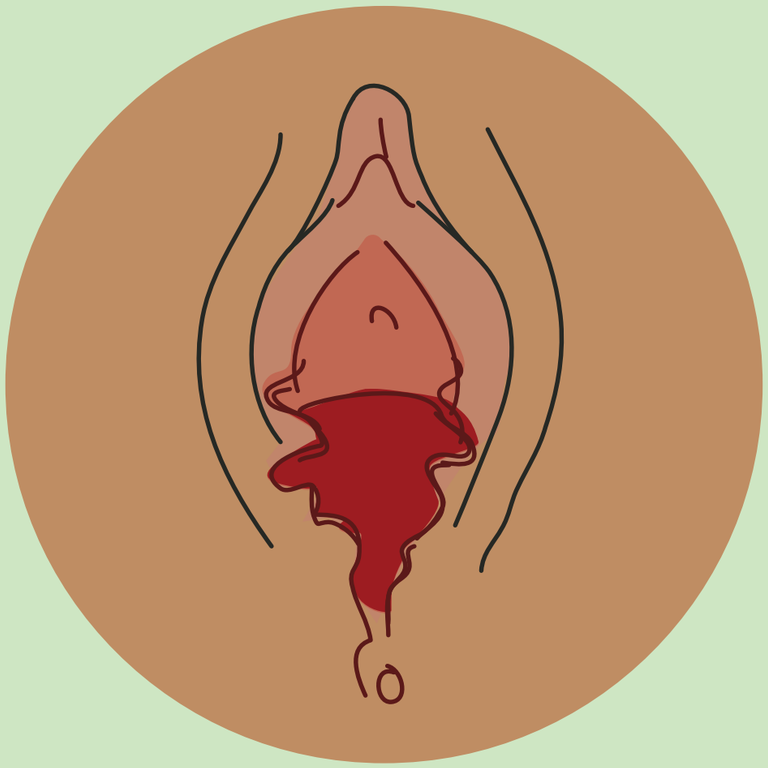
Ask

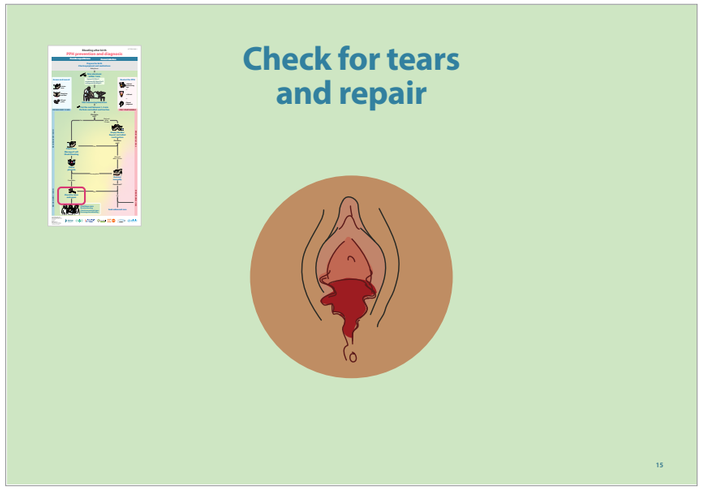
1. If a piece of the placenta or membrane stays in the uterus, why does the woman bleed more?

   — The uterus cannot contract to squeeze the blood vessels and stop the bleeding.



Check for tears and repair





Explain

- Trauma results in tears – another cause of PPH.
- If there is bleeding but the uterus is firm and the placenta is complete, tears are likely the cause.
- Assess gently and thoroughly – tears can be difficult to see.
- The risk of bleeding from tears increases in case of:
  - female genital mutilation or cutting
  - episiotomy
  - large baby
  - augmented or obstructed labour
  - assisted vaginal birth.

If PPH, start treatment bundle immediately.

Act within your scope of practice. Refer to more experienced health workers when tears are beyond your scope.

Demonstrate

Demonstrate using the simulator and a volunteer as the woman.

Provide respectful care

- ☐ Obtain consent and ensure privacy.
- ☐ Explain your actions and be gentle.

Assess for tears

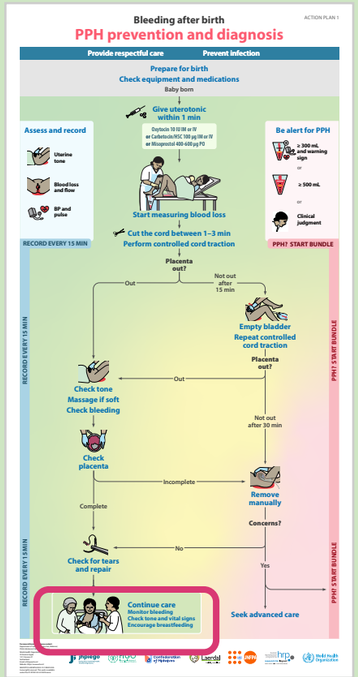
- ☐ Ensure good lighting.
- ☐ Gently wipe away blood and clots from the vagina and cervix.
- ☐ Assess the extent of the tears and related bleeding.
- ☐ Respond based on your scope of practice.

Manage tears

- ☐ Follow local protocols.
- ☐ Apply firm pressure with a sterile gauze.
- ☐ Continue pressure until bleeding stops, or suturing starts.
- ☐ Do not place soaked gauze in the tool.

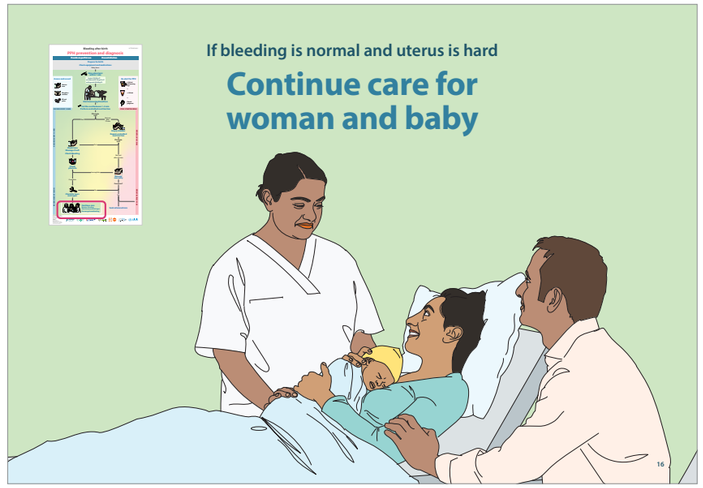
Discuss

1. How do you manage serious lacerations in your facility?
2. Is someone available at all times to repair tears if you cannot?



If bleeding is normal and uterus is hard

Continue care for woman and baby



Explain

Continue monitoring both woman and baby for quality care.

Assess the baby

- Check colour, temperature, breathing, and cord.
- Record on the newborn chart.
- Start skin-to-skin and encourage breastfeeding.

Assess the woman

- Check uterine tone, blood loss, flow, pulse and BP.
- Record on the Postpartum Monitoring Form.

Repeat assessments and record

- Every 15 min for the first hour.
- Then every 30 min for the next 2 h.

Engage woman to prevent PPH

- Teach the woman how to check and massage her uterus.
- Encourage frequent urination.
- Ask her to speak up if she notices bleeding.

After one hour

- If stable and no bleeding:
  - remove the blood loss measurement tool
  - move her to the postpartum ward.
- Women and babies should stay together, in the facility, for 24 h after a normal birth.

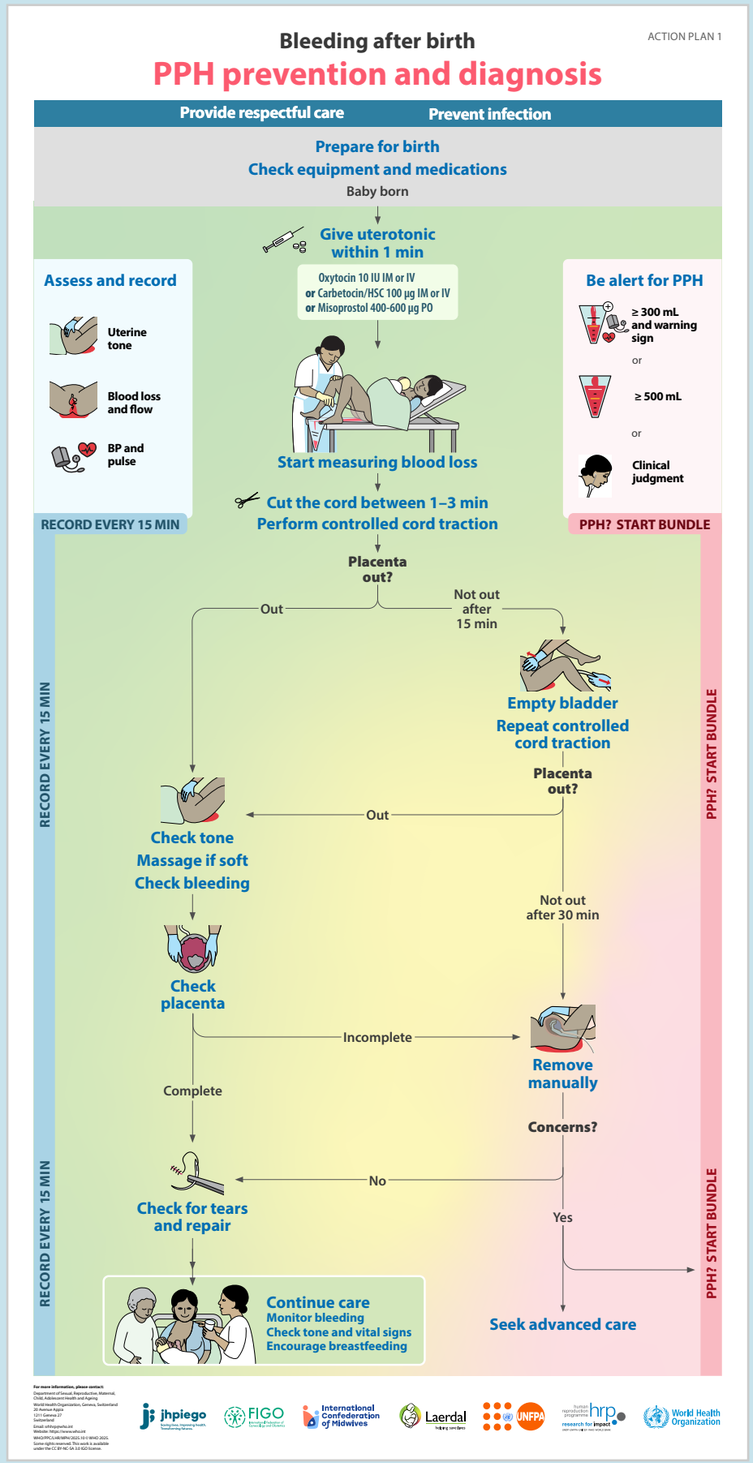
If PPH, start treatment bundle immediately.

Discuss

1. Why should you keep woman and baby under your care for at least an hour?

Ask

1. Why should you assess and record repeatedly, as you give care?
   - Her status can change quickly – check often to respond fast.
   - To see trends and diagnose PPH early.
   - To refer to the record when communicating with team.
   - So team members can quickly see the woman's status and know what care has been given.
2. If Mrs. A's BP is 110/74 and her pulse is 90, what is her SI?
   - 90/110 = SI 0.81 = <1.
3. If Mrs. B's BP is 110/74 and her pulse is 112, what is her SI?
   - 112/110 = SI 1.02 = >1.
4. Which woman has a SI that is a warning sign and why?
   - Mrs. B because her pulse is higher than her systolic BP, meaning her SI is >1.



Scenario practice 1

PPH prevention

Scenario practice 1

PPH prevention



1. Prepare

- Read the guidance on page 1b to prepare.
- Make sure all equipment is ready. See list on page 1b.
- Prepare simulator with:
  - full blood tank
  - baby on woman's chest with cord attached
  - be ready for very light bleeding once placenta is delivered.

2. Brief

Share the learning outcomes

- Actively prevent PPH immediately after birth.
- Use Postpartum Monitoring Form to guide care.
- Provide respectful care.

Assign roles

- Health worker
- Woman (facilitator)

Read the case

> "You are caring for me, Mrs. C, a G2P2, and I just gave birth to a healthy baby after an uncomplicated labour. The baby is crying and the placenta is not delivered."

Resolve doubts

Answer any questions participants may have.

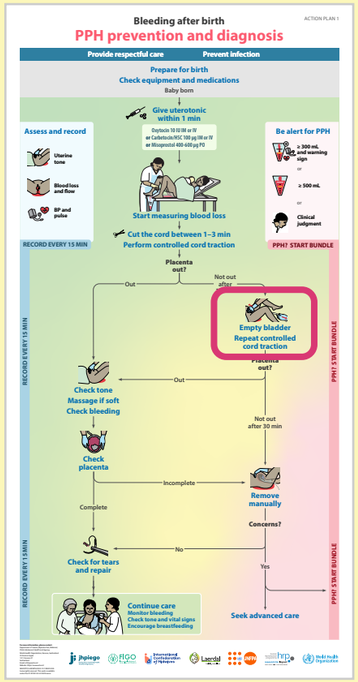
3. Run the scenario

Participants proceed as they would in a real scenario.

4. Debrief

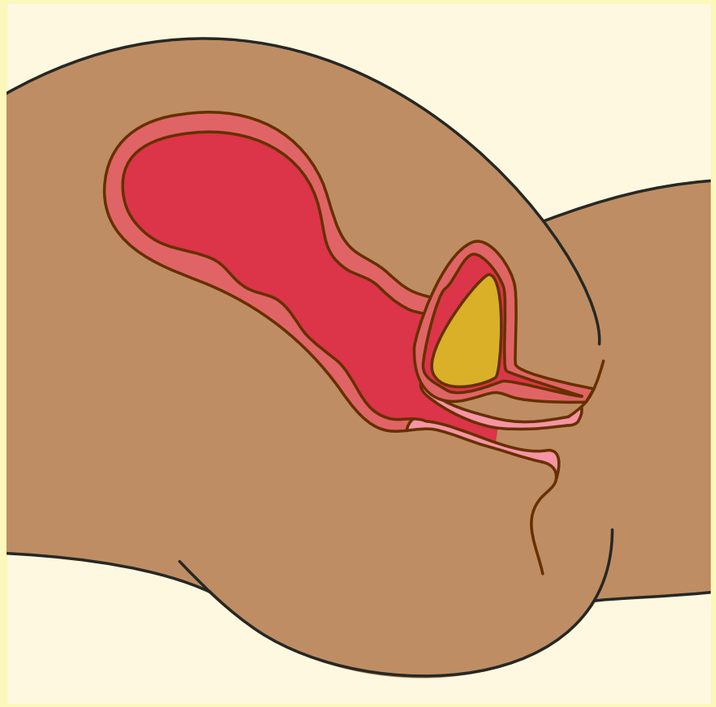
- What did you do?
- What went well?
- What would you do differently?
- Let us look at the Postpartum Monitoring Form.
  - Ensure it is filled in correctly.

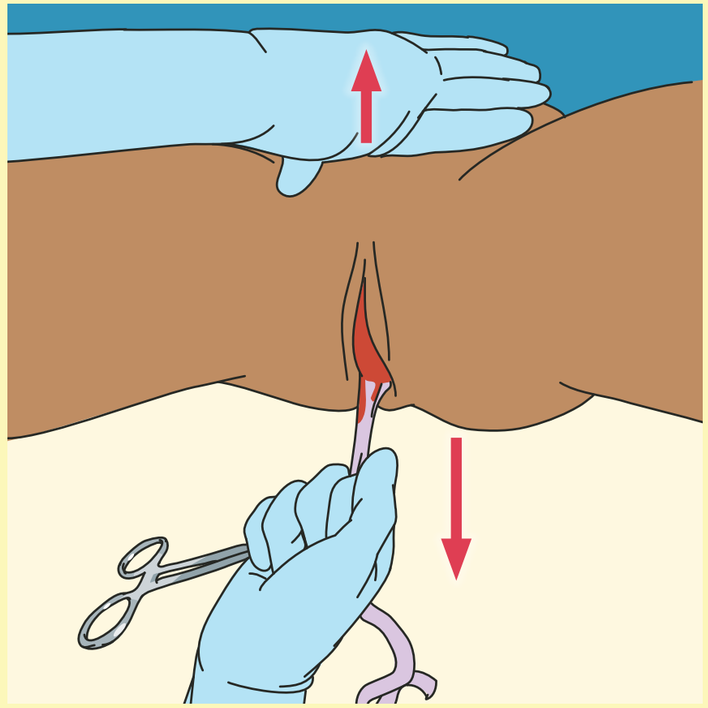
Expected actions	Prompts and info
Immediately after birth ☐ Check for 2nd baby ☐ Give uterotonic within one min ☐ Help woman be comfortable in bed ☐ Place blood loss measurement tool ☐ Clamp and cut cord (1–3 min) ☐ Gentle cord traction during contractions ☐ Hold placenta in both hands ☐ Twist membranes	**Simulator**: • Allow placenta to be delivered intact. • Keep uterus hard and blood flow very light when placenta delivers. **Example prompts if steps missing:** Does she need medication? Is the placenta out?
Check tone, placenta and tears ☐ Communicate with woman ☐ Check tone, massage if soft ☐ Check placenta ☐ Check for tears	**Simulator**: Stop bleeding. **Example prompt:** Did you remember to check tone/placenta/tears?
Assess and record ☐ Check for PPH ☐ Check tone, blood loss and flow ☐ Check BP, pulse and SI ☐ Record	**If assessed:** • Uterine tone: hard • Blood loss: what is in the tool, ensure it remains below 500 mL • Blood flow: no bleeding • Pulse: 95 bpm • BP: 110/65 mmHg **Example prompt:** Does she have PPH?
Say: 15 min have passed. What will you do now?	
Assess and record ☐ Check for PPH ☐ Check tone, blood loss and flow ☐ Check BP, pulse and SI ☐ Communicate with woman ☐ Continue to monitor	**If assessed:** • Uterine tone: hard • Blood loss: less than 300 mL • Blood flow: no bleeding • Pulse: 92 bpm • BP: 116/68 mmHg **Example prompt:** Is she OK? Did you record in the Postpartum Monitoring Form?

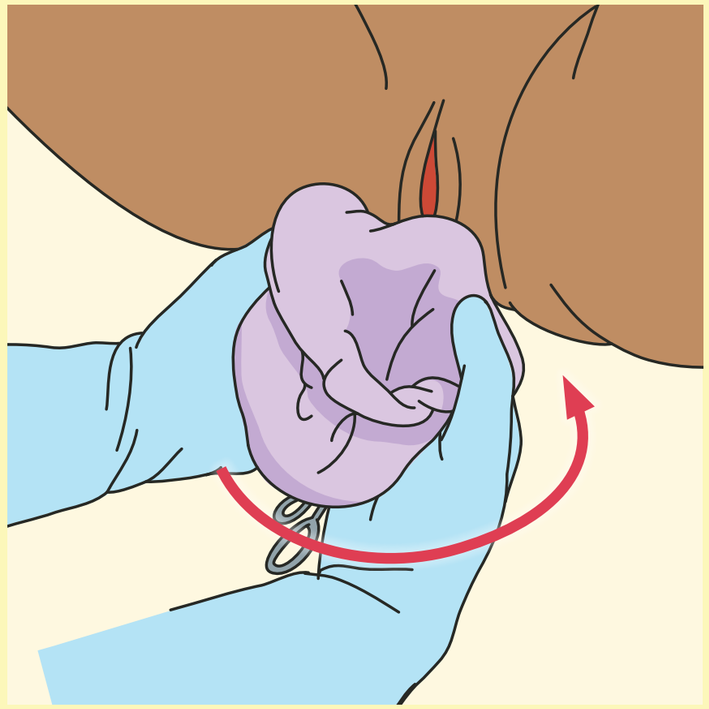


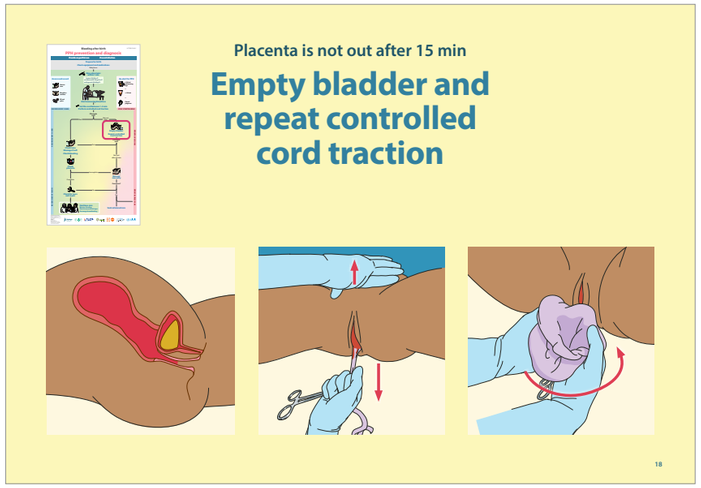
Placenta is not out after 15 min

Empty bladder and repeat controlled cord traction









Facilitation note

Introduce by saying, "We will now review complications on the yellow zone of Action Plan 1"

Explain

- **We will review problems that can happen before PPH.**
- Prompt treatment can prevent PPH.

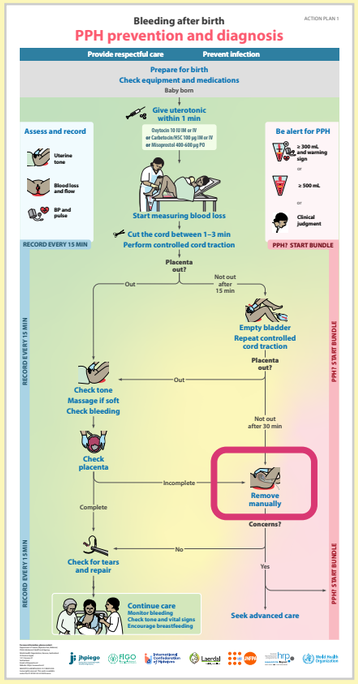
Retained placenta

- The placenta will usually deliver within 15 min, when a uterotonic is given for prevention.
- If it takes longer than 15 min:
  - encourage an empty bladder
  - repeat CCT.

If PPH, but placenta is not out
Start PPH bundle immediately, then move to manual removal.

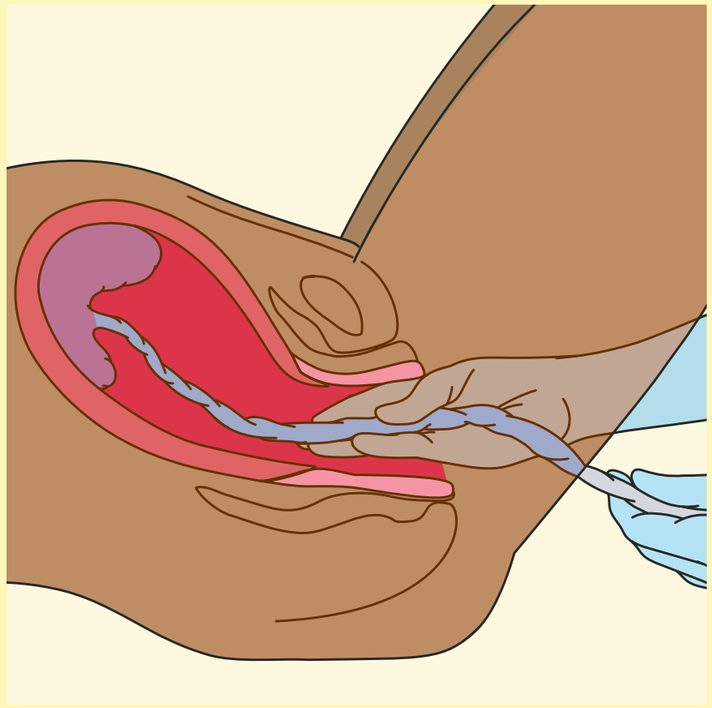
Ask

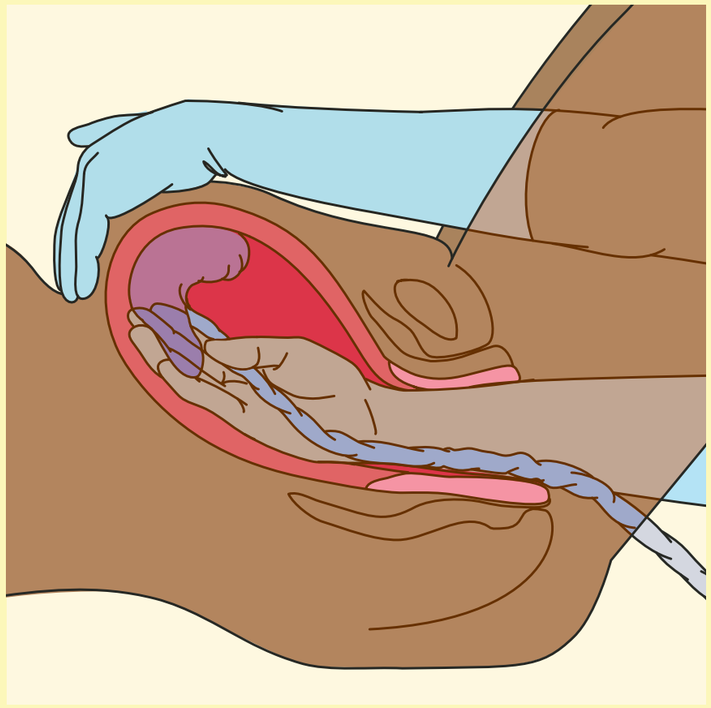
1. Why should you never pull on the placenta or cord when you feel resistance?
   - Because you might tear the cord off or pull the uterus out.
2. How does turning the placenta reduce the risk of leaving tissue behind?
   - It helps it form a rope with the membranes, which is stronger and less likely to leave pieces behind.

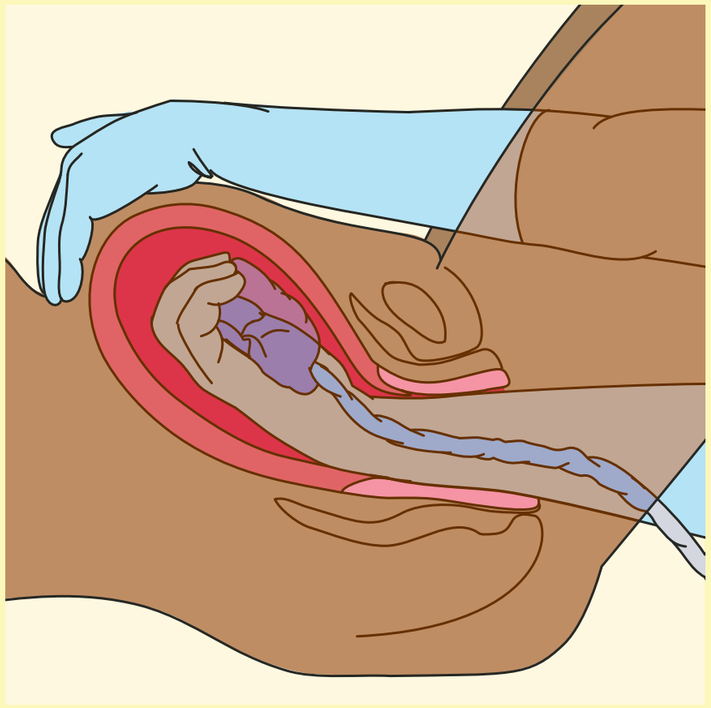


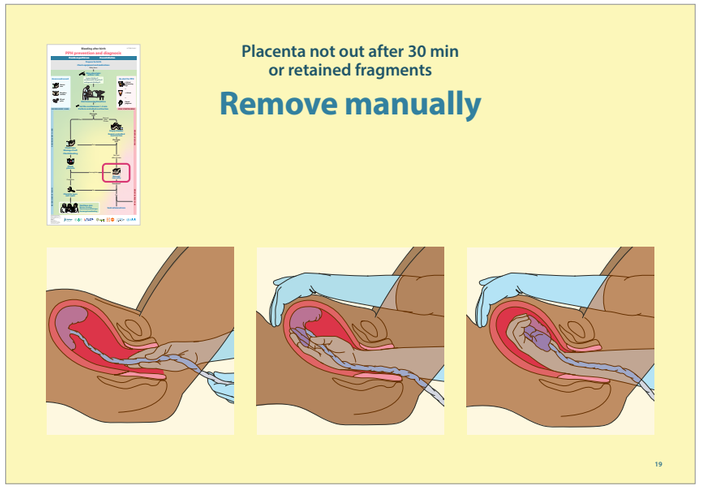
Placenta not out after 30 min or retained fragments

Remove manually









Explain

- Perform manual removal if:
  - the placenta is **not out after 30 min**
  - or the placenta is not complete.
- If PPH before the placenta is out:
  - start PPH bundle immediately,
  - then move to manual removal.

Suspect placenta accreta if unable to remove the placenta or tissue is very adherent. Call for help and advanced care. Proceed to laparotomy and possible total hysterectomy

If PPH start the bundle immediately.

Demonstrate and practise

- Show video **Manual removal of the placenta** (5 min) or demonstrate using the simulator.
- Ask participants to practise in pairs and provide guidance.

Provide respectful care and prepare

- ☐ Obtain consent and ensure privacy.
- ☐ Explain your actions and be gentle.
- ☐ Secure an IV if not in place.
- ☐ Ensure empty bladder.
- ☐ Give pain medication and a single dose of antibiotics – follow protocols.
- ☐ Wash your hands, and put on long sterile gloves.

Perform manual removal

- ☐ Hold umbilical cord with a clamp.
- ☐ Pull gently, using the cord as a guide.
- ☐ Form the hand into a cone and insert it into the uterus.
- ☐ Follow the cord to locate the placenta.
- ☐ Feel for the edge of the placenta.
- ☐ Identify the rough surface behind it.
- ☐ Sweep fingers gently back and forth to separate it from the uterine wall.
- ☐ Confirm total separation of placenta.
- ☐ Provide counter-traction to the fundus abdominally with your other hand.
- ☐ Withdraw your hand bringing the placenta out with it.

Placenta out but incomplete

- ☐ Using the same basic procedures, explore the uterus and remove retained lobes or tissue.

Check tone and placenta

- ☐ Check uterine tone.
- ☐ Massage if soft.
- ☐ Examine placenta for completeness.
- ☐ Check for PPH:
  - if PPH, start treatment bundle
  - if no PPH, give oxytocin with 20 IU in 1 L Ringer's Lactate (RL) or normal saline (NS) IV, over 4 h.
- ☐ Assess and record – tone, total blood lost, blood flow, pulse and BP.

MODULE 3

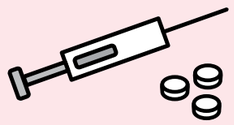
Treatment with the PPH bundle

Diagnose and treat PPH early

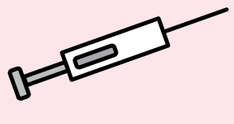
PPH bundle



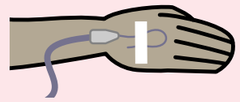
M
Massage



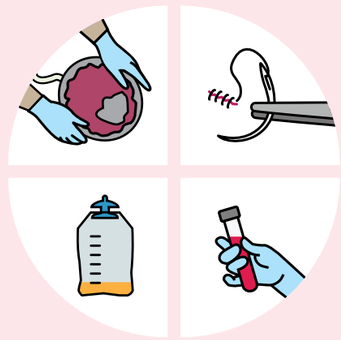
O
Oxytocic



T
TXA



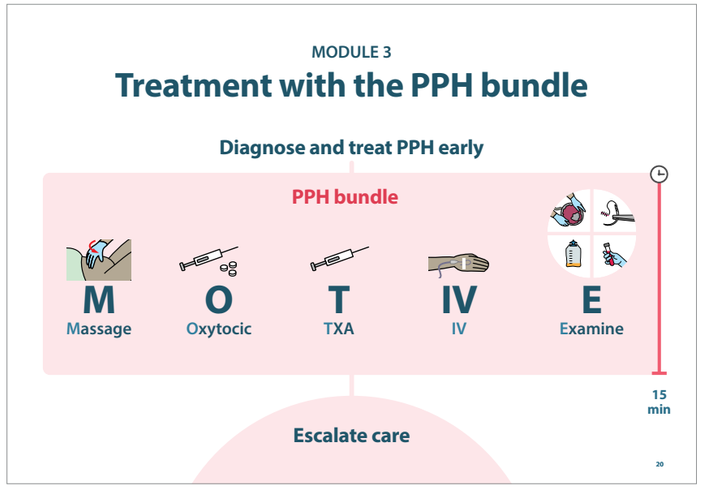
IV
IV



E
Examine

15 min

Escalate care



Facilitation note

Point to each letter on the front of this page as you explain what it stands for. Do not go into detail now, you will explain more later.

Explain

- Many deaths from PPH at both vaginal and caesarean births can be prevented with:
  - **prompt recognition** with the objective blood loss measuring tool and warning signs/vital signs
  - **prompt treatment** with the PPH bundle.

Early diagnosis

Early diagnosis is the first step and it begins with you.

Be alert for PPH

- **Blood loss ≥ 300 mL** with any other warning sign, or
- **Blood loss ≥ 500 mL,** or
- **Clinical judgment.**

If PPH, call for help and start treatment bundle

- A clinical bundle is a set of interventions that, when given together, improves outcomes.
- For the PPH clinical bundle, use the mnemonic **MOTIVE** to remember the interventions:
  - **massage** the uterus to stimulate contractions
  - **oxytocic** (uterotonic) drugs to stimulate contractions
  - **TXA** prevents clots from breaking down
  - **IV** access/fluids
  - **examine** to find the cause of bleeding.

Escalate care

After the PPH bundle, escalate care if bleeding persists, the cause is unclear, or cannot be managed.

When the woman gave birth elsewhere, you may not know the amount of blood lost. Rely on your assessment and clinical judgment. When in doubt, treat.

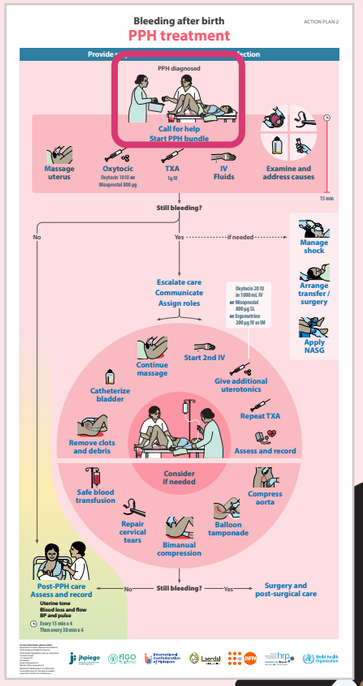
Practice

Taking a "quick pulse"

- Count your heart beat at your neck or wrist for 6 seconds, using a watch or clock.
- Multiply this number by 10 for the quick pulse rate.

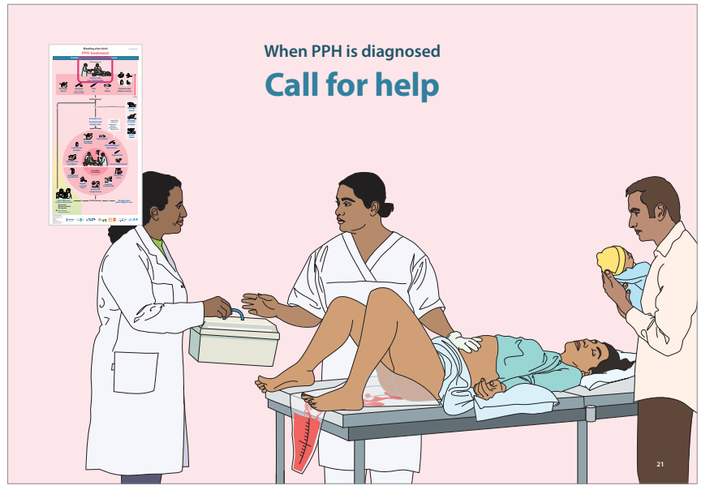
Ask

1. If a woman has lost <500 mL of blood but >300 mL, what are the warning signs that would have you diagnose PPH and begin treatment?
   - Pulse >100, or systolic BP <100, or diastolic BP <60 or a SI >1 (pulse > systolic BP)
   - Rapid blood flow, large clots, history of anaemia.



When PPH is diagnosed

Call for help



Explain

When you diagnose PPH

- Shout for help and emergency equipment and medication.
- Do a rapid assessment for shock.
- Explain your actions to the woman and her companion.
- Communicate with the health team and establish roles.

Tools for team communication

- Good team communication saves lives.
- It helps reduce errors, especially when calling for help or handing over care.
- We will review two techniques:
  - **SBAR** to present a clinical case to team members in a structured way
  - **closed-loop** communication to confirm clear understanding.

Demonstrate

1. SBAR

Act out the following example explaining what the letters stand for.

- **Situation:** What's happening now?
  - Please, come help me with Mariam! She gave birth an hour ago and is now bleeding heavily.
- **Background:** What led up to this?
  - She had a normal bith, but has lost 350 mL of blood. She received oxytocin after birth.
- **Assessment:** What do you think is going on?
  - I think she may be having PPH. Her pulse is 110, BP 90/60, and the uterus is soft.
- **Recommendation:** What do you need?
  - I need you to help me. We need to start the PPH bundle so please bring the emergency trolley.

2. Closed-loop communication

Ask two participants to act out the example of closed-loop on page 10 of the Provider Guide.

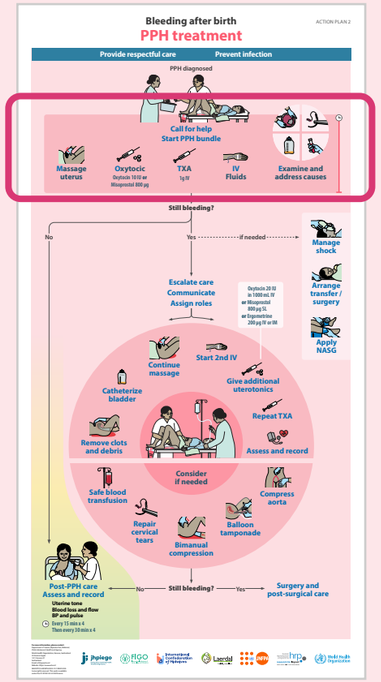
Practice

Read with participants the SBAR case on page 11 of the Provider Guide.

1. **Individual reflection (2 min)**
   - Review the SBAR template in the Provider Guide.
   - Based on the case, how would you use it when calling a colleague?
2. **Share-out (5 min)**
   - One participant demonstrates to the larger group how they would use the tool to communicate this case.

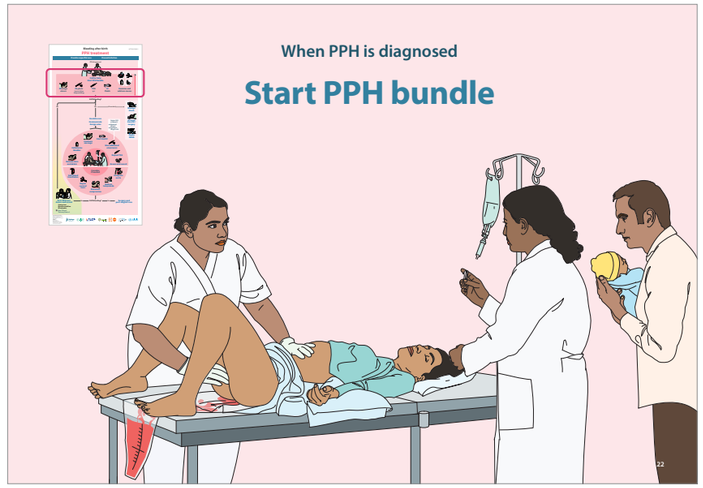
Discuss

- Do you have an emergency trolley or kit? If not, how could you create one?
- What was easy or difficult about using SBAR and closed-loop?
- How could this help in real emergencies?
- Why is it important to establish roles?



When PPH is diagnosed

Start PPH bundle



Explain

Starting the PPH bundle

- Administer all interventions, even if the woman improves after only one or two.
- The order does not matter.
- Aim to finish all interventions in 15 min.

Massage the uterus

- To cause contraction, expel blood and clots, assess for distended bladder.
- If alone, massage first, then move on.

Give oxytocin/uterotonic

- Give oxytocin 10 IU IV:
  - diluted in 100–500 mL of crystalloid fluids as fast as possible (preferable to avoid hypotension)
  - or as bolus over 1–2 min.
- If no IV access:
  - give oxytocin 10 IU IM or misoprostol 800 µg SL
  - call for help to start IV.

Give TXA

- 1 g IV over 10 min. Faster administration can cause hypotension.
- TXA and oxytocin can be given in the same line, but not in the same bag.
- **Do not start TXA if:**
  - 3 h or more since birth
  - or if there is a contraindication or hypersensitivity to TXA, known thromboembolic event, renal failure or history of convulsions.

Start IV

- Use 18 gauge cannula or largest available.
- Collect blood for:
  - bedside haemoglobinometer, or
  - lab testing for Hb, cross-match, and clotting studies if available.
- Begin infusion of crystalloid fluids NS or RL, if not given with oxytocin already.
- If clinically indicated, insert 2nd IV line with crystalloid fluids (NS or RL) for fluid resuscitation.

Examine for causes

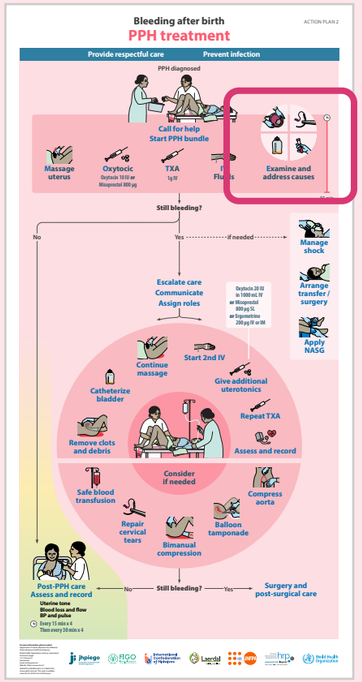
- As you are providing care, notice potential causes such as:
  - atony when you massage
  - tears as you check blood flow.
- We will review examination shortly.

During and after a caesarean section (CS)

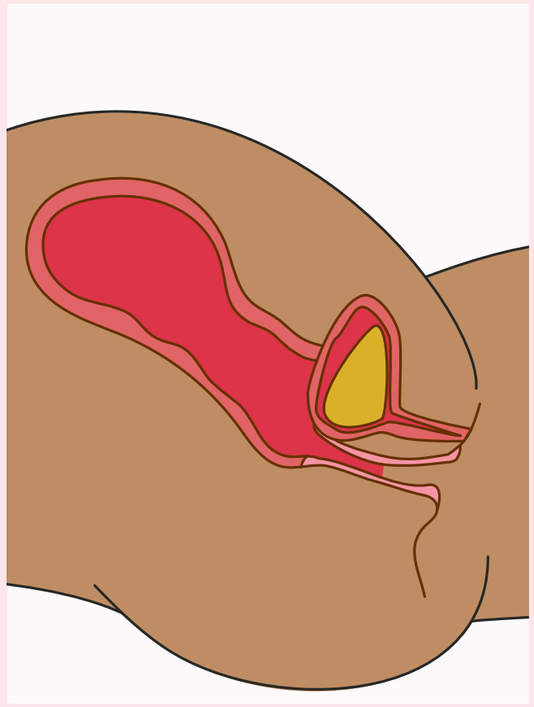
- Challenges:
  - there is often more bleeding
  - interpreting the haemodynamic signs which may be affected by anaesthesia.
- Most PPH bundle components are already given.
- Use clinical judgment to initiate treatment and decide to give TXA.

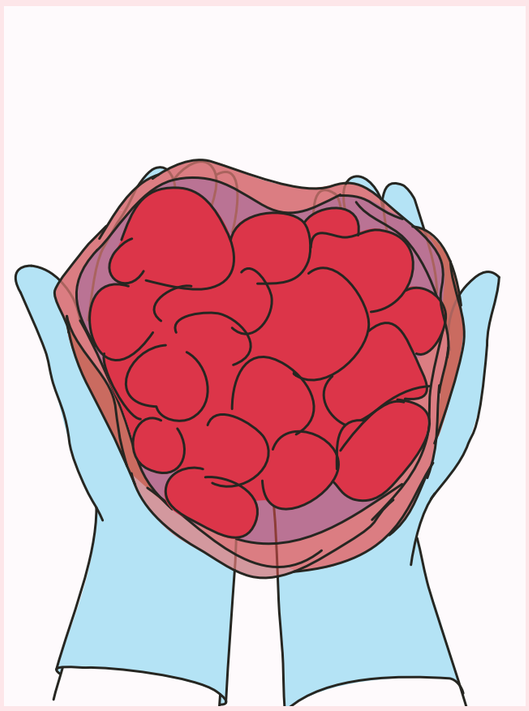
Discuss

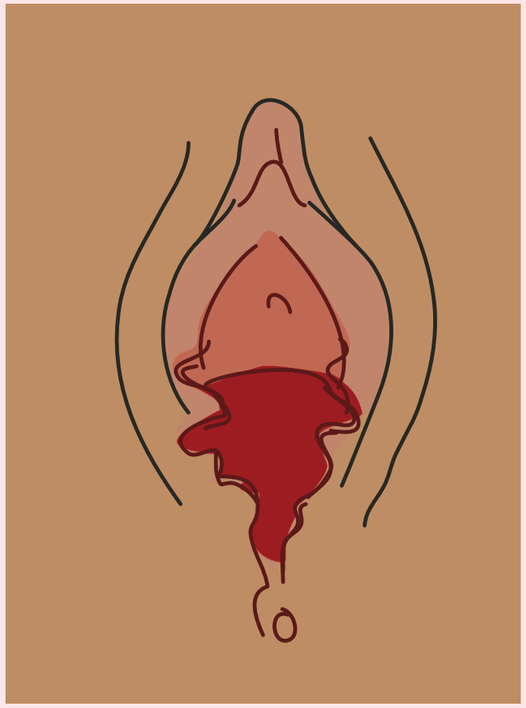
1. How is the PPH bundle different to your current practice?
2. What challenges do you see in this new practice?
3. How do you measure blood loss at CS?



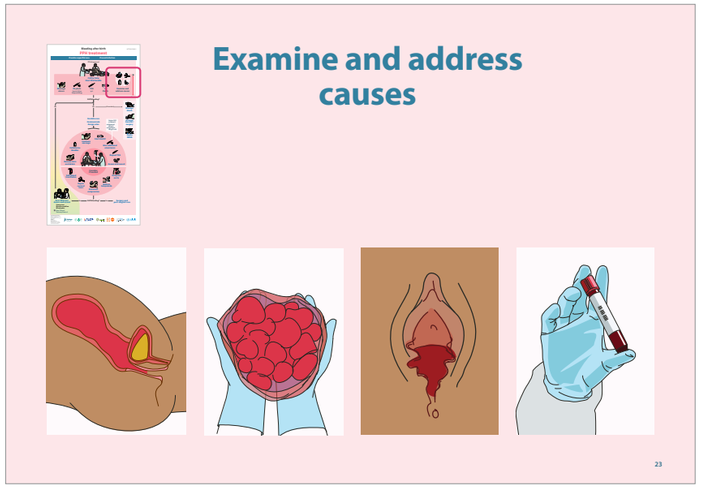
Examine and address causes











Explain

The last **E** in **MOTIVE** is for examination.

Once the other interventions in the bundle have been started:

- Communicate with the woman and companion. Explain respectfully what is happening.
- Examine thoroughly to identify and address all causes of bleeding.
- Begin with tone and work through causes.
- Record findings and actions.

Examine and address

Tone

- Recheck frequently that it remains firm.
- Check if fundus is firm, below umbilicus and midline.
- If the uterus is soft, massage to improve tone and expel clots.
- Assess for full bladder. Encourage frequent urination.
- Catheterize the bladder if she cannot urinate or the bleeding is heavy.

Tissue

- Inspect the placenta carefully for missing lobes, fragments or membranes.
- Inspect vagina, cervix and uterus to remove clots and tissue.
- Remember, if the placenta is adherent, suspect accreta and surgery is needed.

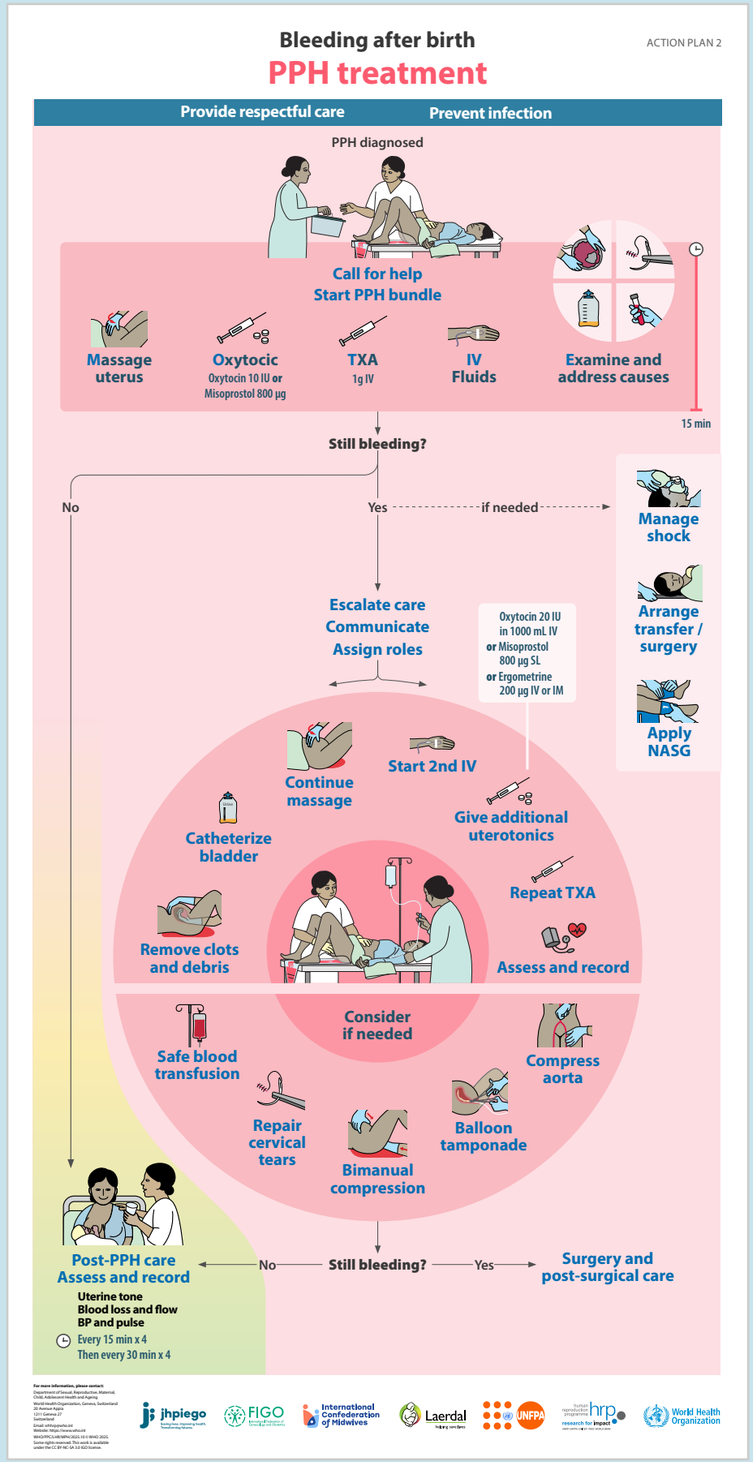
Trauma

- Check for perineal vaginal and cervical tears, and repair.
- Refer if you are unable to do the repair.
- If inverted uterus, replace and follow local protocols.
- If you suspect uterine rupture, seek advanced care.

Thrombin/coagulopathies

- If possible, collect blood for bedside or lab testing for:
  - Hb
  - type and cross match
  - prothrombin time, partial thromboplastin time and platelets
  - fibrinogen
  - coagulopathy.

Advanced Care Note
If within your scope of practice and training, provide additional care per local protocols. If beyond your scope of practice, seek advanced care immediately.

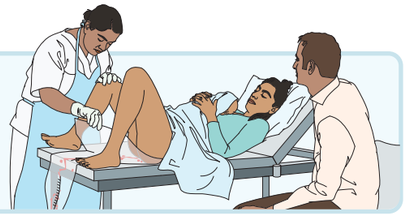


Scenario practice 2

Immediately following birth

Scenario practice 2

Immediately following birth



1. Prepare

- Read the guidance on page 1b to prepare.
- Make sure all equipment is ready. See list on page 1b.
- Prepare simulator with:
  - full blood tank
  - baby on chest and cord cut
  - placenta in tray, intact
  - 500 mL or more blood in measuring tool
  - light blood flow
  - uterus soft.

2. Brief

Share the learning outcomes

- Diagnose PPH at 500 mL with no other clinical signs.
- Treat PPH with the PPH bundle.

Assign roles

- Health worker
- Woman (facilitator)

Read the case

> You are caring for me, an 18 year old G1P1. I gave birth to my first baby without problems 10 min ago. You gave me 10 IU oxytocin IM within 1 min. You delivered the placenta without problems. I am feeling good, breastfeeding my baby. Please care for me.

Resolve doubts

Answer any questions participants may have.

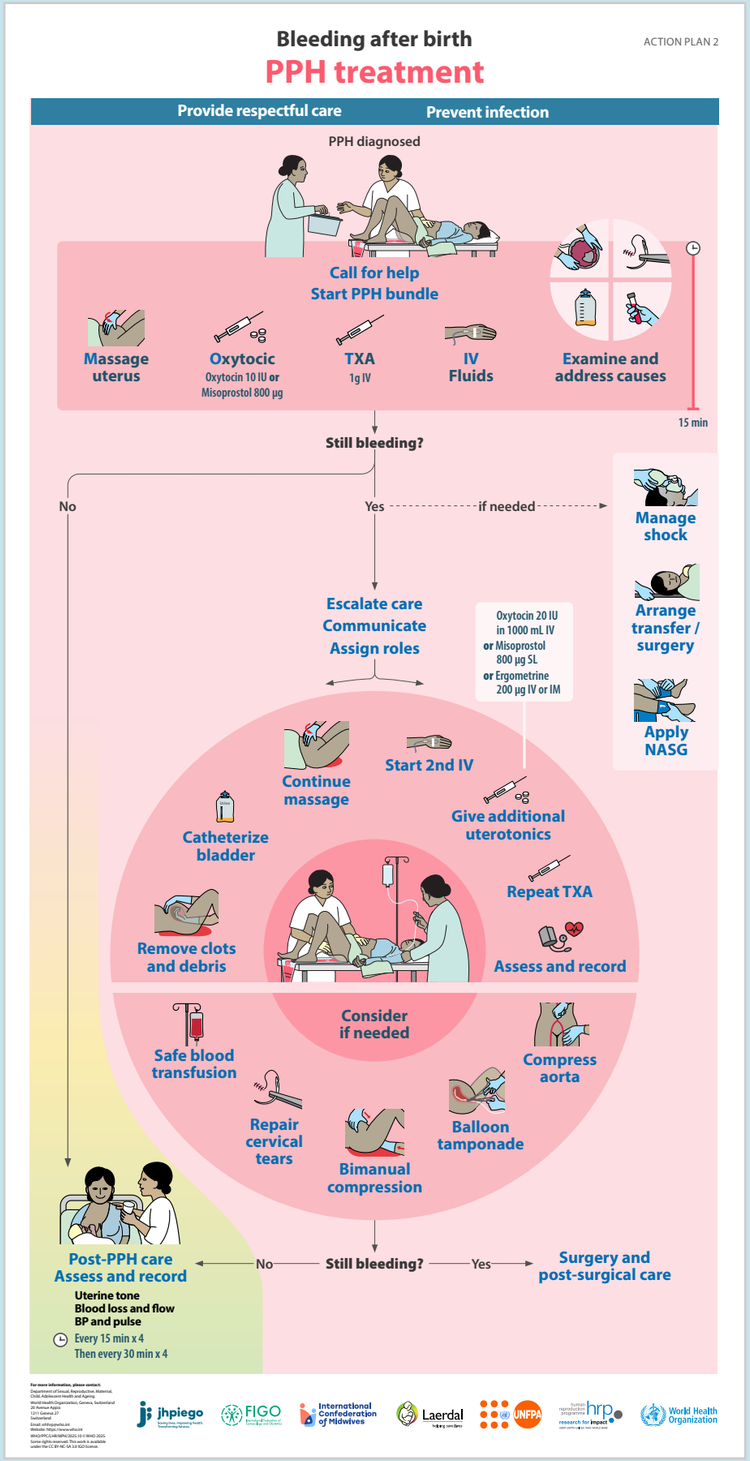
3. Run the scenario

Participants proceed as they would in a real scenario.

4. Debrief

- What did you do?
- What went well?
- What would you do differently?
- Did you diagnose PPH?
  - Yes.
- Why did you diagnose PPH?
  - The blood loss was 500 mL or more, though there were no other warning signs.
- Let's look at the postpartum monitoring form.
  - Ensure it is filled in correctly.

Expected actions	Prompts and info
Assess and communicate ☐ Assess ☐ Explain your actions and be gentle ☐ Call for help and emergency equipment ☐ Communicate using SBAR ☐ Assign roles in the team ☐ Team confirms with closed-loop	**Simulator**: Uterus soft, 500 mL in measurement tool, light blood flow. **Example prompt:** Is her bleeding normal? Do you need help?
Start PPH bundle ☐ Massage uterus ☐ Start IV ☐ Give Oxytocin (10 IU in 100–500 mL IV, as fast as possible) ☐ Give TXA (1 g in 10 mL IV, 10 min)	**Simulator**: Uterus soft, medium blood flow. **Example prompt:** Does she need medication?
Continue PPH bundle (Examine) ☐ Explain your actions and be gentle ☐ Check uterus ☐ Check bladder ☐ Check placenta ☐ Check for tears ☐ Check for clots and fragments	**Results** • Uterus soft • Bladder empty • Placenta intact • No tears • No clots or fragments **Simulator**: Uterus soft, medium blood flow. **Example prompt:** Why is she bleeding? Is the placenta intact?
Assess and record ☐ Check tone, blood loss and flow ☐ Check BP and pulse ☐ Calculate SI ☐ Record	**If assessed:** • Uterine tone: hard • Blood loss: more than 500 mL • Blood flow: stopped • Pulse: 90 bpm • BP: 110/64 mmHg **Example prompt:** How is she now? Did you fill out the Postpartum Monitoring Form?



Scenario practice 3

15 min after birth

Scenario practice 3

15 min after birth



1. Prepare

- Read about facilitating a scenario on page 1b.
- Make sure all equipment is ready. See list on page 1b.
- Prepare simulator with:
  - —full blood tank
  - —baby on chest and cord cut
  - —placenta in tray, intact
  - —300 mL blood in measuring tool
  - —light blood flow
  - —uterus soft.

2. Brief

Share the learning outcomes

- Appropriately use the Postpartum Monitoring Form to diagnose PPH.
- Diagnose PPH < 500 mL with hemodynamic warning signs.
- Treat PPH with the PPH bundle.

Assign roles

- Health worker 1
- Health worker 2
- Woman (facilitator)

Read the case

> You are caring for me, a 29 year old. I gave birth to my third baby without problems 15 min ago. You gave me 10 IU oxytocin IM within 1 min. You delivered my placenta without problems. You have come to check on me. Please care for me.

Resolve doubts

Answer any questions participants may have.

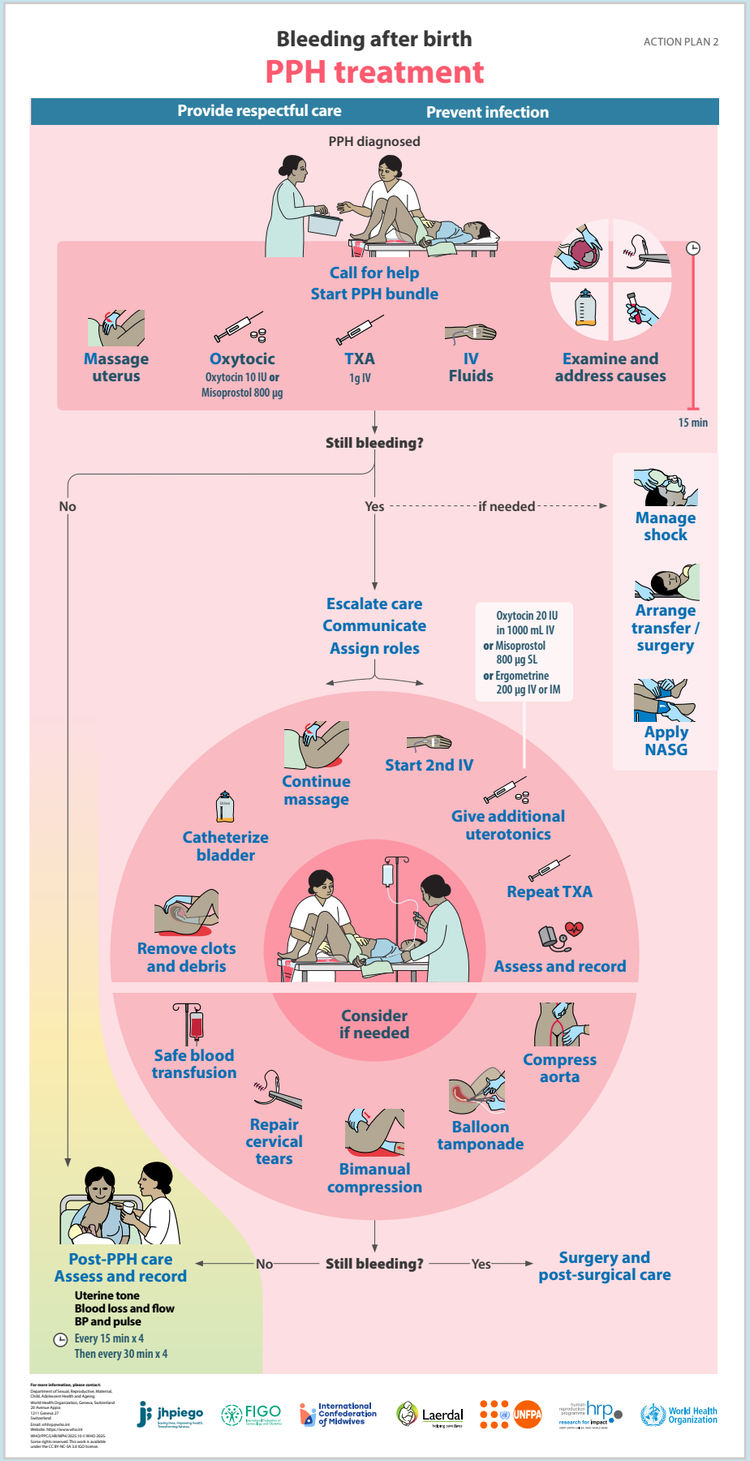
3. Run the scenario

Participants proceed as they would in a real scenario.

4. Debrief

- What did you do?
- What went well?
- What would you do differently?
- Was her SI >1?
  - —Yes.
- Did you diagnose PPH?
  - —Yes.
- Why did you diagnose PPH?
  - —Any of: SI >1, pulse >100, systolic BP <100
- Let's look at the Postpartum Monitoring Form.
  - —Ensure it is filled in correctly.

Expected actions	Prompts and info
Assess, record and communicate ☐ Check tone, blood loss and flow ☐ Check BP and pulse ☐ Calculate SI ☐ Record ☐ Call for help and emergency equipment ☐ Communicate using SBAR ☐ Assign roles in the team ☐ Team confirms with closed-loop	**If assessed:** • Uterine tone: soft • Blood loss: less than 500 mL • Blood flow: light • Pulse: 110 bpm • BP: 90/64 mmHg **Simulator**: Uterus soft, light blood flow. **Example prompts if steps missing:** Does she have a trigger? What is her SI? Is it >1? Can somebody help you?
Start PPH bundle ☐ Massage uterus ☐ Start IV ☐ Give Oxytocin (10 IU in 100–500 mL IV, as fast as possible) ☐ Give TXA (1 g in 10 mL IV, 10 min)	**Simulator**: Uterus soft, light blood flow. **Example prompt:** Does she need medication?
Continue PPH bundle (Examine) ☐ Explain your actions and be gentle ☐ Check uterus ☐ Check bladder ☐ Check placenta ☐ Check for tears ☐ Check for clots and fragments	**Results** • Uterus firm • Bladder empty • Placenta intact • No tears • No clots or fragments **Simulator**: After bundle, uterus hard and stop blood flow. **Example prompt:** Did you check for (tone, tears, placenta?)
Assess and record ☐ Check tone, blood loss and flow ☐ Check BP and pulse ☐ Calculate SI ☐ Record	**If assessed:** • Uterine tone: hard • Blood loss: (see what is in the tool) • Blood flow: stopped • Pulse: 98 bpm • BP: 104/68 mmHg , SI 0.94 or <1 **Example prompt:** How is she now? Did you fill out the Postpartum Monitoring Form?



Scenario practice 4

Retained placenta

Scenario practice 4

Retained placenta



1. Prepare

- Read the guidance on page 1b to prepare.
- Make sure all equipment is ready. See list on page 1b.
- Prepare simulator with:
  - full blood tank
  - baby on chest and cord cut
  - placenta not out
  - 200 mL blood in measuring tool
  - heavy blood flow
  - uterus soft.

2. Brief

Share the learning outcomes

- Identify PPH with retained placenta.
- Start PPH bundle first.
- Manually remove placenta according to protocol.

Assign roles

- Health worker 1
- Health worker 2
- Woman (facilitator)

Read the case

> You are caring for me, Ms. S. and my first baby was born with no problems 20 min ago. You gave me 10 IU oxytocin IM within 1 min. You attempted CCT but my placenta has not delivered. Please care for me.

Resolve doubts

Answer any questions participants may have.

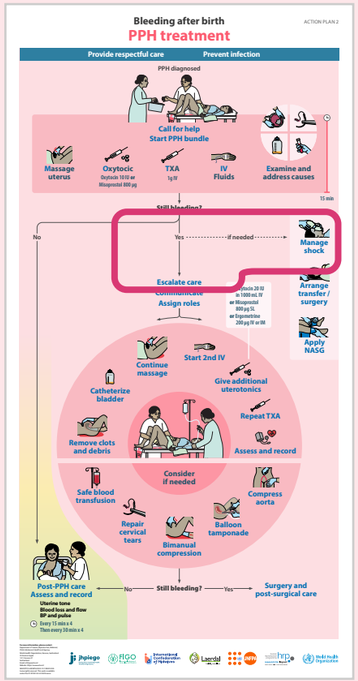
3. Run the scenario

Participants proceed as they would in a real scenario.

4. Debrief

- What did you do?
- What went well?
- What would you do differently?
- Did you diagnose PPH?
  - Yes.
- Why did you diagnose PPH?
  - Retained placenta at 20 min with heavy bleeding.
- Let's look at the Postpartum Monitoring Form.
  - Ensure it is filled in correctly.

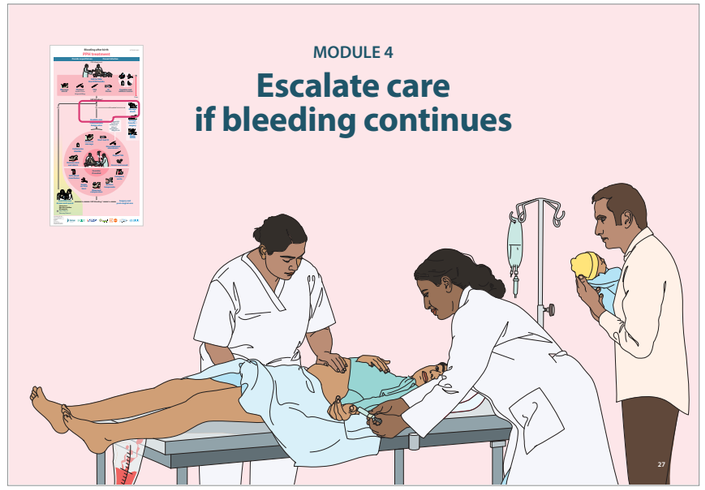
Expected actions	Prompts and info
Assess and communicate ☐ Assess ☐ Explain your actions and be gentle ☐ Call for help and emergency equipment ☐ Communicate using SBAR ☐ Assign roles in the team ☐ Team confirms with closed-loop	**Simulator**: Uterus soft, heavy blood flow. Placenta will not deliver. **Example prompts if steps missing:** Does she have PPH? Can somebody help you? Should you use SBAR?
Start PPH bundle ☐ Massage uterus ☐ Start IV ☐ Give Oxytocin (10 IU in 100–500 mL IV, as fast as possible) ☐ Give TXA (1 g in 10 mL IV, 10 min) ☐ Examine	**Simulator**: Uterus soft, medium blood flow. Placenta not delivered. **Example prompt:** Does she need an intervention?
Prepare for manual removal ☐ Obtain consent and ensure privacy ☐ Explain your actions and be gentle ☐ Give pain relief and antibiotic ☐ Wash hands, wear long sterile gloves	**Simulator**: Uterus soft, medium blood flow. Placenta not delivered.
Manually remove the placenta ☐ Pull cord gently and follow as guide ☐ Insert cone-shaped hand through cervix ☐ Detach placenta gently ☐ Provide counter traction to the uterus ☐ Remove hand with placenta	**Simulator**: Allow placenta removal. Contract uterus and stop bleeding. **Example prompt:** Do you need to remove the placenta?
Once placenta is out ☐ Check tone, massage if soft ☐ Check placenta	**Simulator**: Uterus hard, no blood flow. **Example prompt:** Did you check the placenta?
Assess and record ☐ Check tone, blood loss and flow ☐ Check BP and pulse ☐ Calculate SI ☐ Record	**If assessed:** • Uterine tone: hard • Blood loss: (whatever is in the tool) • Blood flow: stopped • Pulse: 88 bpm • BP: 130/72 mmHg **Example prompt:** How is she now? Did you fill out the Postpartum Monitoring Form?



MODULE 4

Escalate care if bleeding continues





Explain

The last **E** in **MOTIVE** is also for escalation.

Escalate care

- If bleeding:
  - is heavy while giving the bundle, or
  - continues after bundle is complete, or
  - is more than 1000 mL at any time.
- If a woman:
  - has extensive trauma or
  - has uterine inversion or
  - is hypotensive or unresponsive.
- If you are unable to identify or manage the cause.

Act fast. The woman's life is in danger.

Call for more help

- There should always be at least two health workers during an emergency.
- Request the most senior health worker to urgently attend the woman.
- Always continue monitoring and give what treatment you can while waiting for help or transfer.
- Call for the anaesthetist if there is one at this facility.

Communicate and assign roles

- Use **SBAR tool** to communicate clinical status.
- The most senior health worker leads the team and assigns roles.
- Use closed-loop communication to confirm understanding and completion of tasks.

Continually assess for shock

Early signs of shock

- Fast or weak pulse > 100 bpm.
- Systolic BP < 100 mmHg.
- SI >1 (pulse > systolic BP).

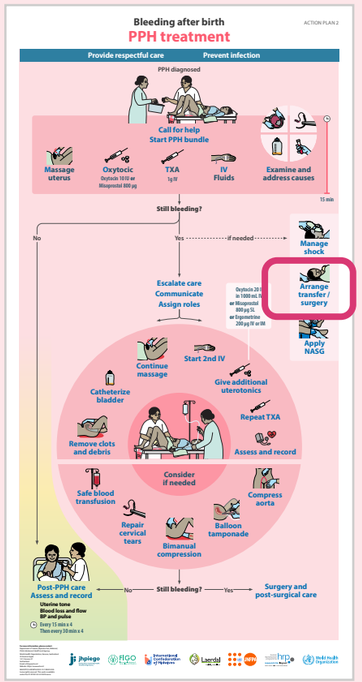
Late signs of shock

- Very low BP and weak pulse.
- Rapid breathing >30 breaths/min.
- Pale skin, including eyelids, mouth, palms.
- Sweating.
- Change in mental status – anxiety, confusion, unconsciousness.
- Scanty urine output (<30 mL/h).

If shock – Check ABCs and manage

- Open airway and ensure breathing.
- If not breathing or in cardiac arrest, begin resuscitation.
- Start oxygen at 6–8 L/min.
- Assess circulation – BP and heart rate.
- Insert second IV for crystalloid fluids and blood transfusion.
- Support/increase blood volume.
- Apply NASG if shock or risk of shock.
- Prepare for blood transfusion.

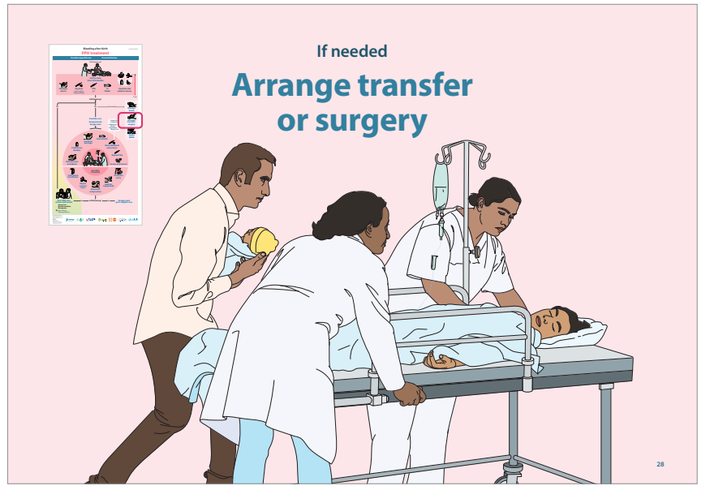
Advanced care note
Transfer immediately while still providing care if what the woman needs is beyond the scope of your facility.



If needed

Arrange transfer or surgery





Explain

All facilities should have a formal referral protocol, including for internal referrals.

- The woman may need to be transferred:
  - to another facility or
  - to the surgical unit in your facility.

Act fast

- Delay in care is a common reason for PPH deaths.
- Don not wait until she is in shock to initiate transfer.

Prepare in advance

- Transfers may be complicated and delay care.
- Visibly post contact information for health workers, facilities, surgical and transport teams.
- Communicate the need for transport or surgery to your team and woman as soon as possible.
- Explain to the woman and her family, what is happening.

Communicate with referral facility or surgical team

- Notify the referral facility or surgical team of the transfer.
- Use SBAR to communicate with the referral health worker.
- Write a note for the receiving facility following national guidelines. Include:
  - assessment
  - diagnosis
  - care provided
  - last vital signs.

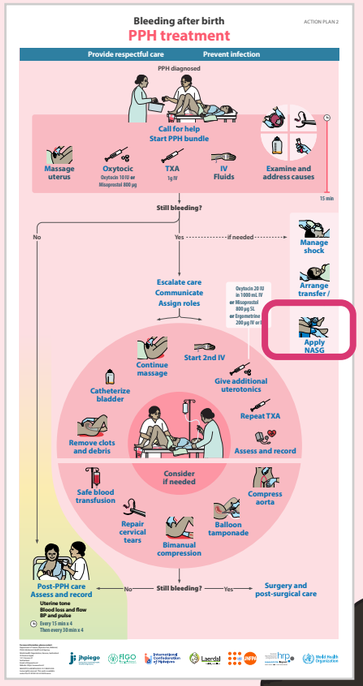
Care before and during transport

- If available, place NASG before transport.
- **Never** leave the woman alone.
- Continue care during referral and transport.
- Ensure all elements of the PPH bundle were given, plus the 2nd dose TXA.
- Continue the oxytocin and crystalloid IV infusion.
- Monitor and record uterine tone, blood flow and loss, pulse and BP.
- Always transfer the baby with the woman. Keep them warm.

Minimize bleeding during transfer

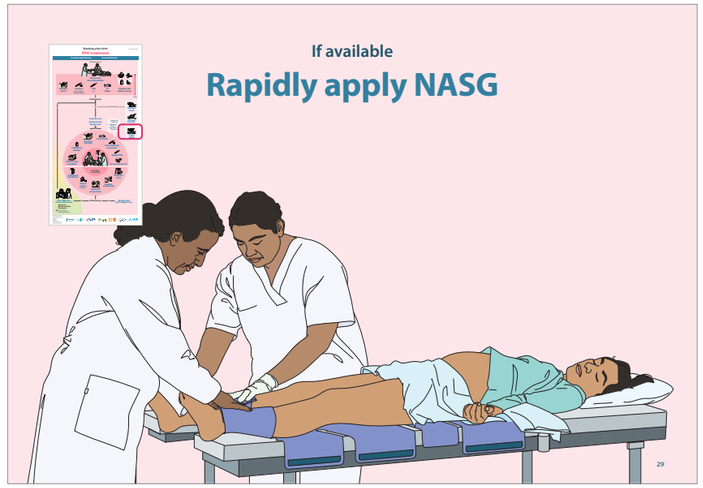
- Insert indwelling catheter to keep the bladder empty.
- For bleeding tears you cannot repair: continue to apply firm, steady pressure with a sterile cloth.
- Continue massage or bimanual uterine compression.
- Use bimanual or aortic compression, as needed.

Advanced Care Note
If participants have additional training and authorization to provide more advanced care, they should do so prior to and during transport.



If available

Rapidly apply NASG



Facilitation note

Before the course day, determine if NASG is available at this facility. If it is, check the protocol for timing of use.
If NASG is not available, skip this page and the next.

Explain

- The NASG applies pressure to the lower body and abdomen.
- It forces blood to the heart, lungs, and brain to prevent shock, or stabilize a woman in shock.
- NASG does not treat PPH but gives you time to seek treatment.

When to use NASG

- Any time there is a delay in care – transfer, waiting for blood, surgery, or procedures.
- Keep it on during all procedures, transport and surgery.
- Remove per protocol, only when bleeding stops and the woman is stable.

Usage tips

- Apply segments with strong pressure, but allow breathing in the abdominal area.
- It's faster with two people, but only one should handle segments 4, 5, and 6 to avoid overtightening.
- Leave the knees exposed for bending.
- For short women, fold segment 1 into segment 2 and start from segment 2.
- Apply the NASG even if the woman is unconscious.

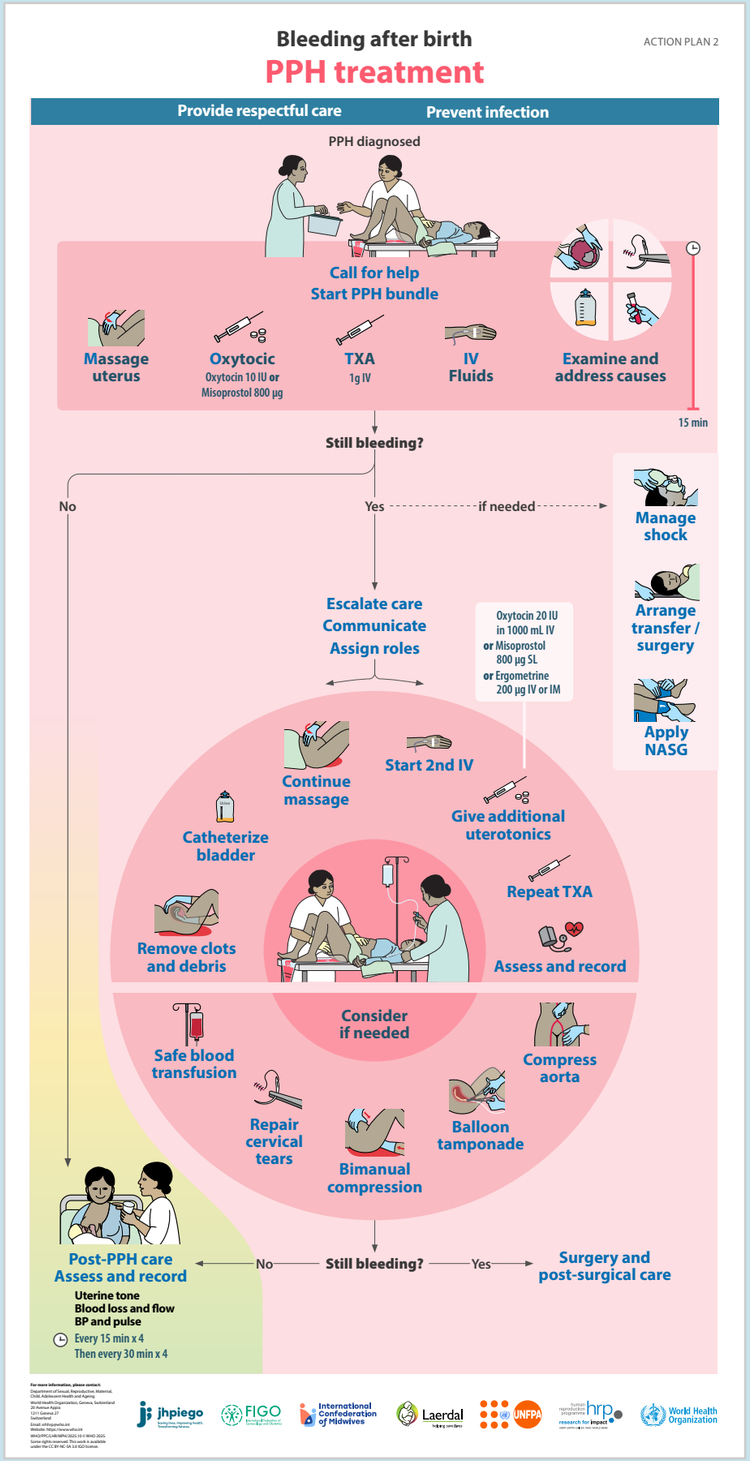
Procedures and treatment while NASG is in place

- The NASG stretches for checking uterine tone and massage.
- It can remain on during vaginal procedures and toileting – slide segment 4 if needed.
- Leave it on during surgery, but remove segments 4, 5, and 6 before incision, replacing them afterward.

Discuss

In your facility, who can:

1. Train staff on cleaning and storage of the NASG?
2. Manage NASG transport and maintenance between referral and referring facilities?
3. Train new health workers on the NASG protocol?



Skills practice – NASG

Skills practice

NASG

Skip this page if NASG is not available in the participants' facility.

1. Prepare

- Divide participants in groups, with one NASG per group.
- Gather necessary equipment and ensure the NASG is laid out flat.
- Refer to page 33 in the Provider Guide.

2. Demonstrate

Show video **Using an anti-shock garment (9:15 min)** or demonstrate with help of a volunteer using the checklist on this page.

3. Practise

Allow participants to practise. Use the checklist, also available in Provider Guide page 33.

Skills practice checklist

Communicate

- ☐ Explain your actions and be gentle.
- ☐ Assign roles in the team.

Place the woman on the NASG

- ☐ Roll her onto her side.
- ☐ Align the top of the NASG at her lowest rib.
- ☐ Align the pressure ball at the level of umbilicus.
- ☐ Align the dotted line with her spine.
- ☐ Roll her to the other side and flatten the garment.
- ☐ Roll her to her back again.
- ☐ Close each segment pair, starting from the ankles and ending with 6th segment.
- ☐ Use sufficient strength while ensuring the woman can breathe normally.

Check fit

- ☐ Place 1–2 fingers under the top of each closed segment.
- ☐ Pull up on the fabric, release and listen for a snapping sound.
- ☐ If no snapping sound, tighten the segment.
- ☐ Check woman can breathe comfortably.
- ☐ Loosen segments 5 and 6 if necessary .

Monitoring during use

- ☐ Continue to check tone, blood loss and flow, pulse and BP, per protocol.
- ☐ Note any signs of shortness of breath or decreased urine output.

Before removal

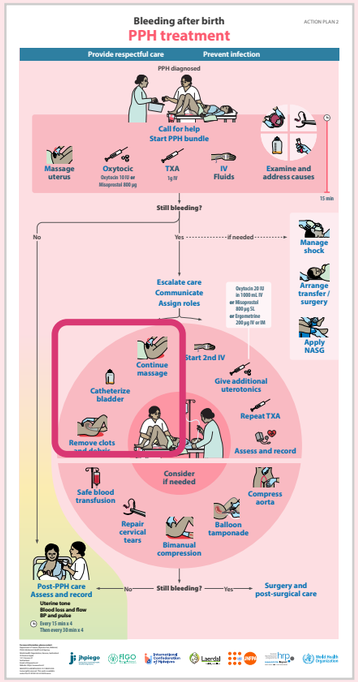
- ☐ Ensure the following has happened for at least 2 h:
  - —pulse is 100 bpm or less
  - —systolic BP is 100 mmHg or greater
  - —bleeding is reduced to 50 mL/h or less.
- ☐ Ensure IV fluids continue.

Removing the NASG

- ☐ Begin removal by opening ankle segments only.
- ☐ Wait 15 min and check that:
  - —pulse does not increase more than 20 bpm
  - —systolic BP does not decrease more than 20 mmHg.
- ☐ If vitals stable, open next segment pair in order, wait 15 min, then check vital signs.
- ☐ Continue in the same way with remaining segments.
- ☐ If any vital signs change beyond limits:
  - —quickly re-secure all segments starting from the last segment opened
  - —check for bleeding source.

After removal

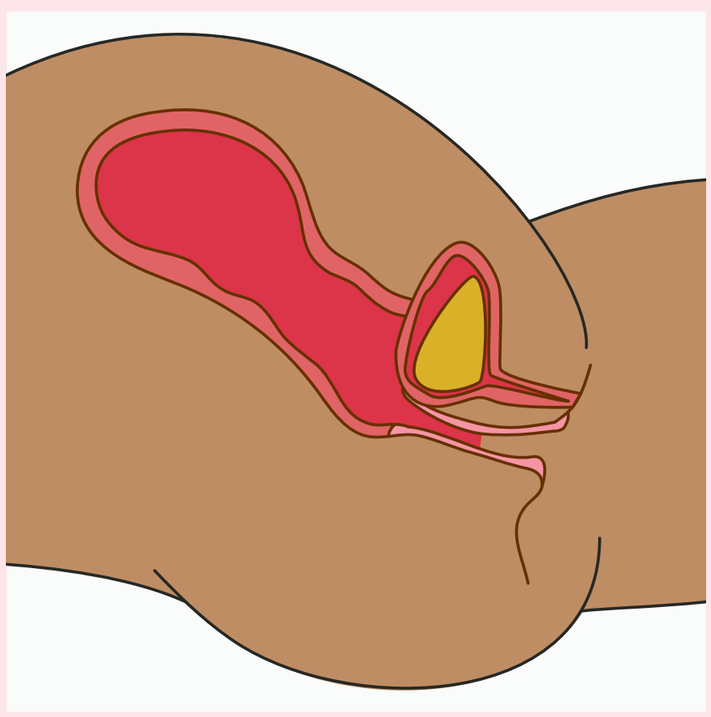
- ☐ Record procedure and observations in the Postpartum Monitoring Form.
- ☐ Continue to assess and record the woman's vital signs and condition.

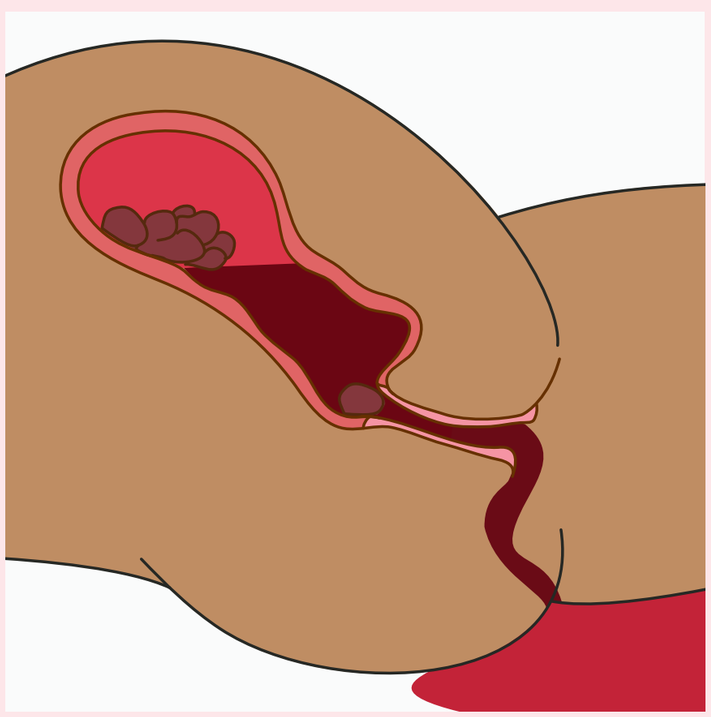


Health worker 1

Continue massage, catheterize bladder, remove clots/debris









Explain

There should always be at least two health workers during an emergency. The first health worker will focus on care at the perineum and abdomen.

Actions include:

1. **Communicate** with the woman, family and team:
   - explain what you are doing, and why
   - let her know that she may feel discomfort.
2. **Massage:**
   - expel clots contract the uterus
   - monitor and report uterine tone and bleeding.
3. **Catheterize the bladder:**
   - use sterile technique and place a catheter to empty the bladder
   - ensure no urine goes into the blood loss measurement tool.
4. **Remove clots and debris:**
   - ensure cervix is not blocked before clearing the uterus
   - remove clots and tissue to aid contraction and prevent infection.

Ensure the uterus is clear before considering UBT. If you cannot clear the uterus, seek advanced care.

Demonstrate

- ☐ Ensure privacy. Explain to the woman what you will do and why.
- ☐ Catheterize the bladder.

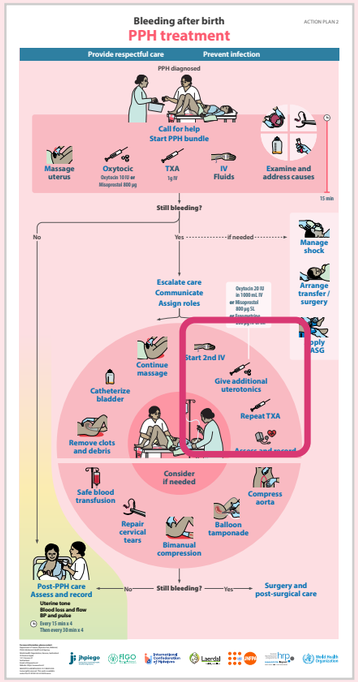
Remove clots and debris from the cervix

- ☐ Use sterile gloves.
- ☐ Gently perform a vaginal exam until you reach the cervix. It will feel swollen and loose.
- ☐ Check for clots and tissue in the cervical os.
- ☐ Gently pull them out.
- ☐ At the same time, massage the fundus with your other hand to assist removal and stimulate contractions.

Remove clots and debris from the uterus

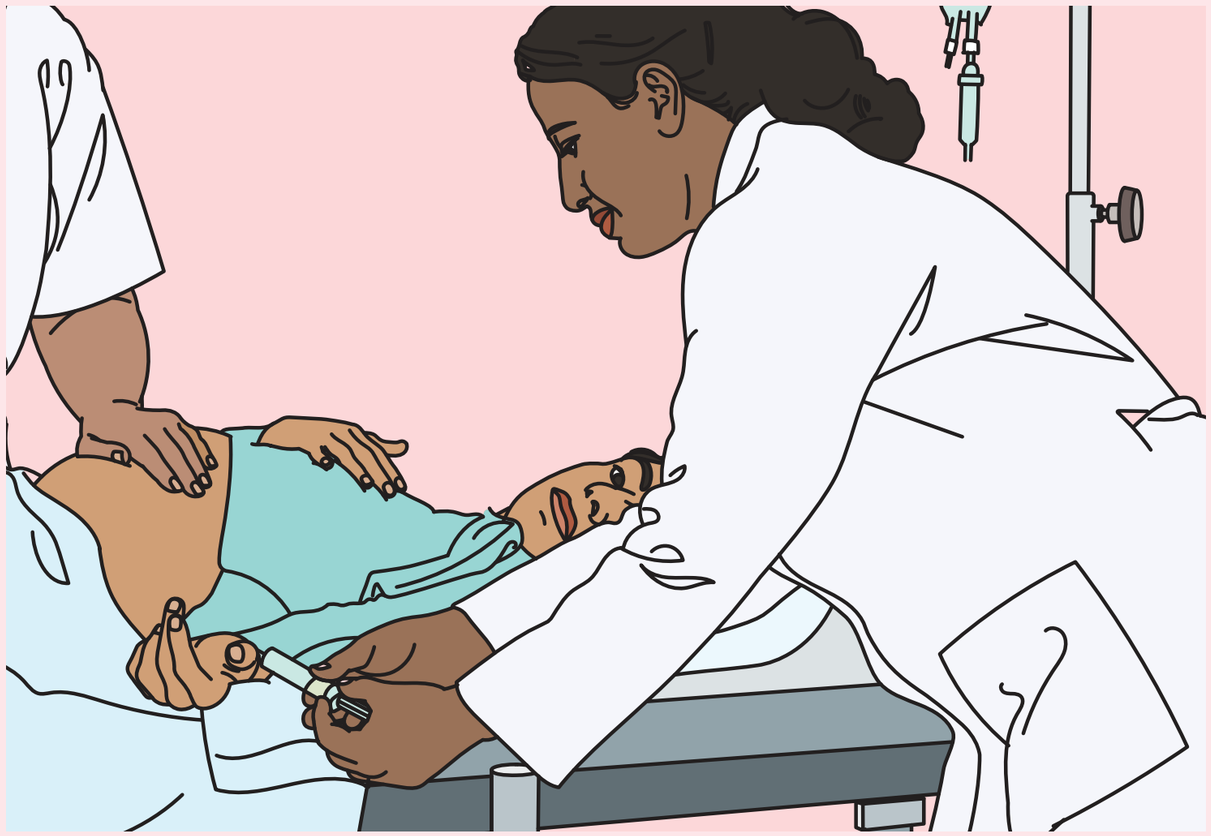
If the uterus is not firm, and it is within your scope of practice.

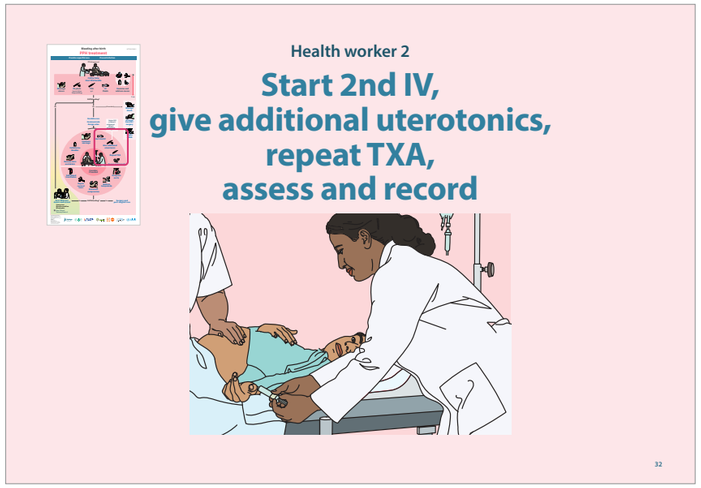
- ☐ Offer pain relief per local protocol, before you begin.
- ☐ Give a single dose of antibiotics:
  - Ampicillin 2 g or
  - Cefazolin 1 g IV.
- ☐ Wash your hands, use long, sterile gloves.
- ☐ Wrap sterile gauze around your index and middle fingers.
- ☐ Similar to manual removal of placenta:
  - form the hand into a cone and
  - gently guide it into the uterus.
- ☐ Wipe the interior of the uterus with the gauze.
- ☐ Bring out all debris as you remove your hand.
- ☐ Ensure you remove all the gauze.
- ☐ Check uterine tone and massage if soft.



Health worker 2

Start 2nd IV, give additional uterotonics, repeat TXA, assess and record





Explain

The second health worker will manage the medications, IV and recording. If you are alone, you will provide all treatments.

Actions include:

1. **Place a second IV** line:
   - for additional IV fluid
   - to administer other medications
   - to allow for blood transfusion
   - to allow for additional laboratory tests per local protocols.
2. **Give additional uterotonics**.
3. **Repeat TXA**.
4. **Assess and record** every 15 min or more often until stable:
   - tone, blood loss and flow, as reported by health worker
   - pulse and BP.

- **Act fast**. If bleeding continues, she may go into shock and die. Delays are a common reason women die from PPH.

Additional uterotonics

The choice of uterotonic should be based on availability and health worker's experience.

Oxytocin

- 20 IU in 1 L crystalloid fluids (NS or RL) IV.
- Infuse at 40 drops/min over 4 h.
- Do not give more than 3 L of IV fluids with oxytocin in 24 h.
- Maximum 60 IU in 24 h.
- Consider other uterotonic if oxytocin already infusing.

Ergometrine or ergometrine/oxytocin fixed dose

- Ergometrine:
  - 200 µg IV or IM in 15 min
  - if needed, repeat every 4 h up to 5 doses (1 mg total) in 24 h.
- Ergometrine/oxytocin fixed dose:
  - 500 µg / 5 IU IM
  - do not repeat.
- Use either if oxytocin unavailable or not effective.
- Do not use either if suspected retained placenta or if hypertension cannot be ruled out.

Misoprostol

- 800 µg SL – or PR if concerned about pyrexia or under anaesthesia.

Do not give heat-stable carbetocin for treatment – only for PPH prevention.

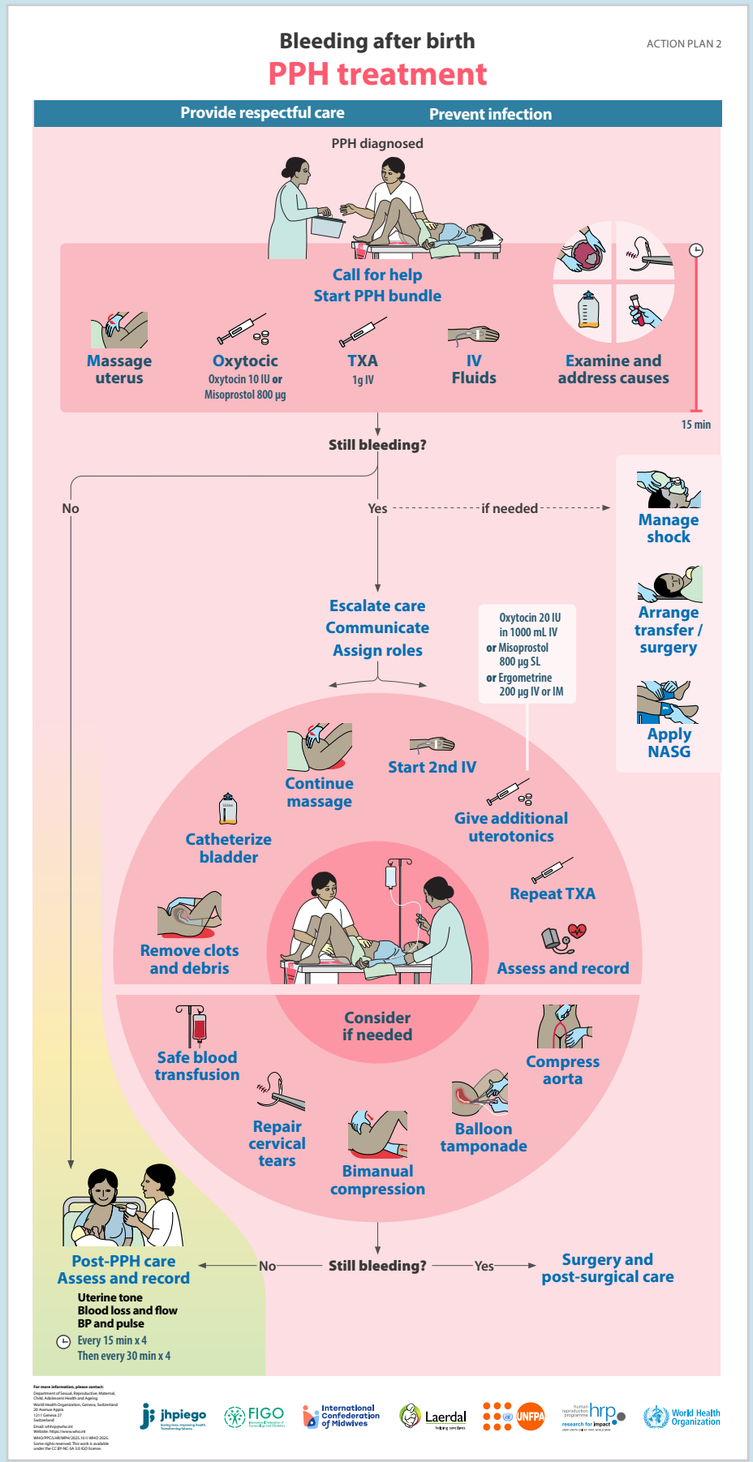
TXA second dose

- 1 g IV over 10 min
- Give if bleeding:
  - continues more than 30 min after first dose
  - or restarts within 24 h after first dose.
- Can be given in the same line as oxytocin but not mixed in the same IV bag.

Pause and reflect
If additional uterotonics and TXA fail, move quickly to transfer/surgery.

Discuss

1. If oxytocin fails or is not available, how do you choose which uterotonic to give?
   - Availablilty and health workers experience.

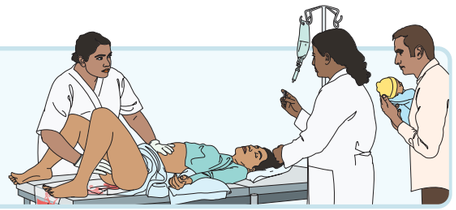


Scenario practice 5

Assessing after the PPH bundle

Scenario practice 5

Assessing after PPH bundle



1. Prepare

- Read the guidance on page 1b to prepare.
- Make sure all equipment is ready. See list on page 1b.
- Prepare simulator with:
  - full blood tank
  - baby on chest and cord cut
  - 550 mL blood in measuring tool
  - medium blood flow
  - uterus soft.

2. Brief

Share the learning outcomes

- Identify the PPH bundle has failed.
- Escalate and manage care for refractory PPH.

Assign roles

- Health worker 1
- Health worker 2
- Woman (facilitator)

Read the case

> You are caring for me, Ms. T. and I gave birth an hour ago. You gave me 10 IU oxytocin IM within 1 min for prevention. My placenta delivered a few min later. My uterus would not contract and I bled heavily. You completed the PPH bundle. Please continue to care for me.

Resolve doubts

Answer any questions participants may have.

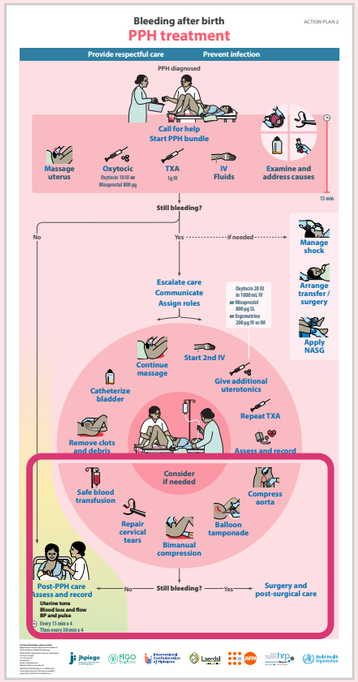
3. Run the scenario

Participants proceed as they would in a real scenario.

4. Debrief

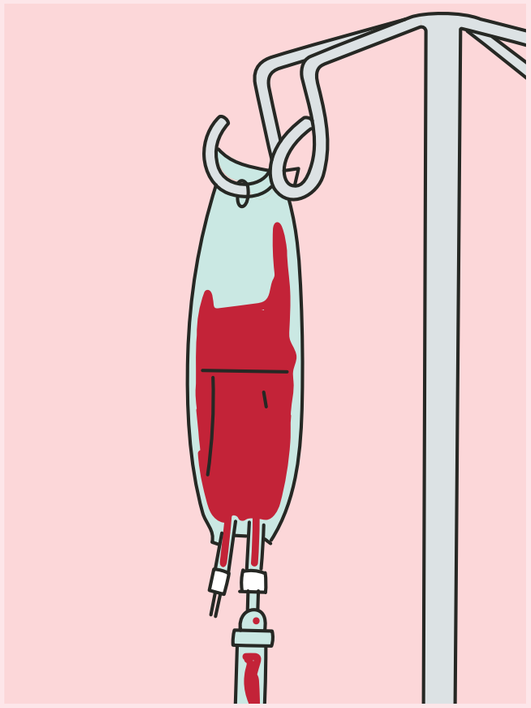
- Why did you escalate her care?
  - She continued to bleed after the bundle was given.
- Let us look at the Postpartum Monitoring Form.
  - Be sure it is filled in and check the shock indices. If not filled, ask what it was
  - First check: 1.87 **or** pulse > systolic BP = warning
  - Second check: 0.91 **or** pulse < systolic BP = OK
- What did you do?
- What went well?
- What would you do differently?

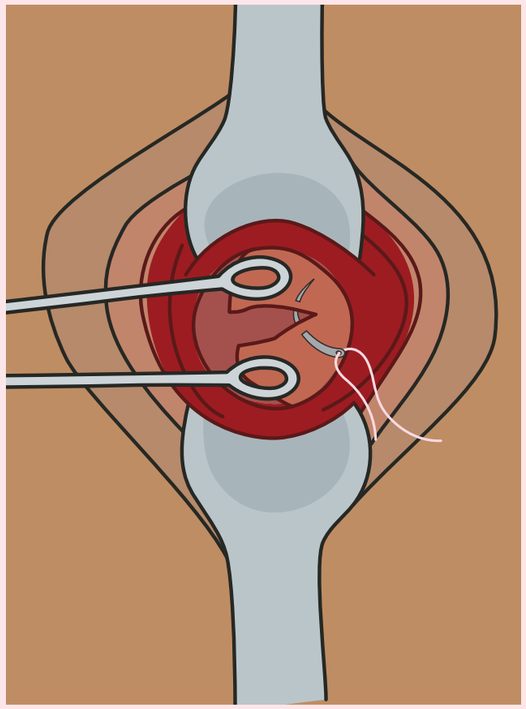
Expected actions	Prompts and info
Act and communicate ☐ Call for more help ☐ Communicate using SBAR ☐ Assign roles in the team ☐ Team confirms with closed-loop ☐ Assess and record: —uterine Tone —blood loss —blood flow —pulse and BP	**Simulator**: Uterus soft, medium blood flow. **If assessed:** • Pulse: 112 bpm • BP: 94/60 mmHg **Example prompt if steps missing:** • Can you get more help? • Should you use SBAR? • Should you take vital signs?
☐ Massage ☐ Catheterize bladder ☐ Clear cervix, sweep uterus, massage ☐ Insert second IV ☐ Give IV fluids ☐ Give additional oxytocic ☐ Assess vital signs ☐ Give 2nd dose of TXA ☐ Closed-loop communication ☐ Record	**Simulator**: After vaginal exam, tell participant they cleared clots from cervix. Make uterus firm and stop bleeding. **Example prompts:** • Do you need another IV? • Can you do anything else to stop the bleeding? • Can you give more medications?
Assess and record ☐ Check tone, blood loss and flow ☐ Check BP and pulse ☐ Record	**If assessed:** • Uterine tone: hard • Blood loss: more than 1000 mL • Blood flow: stopped • Pulse: 98 bpm • BP: 108/60 mmHg **Example prompt:** How is she now?

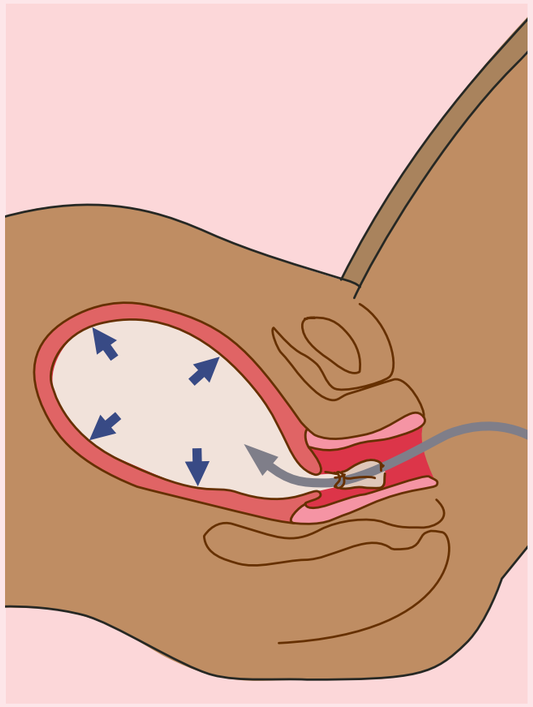


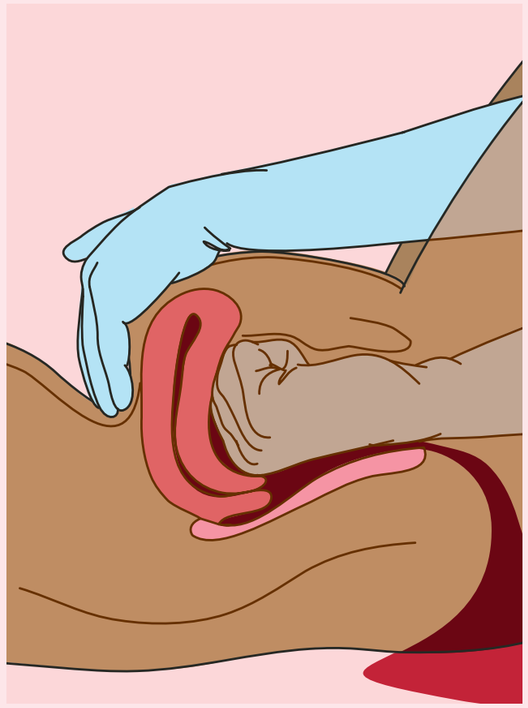
Module 5

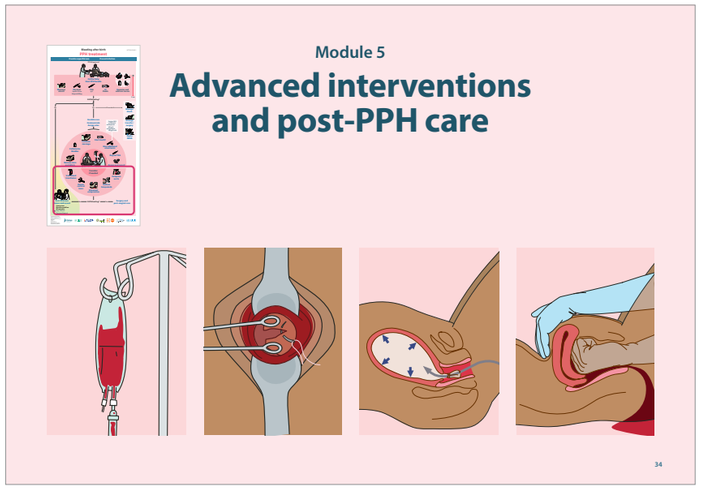
Advanced interventions and post-PPH care











Explain

By now you have:

- Completed the PPH bundle.
- Called for more help, assigned tasks.
- Inserted a 2nd IV and given extra doses of medications.
- Repaired tears, removed the placenta, and clots and tissue as needed.
- Applied NASG, if available.

If bleeding continues, consider:

- More than one cause of bleeding.
- Uterine rupture, especially if augmented labour or prior CS.
- Deep vaginal or cervical laceration.

> Remember – atony is the most common cause of PPH.

Additional care includes

Start safe blood transfusion

- If the woman has severe or ongoing PPH and there is a clinical need.
- Or if the woman is severely anaemic:
  - Hb <7 g/dl, or
  - she looks anaemic clinically.

Repair deep vaginal and cervical tears

- Recheck for vaginal and cervical lacerations that might have been missed.
- Repair if within your scope, or refer.

Apply bimanual compression

- As a temporizing measure.
- Squeeze the uterus to compress vessels, decrease blood flow and express clots.

Insert UBT

- To compress blood vessels from the inside of the uterus.

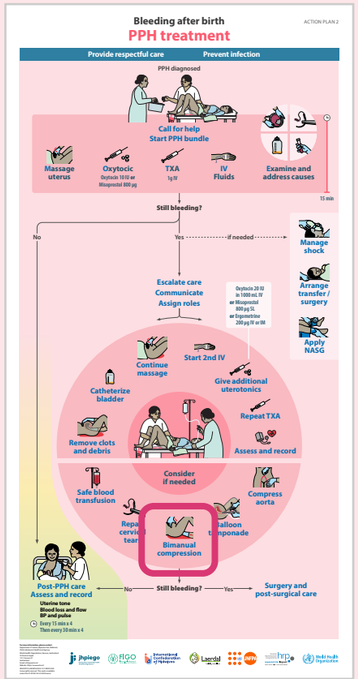
Apply aortic compression

- As a temporizing measure.
- Often done for better visualization when repairing a laceration or inserting a UBT.

Surgical interventions

> These interventions should only be conducted by trained health workers.

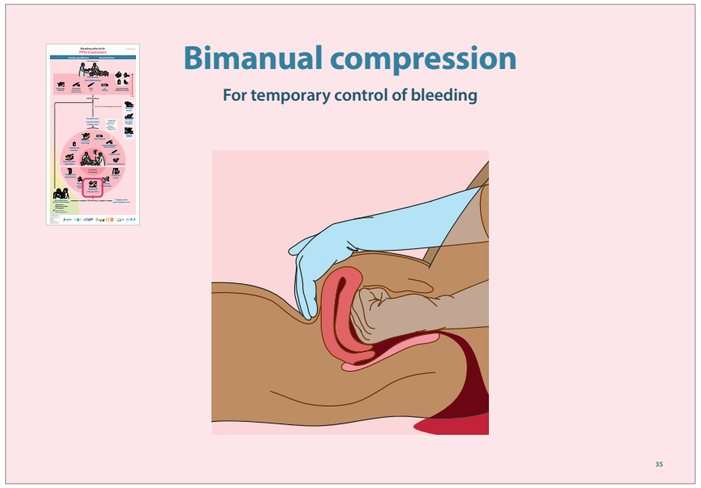
- When all other methods fail, these surgical can interventions address the causes of PPH directly:
  - uterine compression sutures
  - uterine or uteroovarian artery ligation
  - subtotal or total hysterectomy.
- See page 51 in the Provider Guide for details.
- Prepare for surgery and provide additional care in the operating room if:
  - blood loss >1 L, or
  - blood loss >500 mL with warning signs:
    pulse >100 bpm
    systolic BP <100 mmHg
    diastolic BP <60 mmHg
    SI >1.



Bimanual compression

For temporary control of bleeding





Explain

Act quickly. Do not wait until the woman is in shock to compress the uterus.

- Squeeze the uterus between your hands to reduce bleeding and buy time for medications to work.
- Bimanual compression can be used while awaiting placement of UBT or surgery.
- This is painful, but lifesaving. Inform the woman.
- Do not use vaginal packing. It can restrict optimal assessment of blood loss and cause infection.

Demonstrate and practise

- Show video **Uterine compression** (2:40 min) or demonstrate using the simulator.
- Ask participants to practise in pairs and provide guidance.
- As facilitator, wear the simulator with the baby and placenta delivered. Tighten the cervix.

Prepare

☐ Tell the woman what you are about to do, why, and that it will hurt.

☐ Consider pain medication.

☐ Wash hands and put on long sterile gloves.

Bimanual compression

☐ Form the hand into a cone and gently guide it into the vagina.

☐ Make a fist against the upper part of the vagina and the uterus.

☐ Place the other hand on the abdomen at the fundus.

☐ Compress the uterus between your hands for at least 5 min.

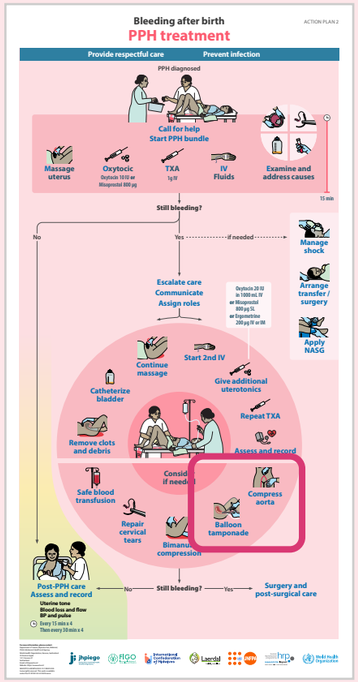
☐ After 5 min, slowly remove hand and check if bleeding restarts.

☐ If bleeding restarts, resume compression.

☐ If no bleeding:

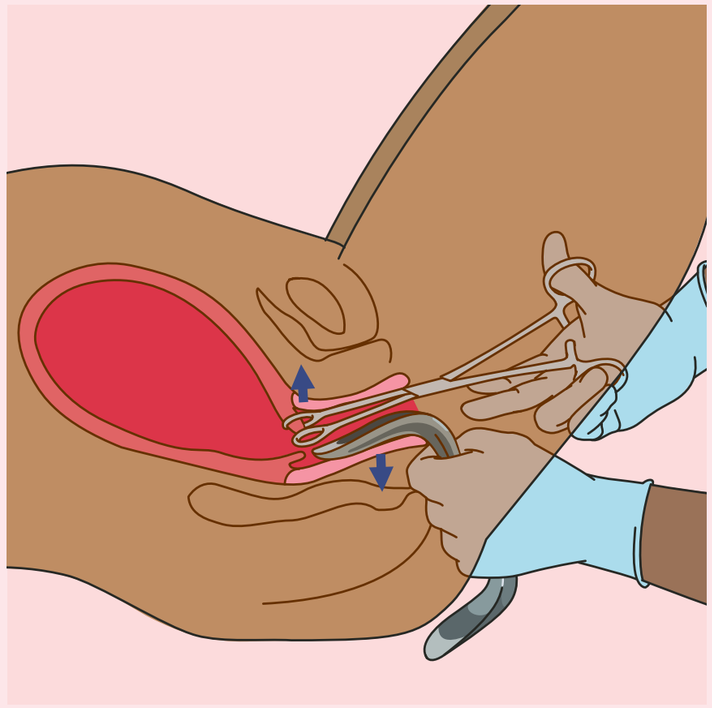
— ensure IV infusion with oxytocin continues

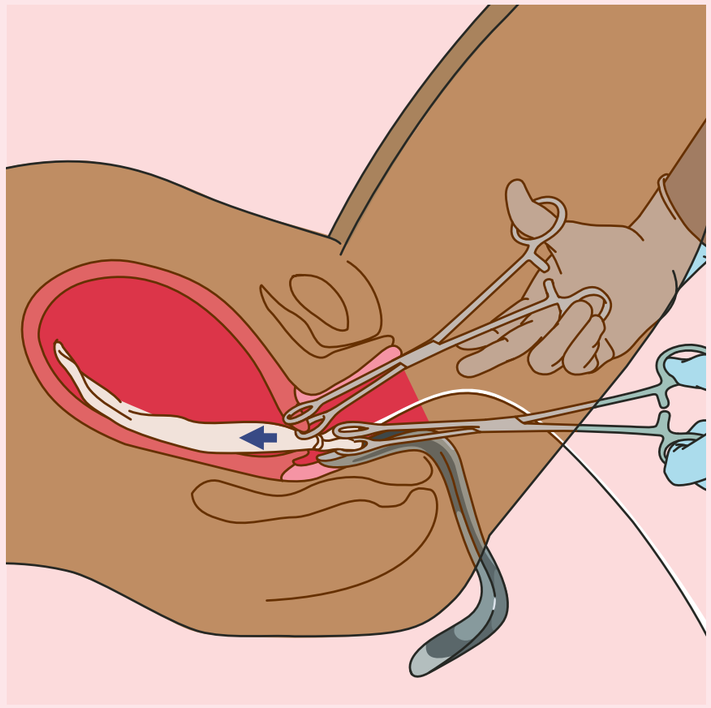
— monitor closely.

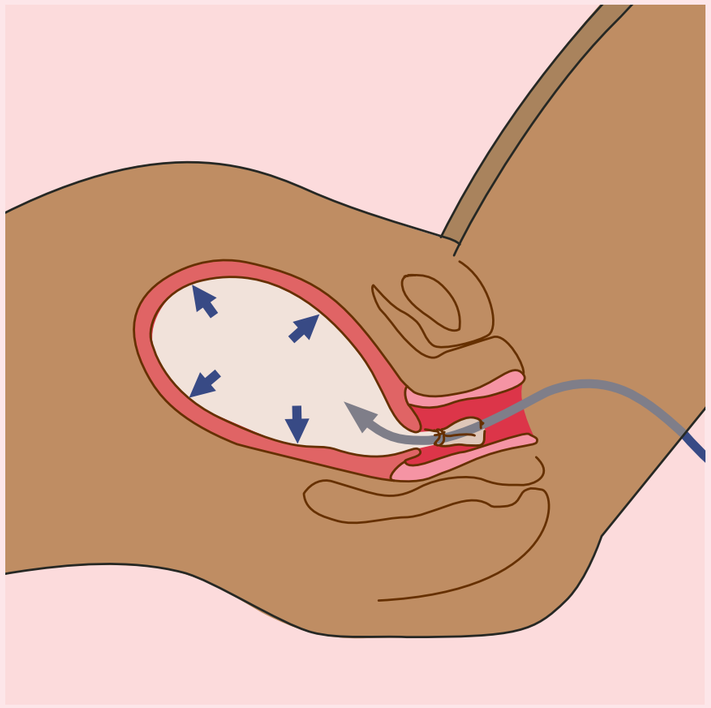


Insert uterine balloon tamponade

While compressing the aorta









Facilitation note

- **Skip this page if**
  - UBT is not in local guidelines, or
  - The facility does not meet WHO recommendations of immediate access to surgery, blood, and PPH bundle components.
- **Use locally preferred UBT.**
  - Have manufacturer instructions available for use during demonstration and practice.

Explain

- Insert a UBT only after completing the PPH bundle and if it fails.
- Do not place UBT until retained tissue and trauma have been ruled out.
- Keep UBT supplies in PPH emergency kit.
- Consider aortic compression before insertion to minimize bleeding for better visualization.

Advanced care note
Health workers should act within their scope of practice. UBT should only be placed by health workers specifically trained to do so.

Demonstrate and practise

- Show video **Aortic compression** (2:50 min) and UBT manufacturer video if available. Or demonstrate using a simulator.
- Use a postpartum uterus model or 500 mL bottle for practice.
- Ask participants to practise in pairs and provide guidance.

Aortic compression

- ☐ Ensure the woman is on a firm surface.
- ☐ Explain the procedure to her.
- ☐ With thumb tucked, make a fist and apply downward pressure on the abdominal aorta – just above and to the left of the umbilicus.
- ☐ Palpate the femoral pulse with your other hand.
- ☐ If still palpable, increase pressure.

UBT insertion prep

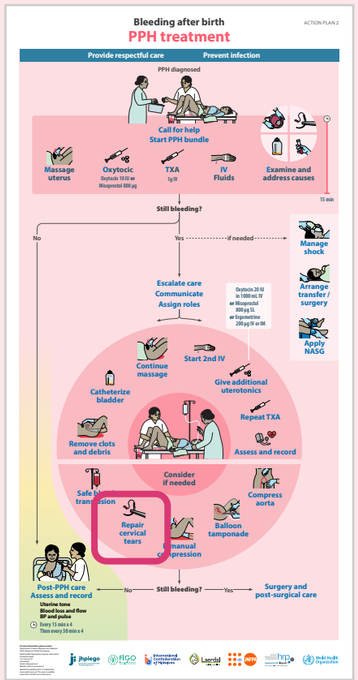
- ☐ Check and remove any retained clots and tissue.
- ☐ Ensure deep tears are repaired.
- ☐ Ensure empty bladder.
- ☐ Administer antibiotics (ampicillin 2 g or cefazolin 1 g IV).
- ☐ Explain the procedure to the woman and ensure privacy.

UBT insertion

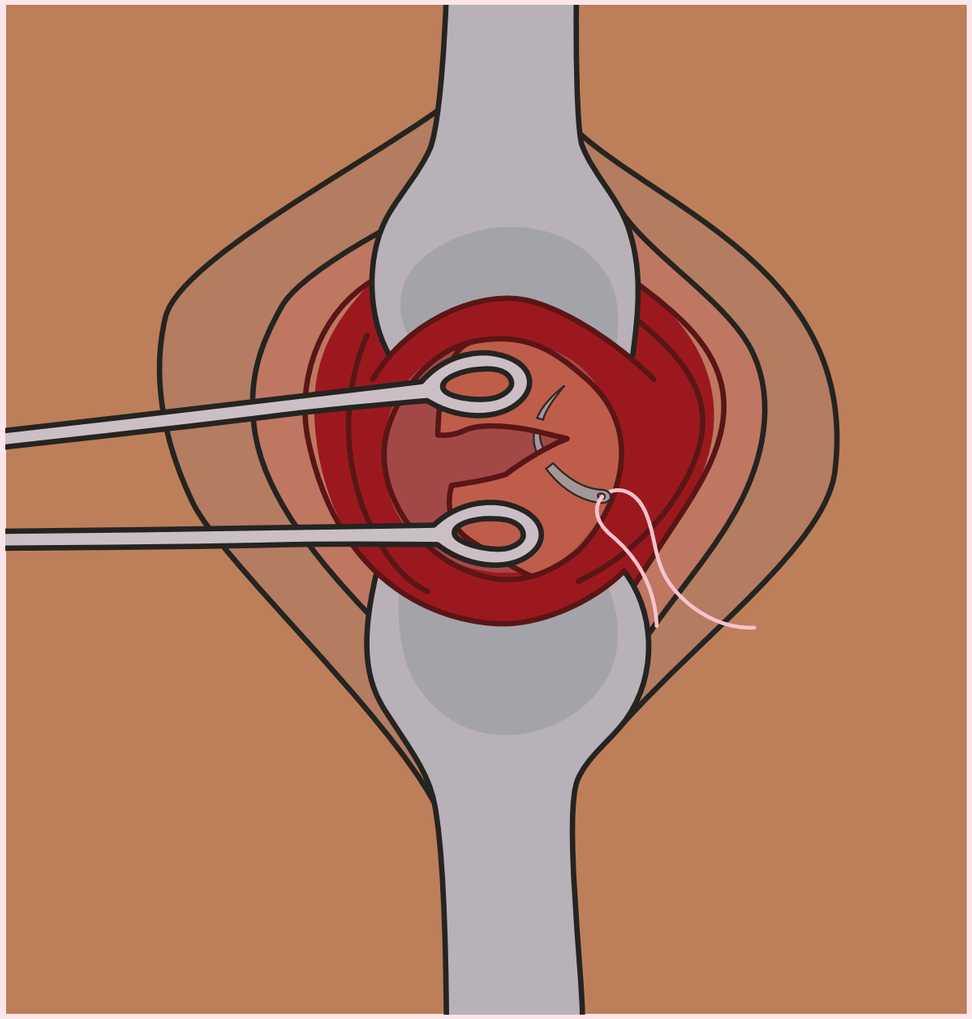
- ☐ Follow manufacturer's checklist for instructions.
- ☐ Monitor blood loss. If still bleeding after 15 min, seek surgical help.
- ☐ Record the procedure in the Postpartum Monitoring Form.
- ☐ Leave the UBT in place per manufacturer's instructions.

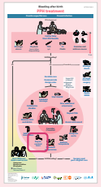
UBT removal

- ☐ Remove per manufacturer's instructions.
- ☐ Monitor closely.
- ☐ Assess and record tone, blood loss and flow, pulse and BP.

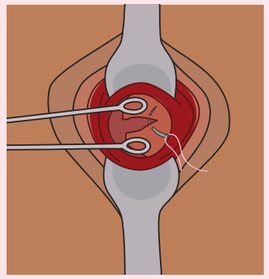


Repair deep tears





Repair deep tears



37

Facilitation note

- Follow local protocols on laceration repair.
- Participants can use foam blocks or towels as tear repair simulators. Encourage participants to create their own simulators for ongoing practice.

Explain

- Tears that cause PPH are most often deep vaginal or cervical tears.
- Cervical tears often occur at 3 and 9 o'clock.
- Bleeding from a tear may be slow, heavy, or spurting from an artery.
- Seek advanced care if repair does not stop bleeding – surgery may be required.

Demonstrate and practise

- Show video **Cervical tears** (5 min) or demonstrate. Then practise.

Vaginal tears – Identify

☐ Ensure privacy and inform the woman.

☐ Ensure good lighting.

☐ Give IV sedative per protocol. Do not give if in shock.

☐ Wash hands and wear sterile gloves.

☐ Clean perineum, vulva, and vagina with antiseptic solution.

☐ Gently separate the labia and assess completely.

☐ Wrap fingers in gauze and press on the vaginal floor to expose the entire sulcus.

Vaginal tears – Repair

☐ Draw 10 mL of 1% lignocaine into syringe.

☐ Insert the needle from the bottom to the apex on one side of the tear.

☐ Withdraw plunger to ensure needle is not in a blood vessel.

☐ Inject medication as needle is withdrawn.

☐ Repeat on other side of the tear. Wait 2 min for effect.

☐ Repair in 2 layers:

- — interrupted sutures to close deep layer first
- — continuous suturing starting just beyond the apex to close the vaginal mucosa.

☐ Avoid suturing into rectum.

☐ Review wound care and hygiene.

Cervical tears – Identify

☐ Have an assistant apply fundal pressure to help visualize the cervix.

☐ Use a sponge forceps to grasp cervix at 12 o'clock position and another at 3 o'clock.

☐ Inspect between the forceps for tears.

☐ Move the first forceps to 6 o'clock and inspect again.

☐ Continue to rotate forceps until full cervix is inspected.

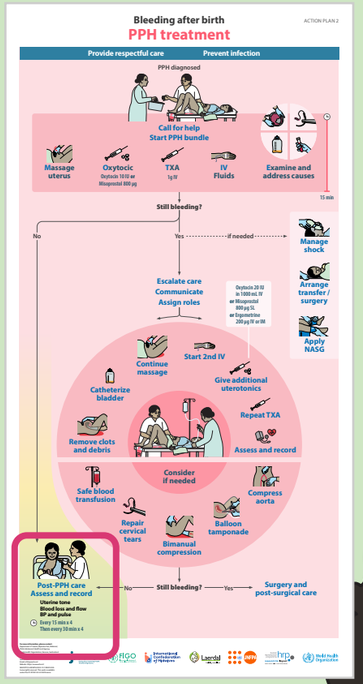
Cervical tears – Repair

☐ Gently hold both forceps in one hand.

☐ Use 0 Vicryl suture.

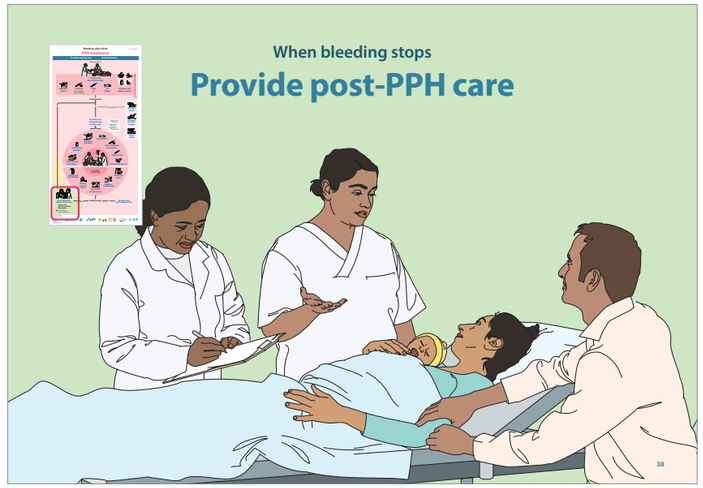
☐ Place first suture above the tear.

☐ Close tear with continuous suture.



When bleeding stops

Provide post-PPH care



Explain

Special care after PPH

- Monitor closely until stable.
- Assess and record tone, blood loss and flow, pulse and BP:
  - at least every 15 min until stable
  - then every 30 min for 2 h
  - then as per routine.
- A maintenance infusion of oxytocin 10–20 IU in 500–1000 mL may be needed in the 4 h after treatment to prevent recurrence.
- Increase PO hydration and remove urinary catheter.
- Decrease oxygen per protocol.
- Support breastfeeding.
- Ensure the woman's record is complete with final measured blood loss, timing of escalation, all assessments and interventions.

When stable

- Repeat lab tests – FBC/Hb per protocol.
- Gradually increase physical activity.

Home readiness

- Before discharge, ensure:
  - normal vital signs, urine output, and mental state
  - no dizziness with walking
  - she can care for herself and her baby
  - no sign of infection.
- Continue or begin antibiotics per protocol.

Discharge instructions

Counsel the woman and her family

- Ensure rest and good nutrition to support breastfeeding and recovery.
- Teach signs of anaemia, bleeding, and infection, and to seek care right away.
- Provide routine postpartum care including family planning.
- Counsel on the risk of PPH in future and importance of a facility birth.
- Recommend non-steroidal anti-inflammatory drugs or paracetamol for pain relief.
- Plan follow up for care at 48 h, 10 days, and 6 weeks.

In case of anaemia

- If Hb was less than 7 g/dl:
  - provide blood transfusion and consider IV iron
  - do not discharge until Hb >7 g/dl.
- If Hb was 7–11 g/dl discharge with at least 3 months of:
  - iron supplement 120 mg and
  - folic acid 400 µg daily.

Instructions in case of perineal tears

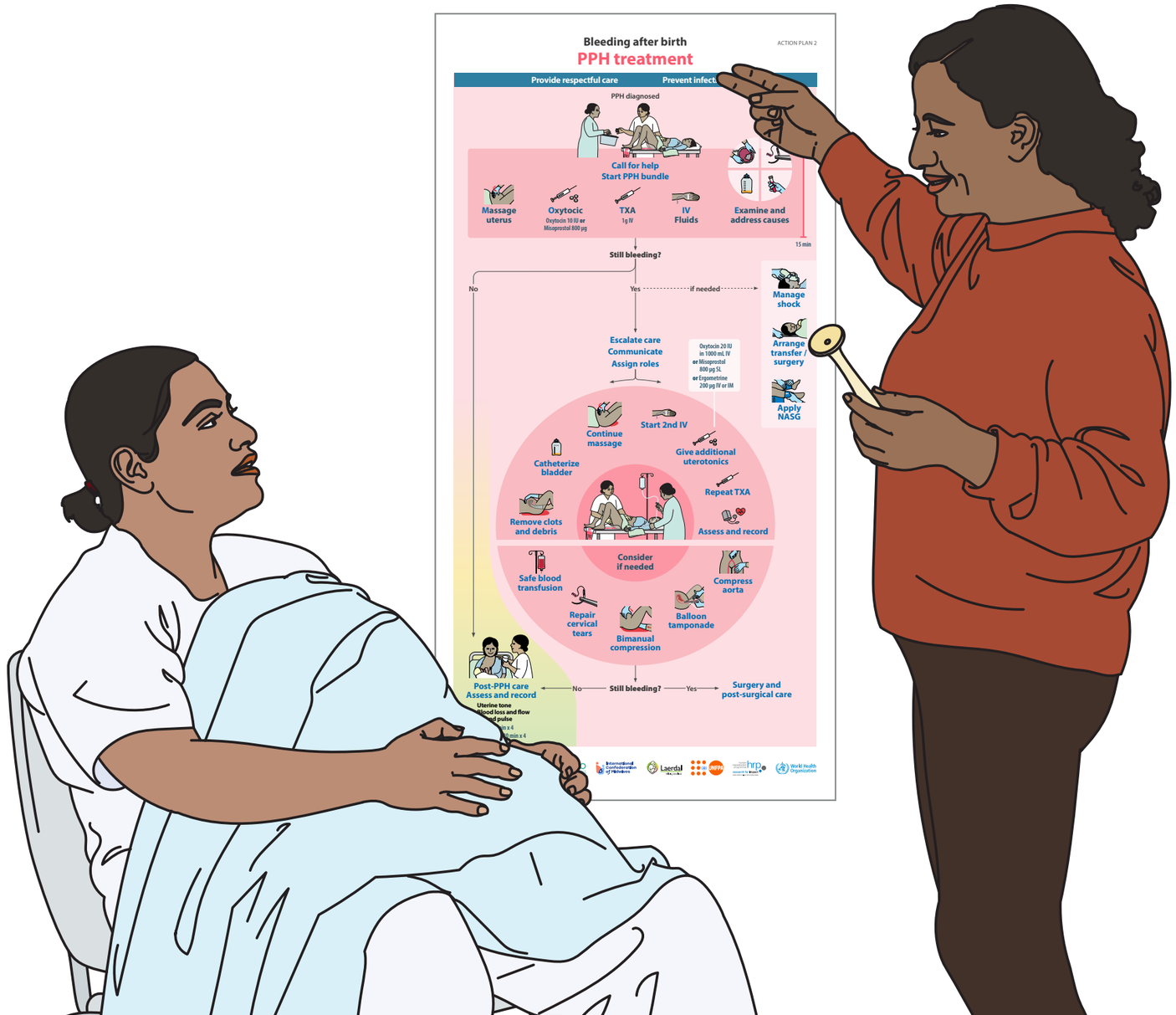
- Wash the perineum gently front to back, twice daily and after bowel movements.
- Change pads frequently.
- Wash hands before and after self-care.

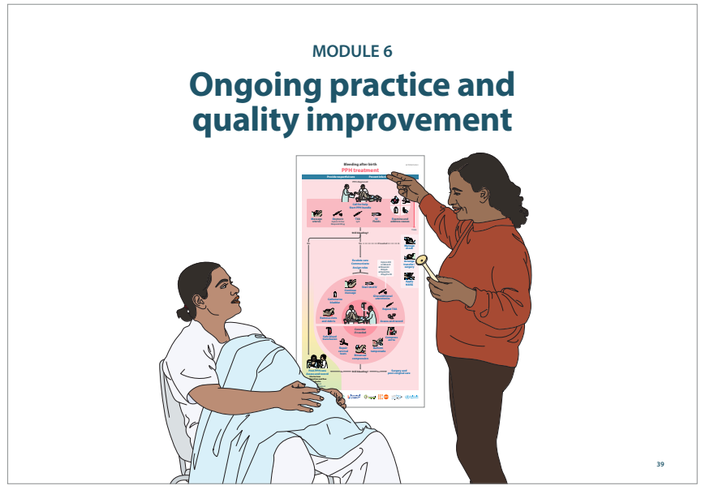
Debrief after PPH

- PPH can be a traumatic experience – for the woman, her family and the healthcare team.
- To learn how to debrief and support each other, review page 53 in the Provider Guide with your team.

MODULE 6

Ongoing practice and quality improvement





Facilitation note – assessments

- When you are done with this page you will move on to assessments:
  - post-course knowledge check
  - OSCEs.
- Have sheets and equipment ready. See page 1b.

Explain

- LDHF is an approach for training:
  - on-site
  - hands-on
  - with the whole team.
- It ensures continuous improvement, better retention of skills, and sustainable behavioural change.

Key Points of LDHF

- **Competency focused** on mastering clinical decision-making, essential skills and behaviours.
- **Case-based training** with simulations and hands-on practice.
- **Team and facility-based** to support multidisciplinary teamwork.
- **Brief ongoing activities** after the initial course:
  - brief skills practice and drills
  - quality improvement activities
  - based on activities outlined in the Provider Guide
  - led by peers selected from your facility.
- **Measuring** quality of care indicators, clinical performance, and outcomes to **identify areas for improvement** guided by local needs and adaptation:
  - uterotonic use for prevention
  - facility use of simulation and drills
  - presence of formal protocols for PPH prevention, treatment, blood transfusion and referral
  - PPH cases, severe PPH cases (>1000 mL) cases treated, and deaths from PPH.

Discuss

Have health workers turn to the table of contents in the Provider Guide to review practice sessions.

- What do you think you need to practise at your facility?
- Do you do regular drills at your facility already?

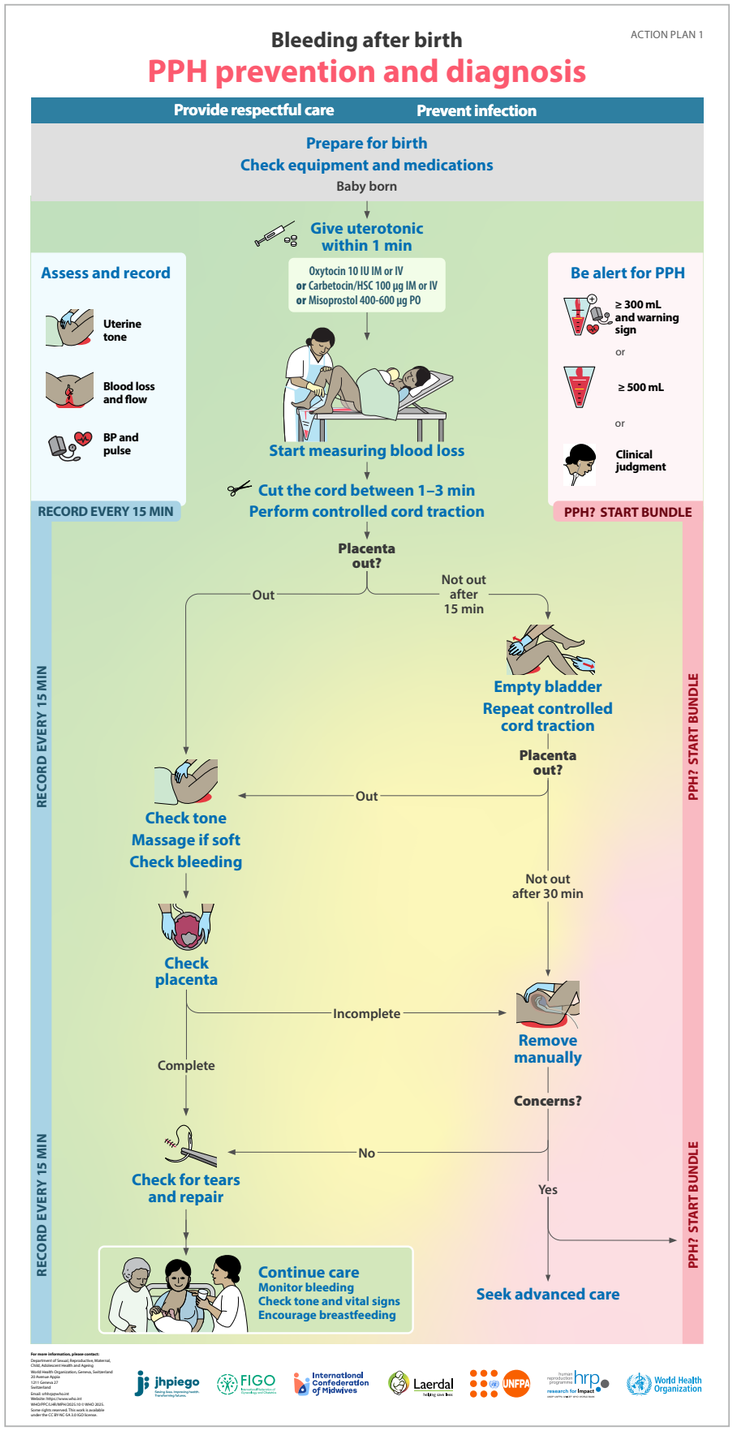
References

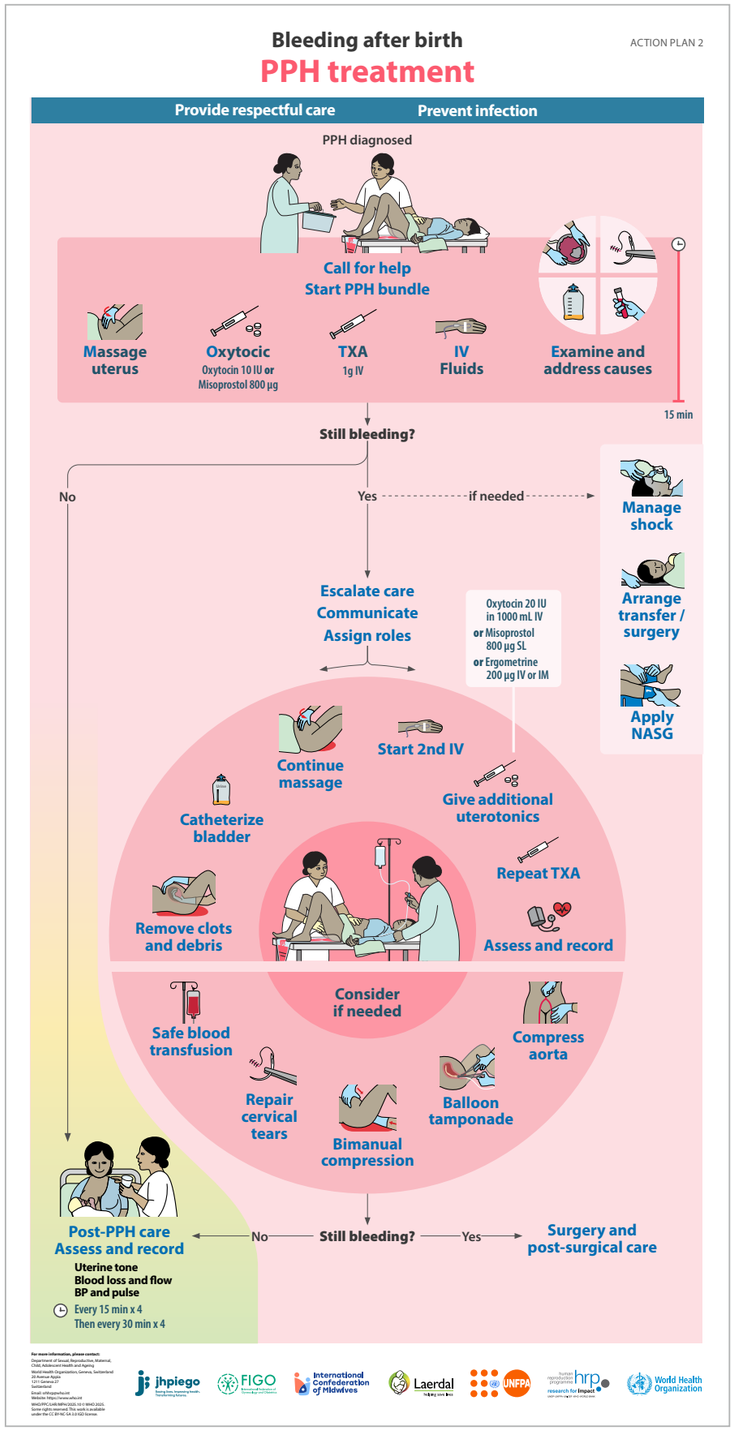
1. Consolidated guidelines for the prevention, diagnosis and treatment of postpartum haemorrhage. Web Annexes: Evidence base. Geneva: World Health Organization; 2025. Licence: CC BY-NC-SA 3.0 IGO.
2. Consolidated guidelines for the prevention, diagnosis and treatment of postpartum haemorrhage. Geneva: World Health Organization; 2025. Licence: CC BY-NC-SA 3.0 IGO.
3. WHO labour care guide: user's manual. Geneva: World Health; 2020. Organization. License: CC BY-NC-SA 3.0 IGO

Additional resources

Find all learning materials, job aids and planing tools at www.who.int/tools/bleeding-after-birth

- Action Plans
- Provider Puide
- postpartum monitoring form
- knowledge assessment
- OSCEs
- videos
- planing templates.





The Bleeding after birth training course aims to translate standards into care. It incorporates the latest WHO recommendations related to PPH[1,2] and includes links to key references and resources for training.

www.who.int/tools/bleeding-after-birth

Course development methods. The development of this training product followed a rigorous, evidence-based, and iterative process. The training package was validated using the Kirkpatrick framework, assessing feasibility, acceptability, changes in provider performance, and improvements in health outcomes. The materials and teaching methods are grounded in adult learning principles and clinical best practices, ensuring relevance and effectiveness in real-world settings. Beta versions of the package were piloted in Afghanistan, Democratic Republic of the Congo, Guinea, India, Kenya, Malawi, Nigeria, United Republic of Tanzania and Zambia with adaptations made based on feedback and implementation learnings from these contexts. The training package was used in the EMOTIVE trial, providing robust evidence for its effectiveness and impact on maternal and newborn health outcomes. The development process involved collaboration with global health experts and institutions to ensure alignment with international standards and contextual relevance.

Declaration of interest. All external experts signed declaration of interest to WHO, disclosing any potential conflicts of interest that might influence or could reasonably be perceived to influence their objectivity and independence in relation to the guide's content. WHO thoroughly reviewed each declaration and determined that none posed an actual or reasonably perceived conflict of interest concerning any aspect of the course.

For more information, please contact:

Department of Sexual, Reproductive, Maternal, Child, Adolescent Health and Ageing

World Health Organization, Geneva, Switzerland
20 Avenue Appia
1211 Geneva 27
Switzerland

Email: srhhrp@who.int
Website: https://www.who.int